

AGIP GAS BV

Libyan Branch

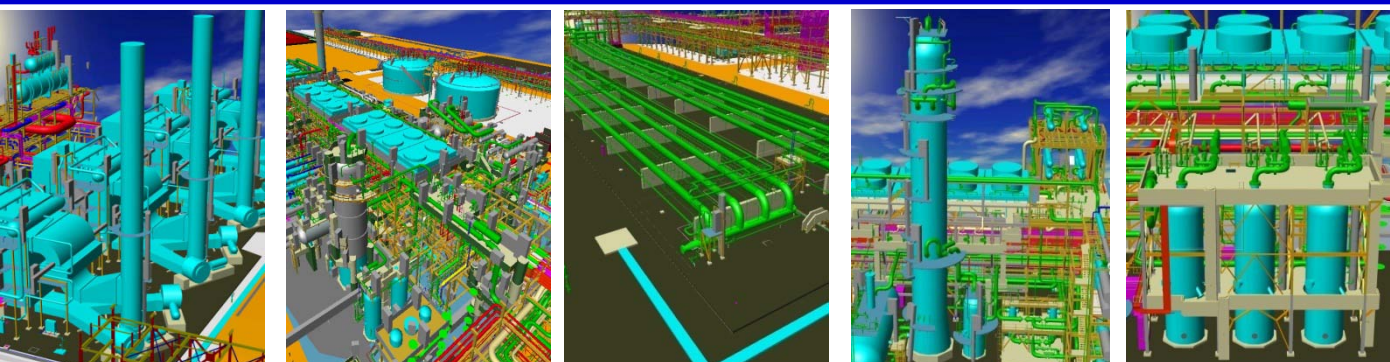
WESTERN LIBYA GAS PROJECT MELLITAH PLANT

OPERATING MANUAL Gas Treatment Units

VOLUME IV

June, 2009

AGBV Job No. 0177/NC/02



Snamprogetti Lummus Gas Ltd



Snamprogetti



ABB Lummus Malta



AgipGas BV LIBYAN BRANCH		WESTERN LIBYA GAS PROJECT		AGBV Job No.		AGBV Doc. No.	
				0177/NC/02		G-50-P-Z-0006 00	
Snamprogetti Lummus Gas Ltd  Eni ENI  ABB Lummus Malta  HYUNDAI ENGINEERING & CONSTRUCTION CO., LTD.	Plant		CTR Job No.		CTR Doc. No.		
	MELLITAH PLANT		310300		50 000-ZA-E-03073		
	Location		GSPLAJ		i		Rev.

General Index

– Volume I	◆ Gas Treatments Units Operating Manual Description
– Volume II	◆ Attachment 11.1 – Block Flow Diagram (PFD), Process Flow Diagram (PFD), Process Safeguarding Diagram (PSD) ◆ Attachment 11.2 – Symbols and Legend P&ID's
– Volume III	◆ Attachment 11.2 – P&ID's Units 50-190, 50-210, 51-210
–	◆
– Volume IV	◆ Attachment 11.2 – P&ID's Units 51-311, 51-320, 51-330, 51-340, 50-550, 51-550
– Volume V	◆ Attachment 11.2 – Process Interconnecting P&ID, Utility Distribution P&ID
– Volume VI	◆ Attachment 11.3 – Plot Plans ◆ Attachment 11.4 – Electrical One-Line Diagram
– Volume VII	◆ Attachment 11.5 – Process Material Balance
– Volume VIII	◆ Attachment 11.6 – Design Premises ◆ Attachment 11.7 – Tie-in list with WAFA Plant ◆ Attachment 11.8 – Utilities Consumption Summary ◆ Attachment 11.9 – Chemical Consumption Summary Mellitah Plant ◆ Attachment 11.10 – Effluent Summary Report ◆ Attachment 11.11 – Electrical Load Summary ◆ Attachment 11.12 – Equipment Summary ◆ Attachment 11.13 – Equipment Process Data Sheets, Units 50-190, 50-210, 51-210, 51-311
– Volume IX	◆ Attachment 11.13 – Equipment Process Data Sheets, Units 51-320, 50-330, 51-330, 51-340, 50-550, 51-550
– Volume X	◆ Attachment 11.14 – Instrument Data Sheets for Control Valves, Safety Valves, Rupture Disks and Analyzers
– Volume XI	◆ Attachment 11.15 – Cause & Effect Diagram, ESD Diagram (Shut Down logic diagram), Specification of Logic Sequence
– Volume XII	◆ Attachment 11.16 – Control Narratives ◆ Attachment 11.17 – DCS Graphic Display

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				0177/NC/02		G-50-P-Z-0006 00													
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		MELLITAH PLANT		310300		50 000-ZA-E-03073													
		Location		GSPLAJ		ii		Rev.	<table><tr><td>P1</td><td>A1</td><td>E1</td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td></tr></table>	P1	A1	E1							
P1	A1	E1																	

Volume II A Index

Attachment 11.1

◆ Block Flow Diagram

BFD Title	BFD Number
Block Flow Diagram	G-65-D-O-2001-01

◆ Process Flow Diagram

PFD Title		PFD Number
Unit 50-190	Receiving Trap & Slug Catcher	G-65-D-O-1119-01
Unit 51-210	Condensate Stabilization Unit	G-65-D-O-1121-01
Unit 50-210	Condensate Stabilization Common Part	G-65-D-O-1121-02
Unit 50-210	Condensate Off-Spec Storage	G-65-D-O-1121-03
Unit 51-311	Gas Dehydration	G-65-D-O-1131-01
Unit 51-320	NGL Fractionation	G-65-D-O-1132-01
Unit 51-330	Gas Sweetening H ₂ S and CO ₂ Absorption	G-65-D-O-1133-01
Unit 51-330	Gas Sweetening Amine Regeneration	G-65-D-O-1133-02
Unit 50-330	Gas Sweetening Common Part – Amine Storage & Recovery	G-65-D-O-1133-03
Unit 51-340	Gas Dew Point Control	G-65-D-O-1134-01
Unit 50-550	Closed Drains NGL Fract. & Flare Area	G-65-D-O-1155-01
Unit 50-550	Closed Drains Inlet Reception & Common Facilities	G-65-D-O-1155-02
Unit 51-550	Closed Drains Gas Train Dehydration Drain Facilities	G-65-D-O-1155-03

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	MELLITAH PLANT		310300		50 000-ZA-E-03073						
	Location		GSPLAJ		iii		Rev.	P1	A1	E1	

◆ Process Safeguarding Diagram

PSD Title		PSD Number
Unit 50-190	Receiving Trap & Slug Catcher	G-65-D-O-1419-01
Unit 51-210	Condensate Stabilization Unit	G-65-D-O-1421-01
Unit 50-210	Condensate Stabilization Common Part	G-65-D-O-1421-02
Unit 50-210	Condensate Off-Spec Storage	G-65-D-O-1421-03
Unit 51-311	Gas Dehydratation	G-65-D-O-1431-01
Unit 51-330	Gas Sweetening H ₂ S and CO ₂ Absorption	G-65-D-O-1433-01
Unit 51-330	Gas Sweetening Amine Regeneration	G-65-D-O-1433-02
Unit 51-340	Gas Dew Point Control	G-65-D-O-1434-01

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		0177/NC/02	G-50-P-Z-0006 00			
 Snamprogetti  Eni  ABB ABB Lummus Malta  HYUNDAI ENGINEERING & CONSTRUCTION CO., LTD.	Plant	CTR Job No.	CTR Doc. No.			
	MELLITAH PLANT	310300	50 000-ZA-E-03073			
	Location	ii	Rev.	P1	A1	E1

Volume II B Index

Attachment 11.2

◆ Symbols and Legend P&ID

Symbols and Legend P&ID Title	P&ID Number
Lines ID., Fluid type, Instr. Primary Elem. and Valves Symbols	G-00-D-O-1000-01
Instrument/Function Symbols, Letter Combination & Inst. Line Symb.	G-00-D-O-1000-02
Valves & Mech. Elem. Conn. Symbols, Abbrev., Size for C.V., Equipments Isolation	G-00-D-O-1000-03
MOV-EBV-BDV-UV Typical Arrangements	G-00-D-O-1000-04
Typical for Press, Temp. & Flow Instruments	G-00-D-O-1000-05
Level Instruments Install. & Steam Trap Assembly	G-00-D-O-1000-06
Typical for Sample Connection & Analyzers	G-00-D-O-1000-07
Sampling Connection List	G-00-D-O-1000-08
List & Typical Vent/Drain for AFC	G-00-D-O-1000-09
Equipment Symbology for PFD	G-00-D-O-1000-10
Equipment Symbology for P&ID	G-00-D-O-1000-11
Typical for Air Cooler	G-00-D-O-1000-12
Typical for Electric Motors	G-00-D-O-1000-13
Symbols and Legend P&ID Title	P&ID Number
Typical for Electric Motors & Air Fin Cooler	G-00-D-O-1000-14
Pump Seal Details	G-00-D-O-1000-15
General Notes	G-00-D-O-1000-16
Typical for Electric Motors	G-00-D-O-1000-17
Analyzer List	G-00-D-O-1000-18
Sampling Connection List	G-00-D-O-1000-19
Pump Seal List	G-00-D-O-1000-25

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	Location GSPLAJ	<i>ii</i>	Rev.	P1	A1
				E1	

Volume III Index

Attachment 11.2

♦ P&ID

P&ID Title		P&ID Number
Unit 50-190	Receiving trap & slug catcher gas pipeline receiving pig trap	G-65-D-O-1019-01
Unit 50-190	Receiving trap & slug catcher cond. pipeline receiving pig trap	G-65-D-O-1019-02
Unit 50-190	Receiving trap & slug catcher gas slug catcher manifold	G-65-D-O-1019-03
Unit 50-190	Receiving trap & slug catcher slug catcher outlet lines	G-65-D-O-1019-04
Unit 50-190	Receiving trap & slug catcher KO drum	G-65-D-O-1019-05
Unit 50-190	Receiving trap & slug catcher gas slug catcher	G-65-D-O-1019-06
Unit 50-210	Condensate stabilization inlet common header	G-65-D-O-1021-01
Unit 51-210	Condensate stabilization surge drum	G-65-D-O-1021-02
Unit 51-210	Condensate stabilization stabilizer feed/bottom exch	G-65-D-O-1021-03
Unit 51-210	Condensate stabilization condensate stabilizer	G-65-D-O-1021-04
Unit 51-210	Condensate stabilization outlet header	G-65-D-O-1021-05
Unit 50-210	Condensate stabilization off-spec tank	G-65-D-O-1021-06
Unit 50-210	Condensate stabilization offgas compr.suct.& disch. Interconnecting	G-65-D-O-1021-07
Unit 50-210	Condensate stabilization offgas compr.1st stage "A" K.O. Drum	G-65-D-O-1021-08
Unit 50-210	Condensate stabilization offgas compr.2nd stage "A" K.O.Drum	G-65-D-O-1021-09
Unit 50-210	Condensate stabilization offgas compr. Disch. Drum "A"	G-65-D-O-1021-10
Unit 50-210	Condensate stabilization offgas compr. 1st stage "B" K.O.Drum	G-65-D-O-1021-11
Unit 50-210	Condensate stabilization offgas compr. 2nd stage "B" K.O.Drum	G-65-D-O-1021-12
Unit 50-210	Condensate stabilization offgas compr. Disch. Drum "B"	G-65-D-O-1021-13
Unit 50-210	Condensate stabilization offgas compr.(1st/2nd stage) "A"	G-65-D-O-1021-14
Unit 50-210	Condensate stabilization offgas compr.(1st/2nd stage) "B"	G-65-D-O-1021-15

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				0177/NC/02		G-50-P-Z-0006 00					
Snamprogetti Lummus Gas Ltd  Snamprogetti ABB ABB Lummus Malta HYUNDAI ENGINEERING & CONSTRUCTION CO., LTD.	Plant		CTR Job No.		CTR Doc. No.						
	MELLITAH PLANT		310300		50 000-ZA-E-03073						
	Location		GSPLAJ		iii		Rev.	P1	A1	E1	

P&ID Title		P&ID Number
Unit 50-210	Condensate stabilization offgas compr. 1st stage "C" K.O.Drum	G-65-D-O-1021-16
Unit 50-210	Condensate stabilization offgas compr.(1st/2nd stage) "C"	G-65-D-O-1021-17
Unit 50-210	Condensate stabilization offgas compr.2nd stage "C" K.O.Drum	G-65-D-O-1021-18
Unit 50-210	Condensate stabilization offgas compr. Disch. Drum "C"	G-65-D-O-1021-19

- Unit 52 P&ID's

Not included – Refer to Unit 51 drawings

P&ID Title		P&ID Number
Unit 52-210	Condensate stabilization surge drum	G-68-D-O-1021-02
Unit 52-210	Condensate stabilization stabilizer feed/bottom exch	G-68-D-O-1021-03
Unit 52-210	Condensate stabilization condensate stabilizer	G-68-D-O-1021-04
Unit 52-210	Condensate stabilization outlet header	G-68-D-O-1021-05

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 Snamprogetti  Eni  ABB ABB Lummus Malta  HYUNDAI ENGINEERING & CONSTRUCTION CO., LTD.	Plant MELLITAH PLANT	CTR Job No. 310300	CTR Doc. No. 50 000-ZA-E-03073		
	Location GSPLAJ	<i>ii</i>	Rev.	P1	A1
				E1	

Volume IV Index

Attachment 11.2

♦ P&ID

P&ID Title		P&ID Number
Unit 51-311	Gas dehydration coalescer	G-65-D-O-1031-01
Unit 51- 311	Gas dehydration dryer	G-65-D-O-1031-02
Unit 51- 311	Gas dehydration dryers	G-65-D-O-1031-03
Unit 51- 311	Gas dehydration molec. sieves outlet filters	G-65-D-O-1031-04
Unit 51- 311	Gas dehydration regeneration gas heater	G-65-D-O-1031-05
Unit 51- 311	Gas dehydration regeneration gas cooling	G-65-D-O-1031-06
Unit 51- 311	Gas dehydration gas re-compress.aftercooler	G-65-D-O-1031-07
Unit 51- 311	Gas dehydration regeneration gas heater safeguarding system	G-65-D-O-1031-08
Unit 51- 311	Gas dehydration gas re-compress. "A"	G-65-D-O-1031-09
Unit 51- 311	Gas dehydration gas re-compress. "B"	G-65-D-O-1031-10
Unit 51- 320	NGL Fractionation deethanizer	G-65-D-O-1032-01
Unit 51- 320	NGL Fractionation product cooling	G-65-D-O-1032-02
Unit 51- 320	NGL Fractionation debutanizer	G-65-D-O-1032-03
Unit 51- 320	NGL Fractionation debutanizer reflux system	G-65-D-O-1032-04
Unit 50- 330	Gas sweetening amine recovery system	G-65-D-O-1033-01
Unit 50- 330	Gas sweetening amine prep. & storage system	G-65-D-O-1033-02
Unit 51- 330	Gas sweetening raw gas inlet manifold	G-65-D-O-1033-03
Unit 51- 330	Gas sweetening raw gas preheating	G-65-D-O-1033-04
Unit 51- 330	Gas sweetening HP absorber	G-65-D-O-1033-05
Unit 51- 330	Gas sweetening treated gas KO drum	G-65-D-O-1033-06
Unit 51- 330	Gas sweetening amine flash drum	G-65-D-O-1033-07
Unit 51- 330	Gas sweetening rich/lean amine exchangers	G-65-D-O-1033-08
Unit 51- 330	Gas sweetening amine regenerator	G-65-D-O-1033-09
Unit 51- 330	Gas sweetening amine regenerator reflux section	G-65-D-O-1033-10
Unit 51- 330	Gas sweetening amine pumps and storage	G-65-D-O-1033-11

AgipGas BV LIBYAN BRANCH		WESTERN LIBYA GAS PROJECT		AGBV Job No. 0177/NC/02		AGBV Doc. No. G-50-P-Z-0006 00				
Snamprogetti Lummus Gas Ltd  Eni GROUP  ABB Lummus Malta  HYUNDAI ENGINEERING & CONSTRUCTION CO., LTD.	Plant MELLITAH PLANT		CTR Job No. 310300		CTR Doc. No. 50 000-ZA-E-03073					
	Location GSPLAJ		iii		Rev.	P1	A1	E1		

P&ID Title		P&ID Number
Unit 51- 330	Gas sweetening antifoam package & sump drum	G-65-D-O-1033-12
Unit 51- 330	Gas sweetening lean amine filtration	G-65-D-O-1033-13
Unit 51- 330	Gas sweetening steam & condensate header	G-65-D-O-1033-14
Unit 51- 330	Gas sweetening raw gas inlet filters	G-65-D-O-1033-15
Unit 51- 340	Gas HC dew point control dry gas cooling	G-65-D-O-1034-01
Unit 51- 340	Gas HC dew point control dry gas cooling	G-65-D-O-1034-02
Unit 51- 340	Gas HC dew point control condensate separator	G-65-D-O-1034-03
Unit 51- 550	Closed drain gas trains area	G-65-D-O-1055-01
Unit 50- 550	Closed drain slug catcher area & condensate stabilization area	G-65-D-O-1055-02
Unit 50- 550	Closed drain NGL fractionation area	G-65-D-O-1055-03
Unit 50- 550	Closed drain HP flare area	G-65-D-O-1055-05

- Units 52/53 P&ID's

Not included – Refer to Unit 51 drawings

P&ID Title		P&ID Number
Unit 52-311	Gas dehydration coalescer	G-68-D-O-1031-01
Unit 52-311	Gas dehydration dryer	G-68-D-O-1031-02
Unit 52-311	Gas dehydration dryers	G-68-D-O-1031-03
Unit 52-311	Gas dehydration molec. sieves outlet filters	G-68-D-O-1031-04
Unit 52-311	Gas dehydration regeneration gas heater	G-68-D-O-1031-05
Unit 52-311	Gas dehydration regeneration gas cooling	G-68-D-O-1031-06
Unit 52-311	Gas dehydration gas re-compress.aftercooler	G-68-D-O-1031-07
Unit 52-311	Gas dehydration regeneration gas heater safeguarding system	G-68-D-O-1031-08
Unit 52-311	Gas dehydration gas re-compress. "A"	G-68-D-O-1031-09
Unit 52-311	Gas dehydration gas re-compress. "B"	G-68-D-O-1031-10
Unit 52-320	NGL Fractionation deethanizer	G-68-D-O-1032-01
Unit 52-320	NGL Fractionation product cooling	G-68-D-O-1032-02
Unit 52-320	NGL Fractionation debutanizer	G-68-D-O-1032-03

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				0177/NC/02		G-50-P-Z-0006 00	
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	MELLITAH PLANT		310300		50 000-ZA-E-03073		
	Location		GSPLAJ		iv		Rev.

P&ID Title		P&ID Number
Unit 52-320	NGL Fractionation debutanizer reflux system	G-68-D-O-1032-04
Unit 52-330	Gas sweetening raw gas inlet manifold	G-68-D-O-1033-03
Unit 52-330	Gas sweetening raw gas preheating	G-68-D-O-1033-04
Unit 52-330	Gas sweetening HP absorber	G-68-D-O-1033-05
Unit 52-330	Gas sweetening treated gas KO drum	G-68-D-O-1033-06
Unit 52-330	Gas sweetening amine flash drum	G-68-D-O-1033-07
Unit 52-330	Gas sweetening rich/lean amine exchangers	G-68-D-O-1033-08
Unit 52-330	Gas sweetening amine regenerator	G-68-D-O-1033-09
Unit 52-330	Gas sweetening amine regenerator reflux section	G-68-D-O-1033-10
Unit 52-330	Gas sweetening amine pumps and storage	G-68-D-O-1033-11
Unit 52-330	Gas sweetening antifoam package & sump drum	G-68-D-O-1033-12
Unit 52-330	Gas sweetening lean amine filtration	G-68-D-O-1033-13
Unit 52-330	Gas sweetening steam & condensate header	G-68-D-O-1033-14
Unit 52-330	Gas sweetening raw gas inlet filters	G-68-D-O-1033-15
Unit 52-340	Gas HC dew point control dry gas cooling	G-68-D-O-1034-01
Unit 52-340	Gas HC dew point control dry gas cooling	G-68-D-O-1034-02
Unit 52-340	Gas HC dew point control condensate separator	G-68-D-O-1034-03
Unit 52-550	Closed drain gas trains area	G-68-D-O-1055-01
Unit 53-311	Gas dehydration coalescer	G-69-D-O-1031-01
Unit 53-311	Gas dehydration dryer	G-69-D-O-1031-02
Unit 53-311	Gas dehydration dryers	G-69-D-O-1031-03
Unit 53-311	Gas dehydration molec. sieves outlet filters	G-69-D-O-1031-04
Unit 53-311	Gas dehydration regeneration gas heater	G-69-D-O-1031-05
Unit 53-311	Gas dehydration regeneration gas cooling	G-69-D-O-1031-06
Unit 53-311	Gas dehydration gas re-compress.aftercooler	G-69-D-O-1031-07
Unit 53-311	Gas dehydration regeneration gas heater safeguarding system	G-69-D-O-1031-08
Unit 53-311	Gas dehydration gas re-compress. "A"	G-69-D-O-1031-09
Unit 53-311	Gas dehydration gas re-compress. "B"	G-69-D-O-1031-10
Unit 53-330	Gas sweetening raw gas inlet manifold	G-69-D-O-1033-03

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	Location GSPLAJ	V	Rev.	P1	A1
				E1	

P&ID Title		P&ID Number
Unit 53-330	Gas sweetening raw gas preheating	G-69-D-O-1033-04
Unit 53-330	Gas sweetening HP absorber	G-69-D-O-1033-05
Unit 53-330	Gas sweetening treated gas KO drum	G-69-D-O-1033-06
Unit 53-330	Gas sweetening amine flash drum	G-69-D-O-1033-07
Unit 53-330	Gas sweetening rich/lean amine exchangers	G-69-D-O-1033-08
Unit 53-330	Gas sweetening amine regenerator	G-69-D-O-1033-09
Unit 53-330	Gas sweetening amine regenerator reflux section	G-69-D-O-1033-10
Unit 53-330	Gas sweetening amine pumps and storage	G-69-D-O-1033-11
Unit 53-330	Gas sweetening antifoam package & sump drum	G-69-D-O-1033-12
Unit 53-330	Gas sweetening lean amine filtration	G-69-D-O-1033-13
Unit 53-330	Gas sweetening steam & condensate header	G-69-D-O-1033-14
Unit 53-330	Gas sweetening raw gas inlet filters	G-69-D-O-1033-15
Unit 53-340	Gas HC dew point control dry gas cooling	G-69-D-O-1034-01
Unit 53-340	Gas HC dew point control dry gas cooling	G-69-D-O-1034-02
Unit 53-340	Gas HC dew point control condensate separator	G-69-D-O-1034-03
Unit 53-550	Closed drain gas trains area	G-69-D-O-1055-01

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				<i>ii</i>	Rev. <table border="1"> <tr> <td>P1</td><td>A1</td><td>E1</td><td></td><td></td><td></td></tr> <tr> <td></td><td></td><td></td><td></td><td></td><td></td></tr> </table>	P1	A1	E1					
P1	A1	E1											

Volume V Index

Attachment 11.2

♦ P&ID's

- Process Interconnecting

P&ID Title		P&ID Number
Unit 160	Zone P2	G-65-D-O-1016-01 Sh. 1 of 2
Unit 160	Zone P2	G-65-D-O-1016-01 Sh. 2 of 2
Unit 160	Zone P2	G-65-D-O-1016-02
Unit 160	Zone P2	G-65-D-O-1016-03
Unit 160	Zone P2 LL-LS-MS-HS-LC-MC-SC-BW	G-65-D-O-1016-04
Unit 160	Zone P2 CW-DW-TW-SW-UW	G-65-D-O-1016-05
Unit 160	Zone P2 IA-UA-UN	G-65-D-O-1016-06
Unit 160	Zone P2 AF-CF-LF-HF	G-65-D-O-1016-07
Unit 160	Zone P2 FG	G-65-D-O-1016-08

- Utility Distribution

P&ID Title	P&ID Number
Zone A2 LS-LC	G-65-D-O-1019-51
Zone A2 DW-UW	G-65-D-O-1019-52
Zone A2 IA-UA-UN	G-65-D-O-1019-53
Zone A2 LF-HF	G-65-D-O-1019-54
Zone A2 SY	G-65-D-O-1019-56
Zone A2 FG	G-65-D-O-1019-57
Zone B2-50 LS-LC	G-65-D-O-1021-51
Zone B2-50 CW-DW-UW	G-65-D-O-1021-52
Zone B2-50 IA-UA-UN	G-65-D-O-1021-53
Zone B2-50 LF	G-65-D-O-1021-54
Zone B2-50 SY	G-65-D-O-1021-56
Zone B2-50 FG	G-65-D-O-1021-57
Zone B2-51 LS-LC-MS-MC	G-65-D-O-1021-58
Zone B2-51 CW-DW-UW	G-65-D-O-1021-59
Zone B2-51 IA-UA-UN	G-65-D-O-1021-60
Zone B2-51 LF	G-65-D-O-1021-61
Zone B2-51 SY	G-65-D-O-1021-63
Zone B2-51 FG	G-65-D-O-1021-64
Zone C2 LL-LS-LC	G-65-D-O-1018-01

AgipGas BV LIBYAN BRANCH		WESTERN LIBYA GAS PROJECT		AGBV Job No.		AGBV Doc. No.	
				0177/NC/02		G-50-P-Z-0006 00	
Snamprogetti Lummus Gas Ltd  Eni G E O P Snamprogetti ABB ABB Lummus Malta HYUNDAI ENGINEERING & CONSTRUCTION CO., LTD.	Plant		CTR Job No.		CTR Doc. No.		
	MELLITAH PLANT		310300		50 000-ZA-E-03073		
	Location		GSPLAJ		iii		Rev.

P&ID Title	P&ID Number
Zone C2 CW-DW-TW-UW	G-65-D-O-1018-02
Zone C2 IA-UA-UN	G-65-D-O-1018-03
Zone C2 AF-CF-LF-HF	G-65-D-O-1018-04
Zone C2 SY-CB-AM	G-65-D-O-1018-06
Zone C2 FG	G-65-D-O-1018-07
Zone L2-1 MS-LS-MC-SC-LC	G-65-D-O-1018-41 Sh. 1 of 2
Zone L2-1 MS-LS-MC-SC-LC	G-65-D-O-1018-41 Sh. 2 of 2
Zone L2-1 IA-UA-FG-UN	G-65-D-O-1018-42 Sh. 1 of 2
Zone L2-1 IA-UA-FG-UN	G-65-D-O-1018-42 Sh. 2 of 2
Zone L2-1 AF-CF-HF-LF	G-65-D-O-1018-43 Sh. 1 of 2
Zone L2-1 AF-CF-HF-LF	G-65-D-O-1018-43 Sh. 2 of 2
Zone L2-1 CW-UW-DW-TW	G-65-D-O-1018-44 Sh. 1 of 2
Zone L2-1 CW-UW-DW-TW	G-65-D-O-1018-44 Sh. 2 of 2
Zone L2-1 SY-MW-CB-AM	G-65-D-O-1018-45 Sh. 1 of 2
Zone L2-1 SY-MW-CB-AM	G-65-D-O-1018-45 Sh. 2 of 2
Zone Q1 IA-UA-UW-LS-LC-CW	G-65-D-O-1021-55

- Zone B2-52 P&ID's

Not included – Refer to Zone B2-51 drawings

P&ID Title	P&ID Number
Zone B2-52 LS-LC-MS-MC	G-68-D-O-1021-58
Zone B2-52 CW-DW-UW	G-68-D-O-1021-59
Zone B2-52 IA-UA-UN	G-68-D-O-1021-60
Zone B2-52 LF	G-68-D-O-1021-61
Zone B2-52 SY	G-68-D-O-1021-63
Zone B2-52 FG	G-68-D-O-1021-64

AgipGas BV LIBYAN BRANCH		WESTERN LIBYA GAS PROJECT		AGBV Job No.		AGBV Doc. No.	
				0177/NC/02		G-50-P-Z-0006 00	
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	MELLITAH PLANT		310300		50 000-ZA-E-03073		
	Location		GSPLAJ		iv		Rev.

- Zone D2/E2 P&ID's

Not included – Refer to Zone C2 drawings

P&ID Title	P&ID Number
Zone D2 LL-LS-LC	G-68-D-O-1018-01
Zone D2 CW-DW-TW-UW	G-68-D-O-1018-02
Zone D2 IA-UA-UN	G-68-D-O-1018-03
Zone D2 AF-CF-LF-HF	G-68-D-O-1018-04
Zone D2 SY-CB-AM	G-68-D-O-1018-06
Zone D2 FG	G-68-D-O-1018-07
Zone E2 LL-LS-LC	G-69-D-O-1018-01
Zone E2 CW-DW-TW-UW	G-69-D-O-1018-02
Zone E2 IA-UA-UN	G-69-D-O-1018-03
Zone E2 AF-CF-LF-HF	G-69-D-O-1018-04
Zone E2 SY-CB-AM	G-69-D-O-1018-06
Zone E2 FG	G-69-D-O-1018-07

- Zone L2-2 P&ID's

Not included – Refer to Zone L2-1 drawings

P&ID Title	P&ID Number
Zone L2-2 MS-LS-MC-SC-LC	G-65-D-O-1018-46 Sh. 1 of 2
Zone L2-2 MS-LS-MC-SC-LC	G-65-D-O-1018-46 Sh. 2 of 2
Zone L2-2 IA-UA-FG-UN	G-65-D-O-1018-47 Sh. 1 of 2
Zone L2-2 IA-UA-FG-UN	G-65-D-O-1018-47 Sh. 2 of 2
Zone L2-2 AF-CF-HF-LF	G-65-D-O-1018-48 Sh. 1 of 2
Zone L2-2 AF-CF-HF-LF	G-65-D-O-1018-48 Sh. 2 of 2
Zone L2-2 CW-UW-DW-TW	G-65-D-O-1018-49 Sh. 1 of 2
Zone L2-2 CW-UW-DW-TW	G-65-D-O-1018-49 Sh. 2 of 2
Zone L2-2 SY-MW-CB-AM	G-65-D-O-1018-50 Sh. 1 of 2
Zone L2-2 SY-MW-CB-AM	G-65-D-O-1018-50 Sh. 2 of 2

AgipGas BV LIBYAN BRANCH	WESTERN LIBYA GAS PROJECT	AGBV Job No.	AGBV Doc. No.			
		0177/NC/02	G-50-P-Z-0006 00			
Snamprogetti Lummus Gas Ltd  Snamprogetti  ABB Lummus Malta  HYUNDAI ENGINEERING & CONSTRUCTION CO., LTD.	Plant	CTR Job No.	CTR Doc. No.			
	MELLITAH PLANT	310300	50 000-ZA-E-03073			
	Location	ii	Rev.	P1	A1	E1

Volume VI Index

Attachment 11.3

◆ Plot Plan

Drawing Title	Dwg Number
General Key Plan	G-60-D-P-3152-01
Process Plot Plan	G-65-D-P-3102-01
Zone A2/1	G-65-D-P-3221-01
Zone A2/2	G-65-D-P-3221-02
Zone B2	G-65-D-P-3222-01
Zone C2	G-65-D-P-3223-01
Zone D2	G-65-D-P-3224-01
Zone E2	G-65-D-P-3225-01
Zone F2	G-65-D-P-3226-01
Zone L2	G-64-D-P-3201-40
Zone Q1	G-65-D-P-3265-01

Attachment 11.4

◆ Electrical One-Line Diagram

Drawing Title	Dwg Number
Key One Line Diagram	G-64-D-E-1001-00

 LIBYAN BRANCH	WESTERN LIBYA GAS PROJECT	AGBV Job No.		AGBV Doc. No.	
		0177/NC/02		G-50-P-Z-0006 00	
		CTR Job No.		CTR Doc. No.	
 Snamprogetti	Plant	MELLITAH PLANT		310300	
	Location	GSPLAJ		50 000-ZA-E-03073	
		ii		Rev.	P1 A1 E1

Volume VII Index

Attachment 11.5

◆ Process Material Balance

Title	Doc. Number
Heat and material balance process Units-Year 2 Summer	G-65-Y-O-0002-00
Heat and material balance process Units-Year 2 Winter	G-65-Y-O-0004-00
Heat and material balance process Units-Year 11 Summer	G-65-Y-O-0001-00
Heat and material balance process Units-Year11 Winter	G-65-Y-O-0003-00
Heat and material balance Unit 330-Year 2 Summer low CO ₂ removal	G-65-Y-O-0007-00
Heat and material balance Unit 330-Year 2 Summer high CO ₂ removal	G-65-Y-O-0008-00
Heat and material balance Unit 330-Year 11 Summer low CO ₂ Removal	G-65-Y-O-0005-00
Heat and material balance Unit 330-Year 11 Summer high CO ₂ removal	G-65-Y-O-0006-00
Heat and material balance Unit 330-Year 2 Summer-Bouri verification case	G-65-Y-O-0028-00
Heat and material balance process Units-Year 2 Summer-Bouri verification case	G-65-Y-O-0029-00
Heat and material balance process Units-Year 2 Winter-Bouri verification case	G-65-Y-O-0030-00

AgipGas BV LIBYAN BRANCH	WESTERN LIBYA GAS PROJECT	AGBV Job No.	AGBV Doc. No.			
		0177/NC/02	G-50-P-Z-0006 00			
	Plant	CTR Job No.	CTR Doc. No.			
	MELLITAH PLANT	310300	50 000-ZA-E-03073			
	Location	ii	Rev.	P1	A1	E1
	GSPLAJ					

Volume VIII Index

Attachment 11.6

◆ Design Premises

Title	Doc. Number
Design Premises for Mellitah Plant	G-65-S-Z-0001-00

Attachment 11.7

◆ Tie-in List with WAFA Plant

Title	Doc. Number
Tie-in List	G-65-L-O-0000-01

Attachment 11.8

◆ Utilities consumption summary

Title	Doc. Number
Utilities consumption summary Mellitah Plant Year 2 Summer	G-67-L-O-0002-00

Attachment 11.9

◆ Chemical consumption summary Mellitah Plant

Title	Doc. Number
Chemical consumption summary Mellitah Plant	G-65-L-O-0004-00

Attachment 11.10

◆ Effluent summary report

Title	Doc. Number
Effluent summary report	G-64-T-Q-0020-00

AgipGas BV LIBYAN BRANCH		WESTERN LIBYA GAS PROJECT		AGBV Job No. 0177/NC/02	AGBV Doc. No. G-50-P-Z-0006 00								
	Plant	MELLITAH PLANT		CTR Job No.	CTR Doc. No.								
	Location	GSPLAJ		310300	50 000-ZA-E-03073								
				<i>iii</i>	Rev. <table border="1"> <tr> <td>P1</td><td>A1</td><td>E1</td><td></td><td></td><td></td></tr> <tr> <td></td><td></td><td></td><td></td><td></td><td></td></tr> </table>	P1	A1	E1					
P1	A1	E1											

Attachment 11.11

♦ Electric load list

Title	Doc. Number
Electric load list	G-64-L-E-0150-00

Attachment 11.12

♦ Equipment summary

Title	Doc. Number
Item index-Mellitah Plant- Process & Utilities units	G-65-L-Z-0001-00

Attachment 11.13

♦ Equipment Process Data Sheets

	Title	Doc. Number
Unit 50-190	Inlet gas slug catcher KO drum	G-65-X-O-19VA-01
Unit 50-190	Gas slug catcher	G-65-X-O-19VQ-01
Unit 50-190	Raw gas pipeline pig trap	G-65-X-O-19VR-01
Unit 50-190	Condensate pipeline pig trap	G-65-X-O-19VR-02
Unit 50-210	Data sheet for Offgas compressor 2ND stage discharge cooler	G-65-X-O-21HA-04
Unit 50-210	Data sheet for Offgas compressor 1st stage discharge cooler	G-65-X-O-21HC-01
Unit 50-210	Offgas compressor	G-65-X-O-21KA-01
Unit 50-210	Offspec condensate pump	G-65-X-O-21PA-01
Unit 50-210	Offspec condensate tank	G-65-X-O-21TA-01
Unit 50-210	Offgas compressor 1st stage KO drum	G-65-X-O-21VA-02
Unit 50-210	Offgas compressor 1st stage discharge drum	G-65-X-O-21VA-03
Unit 50-210	Offgas compressor 2nd stage discharge drum	G-65-X-O-21VA-04
Unit 50-210	Degassing boot	G-65-X-O-21VA-05
Unit 51-210	Condensate filter	G-65-X-O-21CL-01
Unit 51-210	Stabilizer feed bottom exchanger	G-65-X-O-21HA-01
Unit 51-210	Data sheet for stabilized reboiler	G-65-X-O-21HA-02
Unit 51-210	Data sheet for condensate trim cooler	G-65-X-O-21HA-03
Unit 51-210	Condensate air cooler	G-65-X-O-21HC-02
Unit 51-210	Condensate feed pumps	G-65-X-O-21PA-02

AgipGas BV LIBYAN BRANCH		WESTERN LIBYA GAS PROJECT		AGBV Job No.		AGBV Doc. No.											
				0177/NC/02		G-50-P-Z-0006 00											
Snamprogetti Lummus Gas Ltd  Eni GROUP ABB ABB Lummus Malta HYUNDAI ENGINEERING & CONSTRUCTION CO., LTD. Plant		MELLITAH PLANT		CTR Job No.		CTR Doc. No.											
				310300		50 000-ZA-E-03073											
		Location		GSPLAJ		iv		Rev. <table><tr><td>P1</td><td>A1</td><td>E1</td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td></tr></table>		P1	A1	E1					
P1	A1	E1															

Title		Number
Unit 51-210	Offgas condensate surge drum	G-65-X-O-21VA-01
Unit 51-210	Stabilizer	G-65-X-O-21VE-01
Unit 51-311	Specification for molecular sieves (gas Dryer Package item 51-311-XX-001)	G-65-S-O-3100-00
Unit 51-311	Data sheet for ceramic balls for Gas Dryers	G-65-X-O-3102-01
Unit 51-311	Dry Gas filters item 51-311-CL-001 A/B	G-65-X-O-31CL-01
Unit 51-311	Regeneration Gas Heater 51-311-FA-001	G-65-X-O-31FA-01
Unit 51-311	Regeneration Gas Cooler 51-311-HC-001	G-65-X-O-31HC-01
Unit 51-311	Regeneration Gas Compressor Cooler 51-311-HC-002	G-65-X-O-31HC-02
Unit 51-311	Regeneration Gas Compressor 51-311-KB-001 A/B	G-65-X-O-31KB-01
Unit 51-311	Regeneration Gas Compressor KO Drum 51-311-VA-002	G-65-X-O-31VA-01
Unit 51-311	Gas dryers item 51-311-VK-001 A/B/C	G-65-X-O-31VK-01
Unit 51-311	Inlet dryers coalescer	G-65-X-O-31VZ-01

AgipGas BV LIBYAN BRANCH		WESTERN LIBYA GAS PROJECT		AGBV Job No. 0177/NC/02	AGBV Doc. No. G-50-P-Z-0006 00								
	Plant	MELLITAH PLANT		CTR Job No.	CTR Doc. No.								
	Location	GSPLAJ		310300	50 000-ZA-E-03073								
				<i>ii</i>	Rev. <table border="1"> <tr> <td>P1</td><td>A1</td><td>E1</td><td></td><td></td><td></td></tr> <tr> <td></td><td></td><td></td><td></td><td></td><td></td></tr> </table>	P1	A1	E1					
P1	A1	E1											

Volume IX Index

Attachment 11.13

◆ Equipment Process Data Sheets

Title		Doc. Number
Unit 51-320	Deethanizer reboiler	G-65-X-M-32HA-01
Unit 51-320	Deethanizer reboiler	G-65-X-M-32HA-02
Unit 51-320	Debutanizer condenser	G-65-X-M-32HA-03
Unit 51-320	LPG Sub cooler	G-65-X-M-32HA-04
Unit 51-320	Debutanizer bottom cooler	G-65-X-M-32HA-05
Unit 51-320	Debutanizer reflux drum	G-65-X-O-32VA-01
Unit 51-320	Deethanizer	G-65-X-O-32VE-01
Unit 51-320	Debutanizer	G-65-X-O-32VE-02
Unit 51-320	Debutanizer reflux pump	G-65-X-O-32PA-01
Unit 50-330	BFW Cooler	G-65-X-M-33HA-01
Unit 50-330	HP Amine recovery pump	G-65-X-L-33PA-01
Unit 50-330	HP Amine slop oil pump	G-65-X-L-33PA-02
Unit 50-330	HP Amine make-up pump	G-65-X-L-33PA-03
Unit 50-330	HP Amine recovery drum	G-65-X-M-33VA-01
Unit 50-330	Amine drain drum	G-65-X-M-33VA-02
Unit 50-330	HP Fresh amine storage tank	G-65-X-M-33TA-01
Unit 51-330	Specification for Chemical (Antifoam)	G-65-S-O-3300-00
Unit 51-330	Specification for MDEA	G-65-S-O-3300-01
Unit 51-330	Amine unit feedgas filters	G-65-X-O-33CL-01
Unit 51-330	Amine feed/product gas exchanger	G-65-X-O-33HA-01
Unit 51-330	Sweet gas cooler	G-65-X-O-33HA-02
Unit 51-330	Regenerator reboiler	G-65-X-O-33HA-03
Unit 51-330	Regenerator overhead trim cooler	G-65-X-O-33HA-04
Unit 51-330	Lean amine trim cooler	G-65-X-O-33HA-05
Unit 51-330	Rich/lean amine exchangers	G-65-X-O-33HA-06
Unit 51-330	Regenerator overhead condenser	G-65-X-O-33HC-03
Unit 51-330	Lean amine cooler	G-65-X-O-33HC-04
Unit 51-330	Regenerator reflux pump	G-65-X-O-33PA-04
Unit 51-330	Treated gas water wash pumps	G-65-X-O-33PA-06
Unit 51-330	Lean amine circulating pumps	G-65-X-O-33PA-07
Unit 51-330	Lean amine booster pumps	G-65-X-O-33PA-08
Unit 51-330	Lean amine pumps	G-65-X-O-33PA-010
Unit 51-330	Amine sump pump	G-65-X-O-33PH-01
Unit 51-330	Sweet gas KO drum	G-65-X-O-33VA-01
Unit 51-330	Rich amine flash drum	G-65-X-O-33VA-03

AgipGas BV LIBYAN BRANCH		WESTERN LIBYA GAS PROJECT		AGBV Job No. 0177/NC/02		AGBV Doc. No. G-50-P-Z-0006 00				
Snamprogetti Lummus Gas Ltd  Eni GROUP  ABB Lummus Malta  HYUNDAI ENGINEERING & CONSTRUCTION CO., LTD.	Plant MELLITAH PLANT		CTR Job No. 310300		CTR Doc. No. 50 000-ZA-E-03073					
	Location GSPLAJ		iii		Rev.	P1	A1	E1		

	Title	Number
Unit 51-330	Regenerator reflux drum	G-65-X-O-33VA-05
Unit 51-330	Skimming pot	G-65-X-O-33VA-07
Unit 51-330	Sump drum	G-65-X-O-33VA-08
Unit 51-330	Absorber inlet separator	G-65-X-O-33VA-010
Unit 51-330	HP Absorber	G-65-X-O-33VE-01
Unit 51-330	Amine regenerator	G-65-X-O-33VJ-01
Unit 51-330	Lean amine intermediate storage tank	G-65-X-O-33TA-01
Unit 51-330	Lean amine filtration package	G-65-X-O-33XX-01
Unit 51-330	Antifoam injection package	G-65-X-O-33XX-02
Unit 51-340	Data sheet for raw gas/treated gas exchanger	G-65-X-O-34HA-01
Unit 51-340	Data sheet for propane chiller	G-65-X-O-34HA-02
Unit 51-340	Data sheet for NGL condensate separator	G-65-X-O-34VA-01
Unit 51-340	Process duty specification for propane refrigeration unit item 51-340-XX-001 (Mellitah Plant)	G-65-X-O-34XX-01
Unit 51-550	Data sheet for wet hydrocarbons sump pump	G-65-X-O-55PH-01
Unit 50-550	Data sheet for condensate stabilization area sump pump	G-65-X-O-55PH-02
Unit 50-550	Data sheet for inlet reception facilities sump pump	G-65-X-O-55PH-03
Unit 50-550	Data sheet for NGL fractionation hot HC sump pump	G-65-X-L-55PH-04
Unit 50-550	NGL fract. cold HC drain drum	G-65-X-M-55VA-03
Unit 50-550	NGL fract. cold HC sump drum	G-65-X-M-55VA-04
Unit 50-550	HP Flare area sump drum	G-65-X-M-55VA-05
Unit 51-550	Data sheet for wet hydrocarbons sump drum (51-550-VA-002 + 51-550-HN-001 A/B)	G-65-X-O-55VA-02
Unit 50-550	Data sheet for condensate stabilization area sump drum (50-550-VA-001 + 50-550-HN-001 A/B)	G-65-X-O-55VA-03
Unit 50-550	Data sheet for inlet reception facilities sump drum (50-550-VA-02 + 50-550-HN-02 A/B)	G-65-X-O-55VA-04
Unit 51-550	Data sheet for fractionation cold hydrocarbon drain drum (51-550-VA-001)	G-65-X-O-55VA-05

AgipGas BV LIBYAN BRANCH		WESTERN LIBYA GAS PROJECT		AGBV Job No. 0177/NC/02	AGBV Doc. No. G-50-P-Z-0006 00								
	Plant	MELLITAH PLANT		CTR Job No.	CTR Doc. No.								
	Location	GSPLAJ		310300	50 000-ZA-E-03073								
				<i>ii</i>	Rev. <table border="1"> <tr> <td>P1</td><td>A1</td><td>E1</td><td></td><td></td><td></td></tr> <tr> <td></td><td></td><td></td><td></td><td></td><td></td></tr> </table>	P1	A1	E1					
P1	A1	E1											

Volume X Index

Attachment 11.14

◆ Instruments Data Sheets for Control Valves, Safety Valves, Rupture Disks and Analyzers

Title	Doc. Number
Individual specification for critical control valves Units 50-190, 51-311, 51-330, 51-340	G-65-S-I-0128-00
Individual specification for non critical control valves Units 50-190, 50-210, 51-210, 51-311, 51-330, 51-340, 50-160	G-65-S-I-0128-01
Individual specification for control valves Zone H2 & L2 <i>Extract from spec n° G-65-S-I-0128-40 pages 1-3,10,11,15-26 (pages relevant to Gas Treatments Units)</i>	G-65-S-I-0128-40
Individual specification for self actuated valves Units 50-210, 51-311, 51-330, 51-210	G-65-S-I-0128-02
Individual specification for self acting valves Zone H2 & L2 <i>Extract from spec n° G-65-S-I-0128-41 pages 1, 2,10-12 (pages relevant to Gas Treatments Units)</i>	G-65-S-I-0128-41
Individual specification for safety relief valves Units 50-160, 50-190, 50-210, 51-210, 51-311, 51-330, 51-340	G-65-S-I-0132-00
Individual specification for pressure relief valves Zone H2 & L2 - <i>Extract from spec n° G-65-S-I-0132-40 pages 1-3,10-15, 30-46 (pages relevant to Gas Treatments Units)</i>	G-65-S-I-0132-40
Individual specification for rupture disks Units 50-210, 51-210, 51-330	G-65-S-I-0132-01
Individual specification for rupture disks Zone H" & L2 Units 50-330, 51-320, 51-582, 51-585 <i>Extract from spec n° G-65-S-I-0132-41 pages 1, 2, 10, 11, 17-28 (pages relevant to Gas Treatments Units)</i>	G-65-S-I-0132-41
Individual specification for process analyzer Units 50-160, 50-190, 51-210, 51-311, 51-330, 51-340	G-65-S-I-0127-00

AgipGas BV LIBYAN BRANCH		WESTERN LIBYA GAS PROJECT		AGBV Job No.		AGBV Doc. No.					
				0177/NC/02		G-50-P-Z-0006 00					
Snamprogetti Lummus Gas Ltd  Eni G E O P Snamprogetti ABB ABB Lummus Malta HYUNDAI ENGINEERING & CONSTRUCTION CO., LTD.	Plant		CTR Job No.		CTR Doc. No.						
	MELLITAH PLANT		310300		50 000-ZA-E-03073						
	Location		GSPLAJ		ii		Rev.	P1	A1	E1	

Volume XI Index

Attachment 11.15

- ♦ Cause & Effect Diagram, ESD Diagram (Shut Down Logic Diagram), Specification of Logic Sequence

Title	Doc. Number
Specification of dehydration unit logic sequence	G-65-X-O-3101-00
Shutdown Logic Diagram Plant/Zone ESD/PSD Diagram	G-65-D-O-1000-20
Shutdown Logic Diagram Zone A2/B2/Q1 LSD Diagram	G-65-D-O-1000-21
Shutdown Logic Diagram Zone C2 LSD Diagram	G-65-D-O-1000-22
Cause and Effect Diagram TotalPlant ESD & PSD; Units 190, 210, 311, 330, 340, 550	G-65-D-O-8001-02
Cause and Effect Diagram Units 51-320, 50-330, 50-550	G-65-D-O-8001-01

AgipGas BV LIBYAN BRANCH		WESTERN LIBYA GAS PROJECT		AGBV Job No.		AGBV Doc. No.					
				0177/NC/02		G-50-P-Z-0006 00					
Snamprogetti Lummus Gas Ltd Eni Eni Snamprogetti ABB ABB Lummus Malta HYUNDAI ENGINEERING & CONSTRUCTION CO., LTD.	Plant		MELLITAH PLANT		CTR Job No.		CTR Doc. No.				
					310300		50 000-ZA-E-03073				
	Location		GSPLAJ		ii		Rev.	P1	A1	E1	

Volume XII Index

Attachment 11.16

◆ Control Narrative

Title	Doc. Number
DCS Loop control narrative Unit 50-190	G-65-S-I-3084-00
DCS Loop control narrative Unit 50-210	G-65-S-I-3085-00
DCS Loop control narrative Unit 51-210	G-65-S-I-3093-00
DCS Loop control narrative Unit 51-311	G-65-S-I-3094-00
DCS Loop control narrative Unit 51-330	G-65-S-I-3096-00
DCS Loop control narrative Unit 51-340	G-65-S-I-3097-00
DCS Loop control narrative Sulphur Plant Units 51-581, 51-582, 51-583, 51-584, 51-585, 51-586, 51-587 and Process Units 51-230, 51-320, 50-330, 50-550 <i>Extract from spec n° G-65-S-I-3101-00 pages 1-9, 203-211 (pages relevant to Gas Treatment Units)</i>	G-65-S-I-3101-00

Attachment 11.17

◆ DCS Graphic Display

Title	Doc. Number
DCS Graphic Display Units 50-190, 50-210, 51-210, 51-311, 51-330, 51-340, 50-550, 51-550	G-64-D-I-2500-00
DCS Graphic Display Units 50-210, 51-311, 51-330, 51-340, 5E-620	G-64-D-I-2500-02
DCS Graphic Display Units 50-000, 50-190, 50-210, 50-620, 51-210, 51-311, 51-330, 51-340, 5E-631, 50-160	G-64-D-I-2500-04
DCS Graphic Display Units 50-230, 51-320, 50-550, 50-586, 51-587, 51-183 <i>Extract from spec n° G-64-D-I-2500-41 (Process part relevant to Gas Treatment Units)</i>	G-64-D-I-2500-41
DCS Graphic Display Units 50-230, 50-330, 50-550, 50-586, 51-320, 51-581, 51-582, 51-583, 51-584, 51-585, 51-587 <i>Extract from spec n° G-64-D-I-2500-48 (Process part relevant to Gas Treatment Units)</i>	G-64-D-I-2500-48

AgipGas BV LIBYAN BRANCH		WESTERN LIBYA GAS PROJECT		AGBV Job No. 0177/NC/02	AGBV Doc. No. G-50-P-Z-0006 00
 Snamprogetti  Eni  ABB ABB Lummus Malta  HYUNDAI ENGINEERING & CONSTRUCTION CO., LTD.	Plant MELLITAH PLANT	CTR Job No. 310300	CTR Doc. No. 50 000-ZA-E-03073		
	Location GSPLAJ	<i>ii</i>	Rev.	P1	A1
				E1	

Volume IV Index

Attachment 11.2

♦ P&ID

P&ID Title		P&ID Number
Unit 51-311	Gas dehydration coalescer	G-65-D-O-1031-01
Unit 51- 311	Gas dehydration dryer	G-65-D-O-1031-02
Unit 51- 311	Gas dehydration dryers	G-65-D-O-1031-03
Unit 51- 311	Gas dehydration molec. sieves outlet filters	G-65-D-O-1031-04
Unit 51- 311	Gas dehydration regeneration gas heater	G-65-D-O-1031-05
Unit 51- 311	Gas dehydration regeneration gas cooling	G-65-D-O-1031-06
Unit 51- 311	Gas dehydration gas re-compress.aftercooler	G-65-D-O-1031-07
Unit 51- 311	Gas dehydration regeneration gas heater safeguarding system	G-65-D-O-1031-08
Unit 51- 311	Gas dehydration gas re-compress. "A"	G-65-D-O-1031-09
Unit 51- 311	Gas dehydration gas re-compress. "B"	G-65-D-O-1031-10
Unit 51- 320	NGL Fractionation deethanizer	G-65-D-O-1032-01
Unit 51- 320	NGL Fractionation product cooling	G-65-D-O-1032-02
Unit 51- 320	NGL Fractionation debutanizer	G-65-D-O-1032-03
Unit 51- 320	NGL Fractionation debutanizer reflux system	G-65-D-O-1032-04
Unit 50- 330	Gas sweetening amine recovery system	G-65-D-O-1033-01
Unit 50- 330	Gas sweetening amine prep. & storage system	G-65-D-O-1033-02
Unit 51- 330	Gas sweetening raw gas inlet manifold	G-65-D-O-1033-03
Unit 51- 330	Gas sweetening raw gas preheating	G-65-D-O-1033-04
Unit 51- 330	Gas sweetening HP absorber	G-65-D-O-1033-05
Unit 51- 330	Gas sweetening treated gas KO drum	G-65-D-O-1033-06
Unit 51- 330	Gas sweetening amine flash drum	G-65-D-O-1033-07
Unit 51- 330	Gas sweetening rich/lean amine exchangers	G-65-D-O-1033-08
Unit 51- 330	Gas sweetening amine regenerator	G-65-D-O-1033-09
Unit 51- 330	Gas sweetening amine regenerator reflux section	G-65-D-O-1033-10
Unit 51- 330	Gas sweetening amine pumps and storage	G-65-D-O-1033-11

AgipGas BV LIBYAN BRANCH		WESTERN LIBYA GAS PROJECT		AGBV Job No. 0177/NC/02	AGBV Doc. No. G-50-P-Z-0006 00								
	Plant	MELLITAH PLANT		CTR Job No.	CTR Doc. No.								
	Location	GSPLAJ		310300	50 000-ZA-E-03073								
				<i>iii</i>	Rev. <table border="1"> <tr> <td>P1</td><td>A1</td><td>E1</td><td></td><td></td><td></td></tr> <tr> <td></td><td></td><td></td><td></td><td></td><td></td></tr> </table>	P1	A1	E1					
P1	A1	E1											

P&ID Title		P&ID Number
Unit 51- 330	Gas sweetening antifoam package & sump drum	G-65-D-O-1033-12
Unit 51- 330	Gas sweetening lean amine filtration	G-65-D-O-1033-13
Unit 51- 330	Gas sweetening steam & condensate header	G-65-D-O-1033-14
Unit 51- 330	Gas sweetening raw gas inlet filters	G-65-D-O-1033-15
Unit 51- 340	Gas HC dew point control dry gas cooling	G-65-D-O-1034-01
Unit 51- 340	Gas HC dew point control dry gas cooling	G-65-D-O-1034-02
Unit 51- 340	Gas HC dew point control condensate separator	G-65-D-O-1034-03
Unit 51- 550	Closed drain gas trains area	G-65-D-O-1055-01
Unit 50- 550	Closed drain slug catcher area & condensate stabilization area	G-65-D-O-1055-02
Unit 50- 550	Closed drain NGL fractionation area	G-65-D-O-1055-03
Unit 50- 550	Closed drain HP flare area	G-65-D-O-1055-05

- Units 52/53 P&ID's

Not included – Refer to Unit 51 drawings

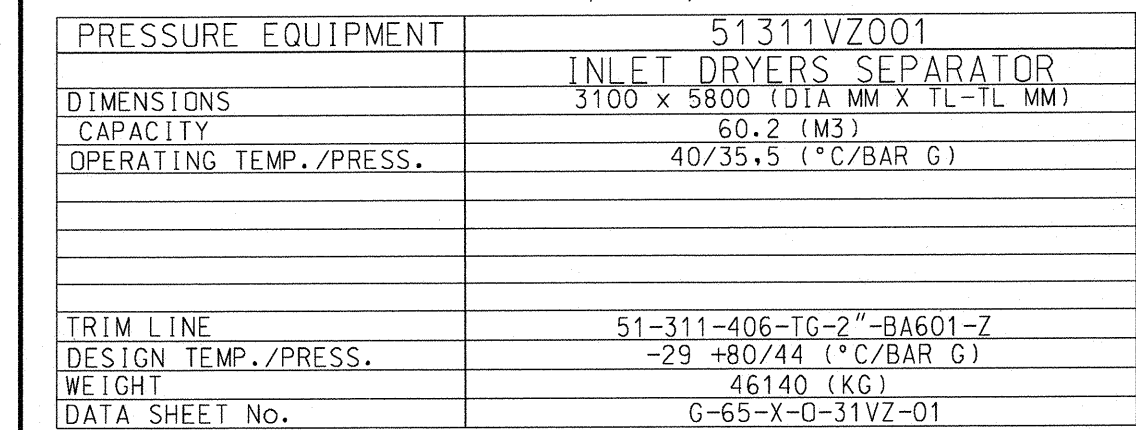
P&ID Title		P&ID Number
Unit 52-311	Gas dehydration coalescer	G-68-D-O-1031-01
Unit 52-311	Gas dehydration dryer	G-68-D-O-1031-02
Unit 52-311	Gas dehydration dryers	G-68-D-O-1031-03
Unit 52-311	Gas dehydration molec. sieves outlet filters	G-68-D-O-1031-04
Unit 52-311	Gas dehydration regeneration gas heater	G-68-D-O-1031-05
Unit 52-311	Gas dehydration regeneration gas cooling	G-68-D-O-1031-06
Unit 52-311	Gas dehydration gas re-compress.aftercooler	G-68-D-O-1031-07
Unit 52-311	Gas dehydration regeneration gas heater safeguarding system	G-68-D-O-1031-08
Unit 52-311	Gas dehydration gas re-compress. "A"	G-68-D-O-1031-09
Unit 52-311	Gas dehydration gas re-compress. "B"	G-68-D-O-1031-10
Unit 52-320	NGL Fractionation deethanizer	G-68-D-O-1032-01
Unit 52-320	NGL Fractionation product cooling	G-68-D-O-1032-02
Unit 52-320	NGL Fractionation debutanizer	G-68-D-O-1032-03

AgipGas BV LIBYAN BRANCH		WESTERN LIBYA GAS PROJECT		AGBV Job No.		AGBV Doc. No.	
				0177/NC/02		G-50-P-Z-0006 00	
Snamprogetti Lummus Gas Ltd  Eni GROUP  ABB Lummus Malta  HYUNDAI ENGINEERING & CONSTRUCTION CO., LTD.	Plant		CTR Job No.		CTR Doc. No.		
	MELLITAH PLANT		310300		50 000-ZA-E-03073		
	Location		GSPLAJ		iv		Rev.

P&ID Title		P&ID Number
Unit 52-320	NGL Fractionation debutanizer reflux system	G-68-D-O-1032-04
Unit 52-330	Gas sweetening raw gas inlet manifold	G-68-D-O-1033-03
Unit 52-330	Gas sweetening raw gas preheating	G-68-D-O-1033-04
Unit 52-330	Gas sweetening HP absorber	G-68-D-O-1033-05
Unit 52-330	Gas sweetening treated gas KO drum	G-68-D-O-1033-06
Unit 52-330	Gas sweetening amine flash drum	G-68-D-O-1033-07
Unit 52-330	Gas sweetening rich/lean amine exchangers	G-68-D-O-1033-08
Unit 52-330	Gas sweetening amine regenerator	G-68-D-O-1033-09
Unit 52-330	Gas sweetening amine regenerator reflux section	G-68-D-O-1033-10
Unit 52-330	Gas sweetening amine pumps and storage	G-68-D-O-1033-11
Unit 52-330	Gas sweetening antifoam package & sump drum	G-68-D-O-1033-12
Unit 52-330	Gas sweetening lean amine filtration	G-68-D-O-1033-13
Unit 52-330	Gas sweetening steam & condensate header	G-68-D-O-1033-14
Unit 52-330	Gas sweetening raw gas inlet filters	G-68-D-O-1033-15
Unit 52-340	Gas HC dew point control dry gas cooling	G-68-D-O-1034-01
Unit 52-340	Gas HC dew point control dry gas cooling	G-68-D-O-1034-02
Unit 52-340	Gas HC dew point control condensate separator	G-68-D-O-1034-03
Unit 52-550	Closed drain gas trains area	G-68-D-O-1055-01
Unit 53-311	Gas dehydration coalescer	G-69-D-O-1031-01
Unit 53-311	Gas dehydration dryer	G-69-D-O-1031-02
Unit 53-311	Gas dehydration dryers	G-69-D-O-1031-03
Unit 53-311	Gas dehydration molec. sieves outlet filters	G-69-D-O-1031-04
Unit 53-311	Gas dehydration regeneration gas heater	G-69-D-O-1031-05
Unit 53-311	Gas dehydration regeneration gas cooling	G-69-D-O-1031-06
Unit 53-311	Gas dehydration gas re-compress.aftercooler	G-69-D-O-1031-07
Unit 53-311	Gas dehydration regeneration gas heater safeguarding system	G-69-D-O-1031-08
Unit 53-311	Gas dehydration gas re-compress. "A"	G-69-D-O-1031-09
Unit 53-311	Gas dehydration gas re-compress. "B"	G-69-D-O-1031-10
Unit 53-330	Gas sweetening raw gas inlet manifold	G-69-D-O-1033-03

AgipGas BV LIBYAN BRANCH		WESTERN LIBYA GAS PROJECT		AGBV Job No.		AGBV Doc. No.													
				0177/NC/02		G-50-P-Z-0006 00													
Snamprogetti Lummus Gas Ltd  Eni GROUP  ABB Lummus Malta  HYUNDAI ENGINEERING & CONSTRUCTION CO., LTD.	Plant		CTR Job No.		CTR Doc. No.														
	MELLITAH PLANT		310300		50 000-ZA-E-03073														
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P1	A1	E1																	

P&ID Title		P&ID Number
Unit 53-330	Gas sweetening raw gas preheating	G-69-D-O-1033-04
Unit 53-330	Gas sweetening HP absorber	G-69-D-O-1033-05
Unit 53-330	Gas sweetening treated gas KO drum	G-69-D-O-1033-06
Unit 53-330	Gas sweetening amine flash drum	G-69-D-O-1033-07
Unit 53-330	Gas sweetening rich/lean amine exchangers	G-69-D-O-1033-08
Unit 53-330	Gas sweetening amine regenerator	G-69-D-O-1033-09
Unit 53-330	Gas sweetening amine regenerator reflux section	G-69-D-O-1033-10
Unit 53-330	Gas sweetening amine pumps and storage	G-69-D-O-1033-11
Unit 53-330	Gas sweetening antifoam package & sump drum	G-69-D-O-1033-12
Unit 53-330	Gas sweetening lean amine filtration	G-69-D-O-1033-13
Unit 53-330	Gas sweetening steam & condensate header	G-69-D-O-1033-14
Unit 53-330	Gas sweetening raw gas inlet filters	G-69-D-O-1033-15
Unit 53-340	Gas HC dew point control dry gas cooling	G-69-D-O-1034-01
Unit 53-340	Gas HC dew point control dry gas cooling	G-69-D-O-1034-02
Unit 53-340	Gas HC dew point control condensate separator	G-69-D-O-1034-03
Unit 53-550	Closed drain gas trains area	G-69-D-O-1055-01



NOTES

- FOR SYMBOLS, LEGEND, TYPICAL AND GENERAL NOTES SEE P&ID'S:
G-00-D-0-1000-01/02/03/04/05/06/07/08/09/10/11/12/13/14/15/16/17/18/19/20
- FOR GENERAL SHUTDOWN LOGIC SEE P&ID'S: G-65-D-0-1000-20/21/22/23/24.
- FOR PIPING ASSEMBLY (I.e., 4D) REFER TO DOC: G-65-S-P-0028-00

- | HOLD INDEX | |
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| 2-- | DELETED |
| 3-- | DELETED |
| 4-- | DELETED |
| 5-- | DELETED |

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A3	25/10/04	ISSUED FOR CONSTRUCTION REVISED WHERE INDICATED	CAREDOU	D. POLI	G. LAMANT	A. MAGDALONI	T.M. SAID
A4	16/04/04	ISSUED FOR CONSTRUCTION	CAREDOU	D. POLI	G. LAMANT	A. MAGDALONI	T.M. SAID
A1	31/03/03	ISSUED FOR CONSTRUCTION	CAREDOU	D. POLI	G. LAMANT	A. MAGDALONI	T. SAID
P2	30/05/03	ISSUED FOR DESIGN	R. CAREDOU	D. POLI	G. LAMANT	A. MAGDALONI	T. SAID
P1	18/02/03	ISSUED FOR REVIEW	MEP	DP	GL		
Revision	DATE	DESCRIPTION	PREP.	CHECK.	APPR.	PM'S CHECK.	AGBY APPR.

AgipGas BV

LIBYAN BRANCH

WESTERN LIBYA GAS PROJECT

MELITIAH PLANT

GSP/LA

AGBY DWG No.

G-65-D-0-1031-01

Snamprogetti

Snamprogetti Engineering & Construction Co., Ltd.

ABB

ABB Power & Automation Ltd.

HYUNDAI

Hyundai Engineering & Construction Co., Ltd.

AGBY JOB No.0177NC02

CTR DWG No.

51-311-GD-B-038101

Sh. _____ of _____

CTR JOB No.

013000

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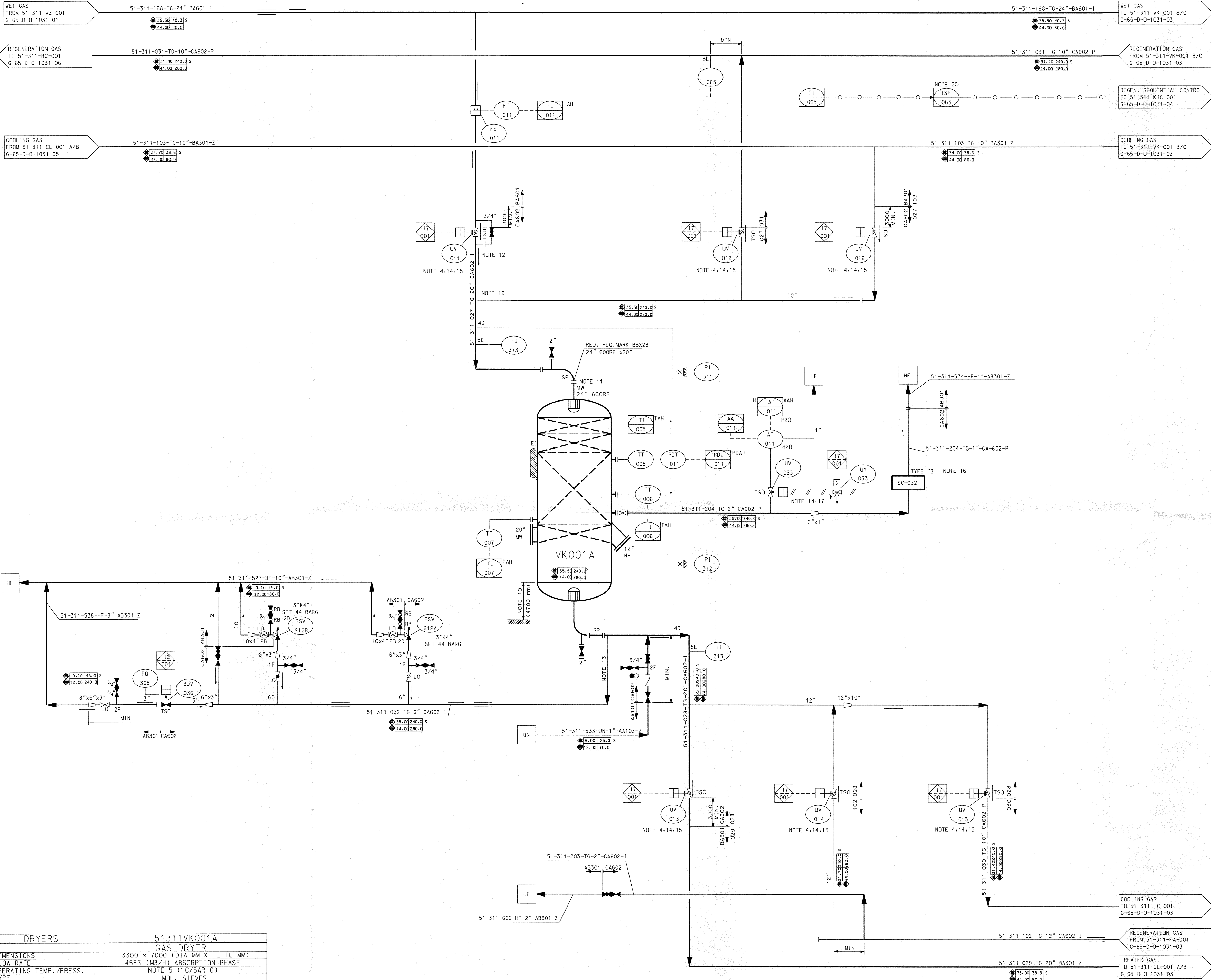
SCALE _____

IN REPLACEMENT OF _____

TITLE: PIPING & INSTRUMENT DIAGRAM

UNIT 311 - GAS DEHYDRATION

INLET DRYER SEPARATOR



DRYERS	
51311VK001A	
GAS DRYER	
DIMENSIONS	3300 x 7000 (DIA MM X TL-TL MM)
FLOW RATE	4553 (M3/H) ABSORPTION PHASE
OPERATING TEMP./PRESS.	NOTE 5 (*C/BAR G)
TYPE	MOL. SIEVES
REGENERATION	HOT GAS
EFFLUENT WATER CONTENT	0.1 ppm
TRIM LINE	51-311-407-TG-2"-CA602-I
DESIGN TEMP./PRESS.	-29 +280/44.0 (*C/BAR G)
WEIGHT	118070 (KG)
DATA SHEET No.	G-65-X-0-31VK-01

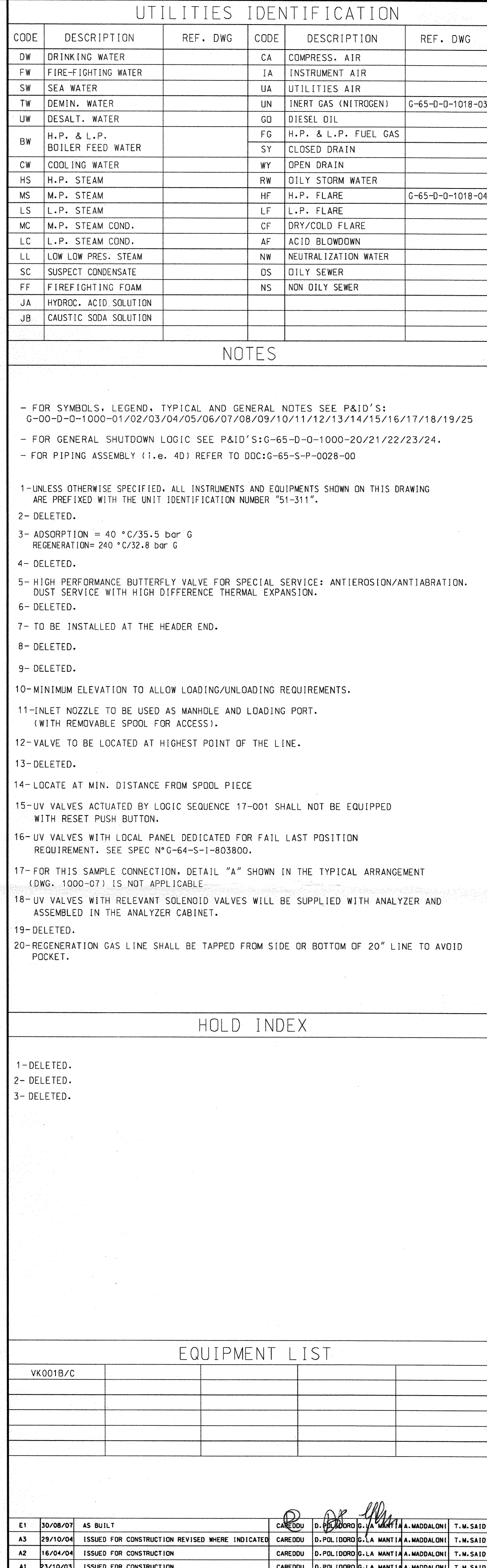
UTILITIES IDENTIFICATION					
CODE	DESCRIPTION	REF. DWG	CODE	DESCRIPTION	REF. DWG
DW	DRINKING WATER		CA	COMPRESS. AIR	
FW	FIRE-FIGHTING WATER		IA	INSTRUMENT AIR	
SW	SEA WATER		UA	UTILITIES AIR	
TW	DEMIN. WATER		UN	INERT GAS (NITROGEN)	G-65-D-0-1018-03
UW	DESALT. WATER		GO	DIESEL OIL	
BW	H.P. & L.P. BOILER FEED WATER		FG	H.P. & L.P. FUEL GAS	
CW	COOLING WATER		SY	CLOSED DRAIN	
HS	H.P. STEAM		WY	OPEN DRAIN	
MS	M.P. STEAM		RW	OILY STORM WATER	
LS	L.P. STEAM		HF	H.P. FLARE	G-65-D-0-1018-04
MC	M.P. STEAM COND.		LF	L.P. FLARE	
LC	L.P. STEAM COND.		CF	DRY/COLD FLARE	
LL	LOW LOW PRES. STEAM		AF	ACID BLOWDOWN	
SC	SUSPECT CONDENSATE		NW	NEUTRALIZATION WATER	
FF	FIRE FIGHTING FOAM				
JA	HYDROC. ACID SOLUTION				
JB	CAUSTIC SODA SOLUTION				

- NOTES
- FOR SYMBOLS, LEGEND, TYPICAL AND GENERAL NOTES SEE P&ID'S:
G-00-D-0-1000-01/02/03/04/05/06/07/08/09/10/11/12/13/14/15/16/17/18/19/25
- FOR GENERAL SHUTDOWN LOGIC SEE P&ID'S: G-65-D-0-1000-20/21/22/23/24.
- FOR PIPING ASSEMBLY (T.e. 4D) REFER TO DOC: G-65-S-P-0028-00
- 1- UNLESS OTHERWISE SPECIFIED, ALL INSTRUMENTS AND EQUIPMENTS SHOWN ON THIS DRAWING ARE PREFIXED WITH THE UNIT IDENTIFICATION NUMBER '51-311'.
- 2- DELETED.
- 3- DELETED.
- 4- HIGH PERFORMANCE BUTTERFLY VALVE FOR SPECIAL SERVICE: ANTIEROSION/ANTIABRASION.
- 5- ADSORPTION= 40 C/ 35.5 BARG
REGENERATION= 240 C/ 32.8 BARG
- 6- DELETED.
- 7- DELETED.
- 8- DELETED.
- 9- DELETED.
- 10- MINIMUM ELEVATION TO ALLOW LOADING/UNLOADING REQUIREMENTS.
- 11- INLET NOZZLE TO BE USED AS MANHOLE AND LOADING PORT (WITH REMOVABLE SPOOL FOR ACCESS).
- 12- VALVE TO BE LOCATED AT HIGHEST POINT OF THE LINE.
- 13- LOCATE AT MIN. DISTANCE FROM SPOOL PIECE
- 14- UV VALVES ACTUATED BY LOGIC SEQUENCE 17-001 SHALL NOT BE EQUIPPED WITH RESET PUSHBUTTON.
- 15- UV VALVES WITH LOCAL PANEL DEDICATED FOR FAIL LAST POSITION REQUIREMENT SEE SPEC. N° G-64-S-1-803800.
- 16- FOR THIS SAMPLE CONNECTION DETAIL "A" SHOWN IN THE TYPICAL ARRANGEMENT (DWG. 1000-07) IS NOT APPLICABLE
- 17- UV VALVE WITH RELEVANT SOLENOID VALVE WILL BE SUPPLIED WITH ANALYZER AT-011 AND ASSEMBLED IN THE ANALYZER CABINET.
- 18- DELETED.
- 19- REGENERATION GAS LINE SHALL BE TAPPED FROM SIDE OR BOTTOM OF 20" LINE TO AVOID POCKET.
- 20- TSH-065 SET AT MINIMUM REGENERATION TEMPERATURE.

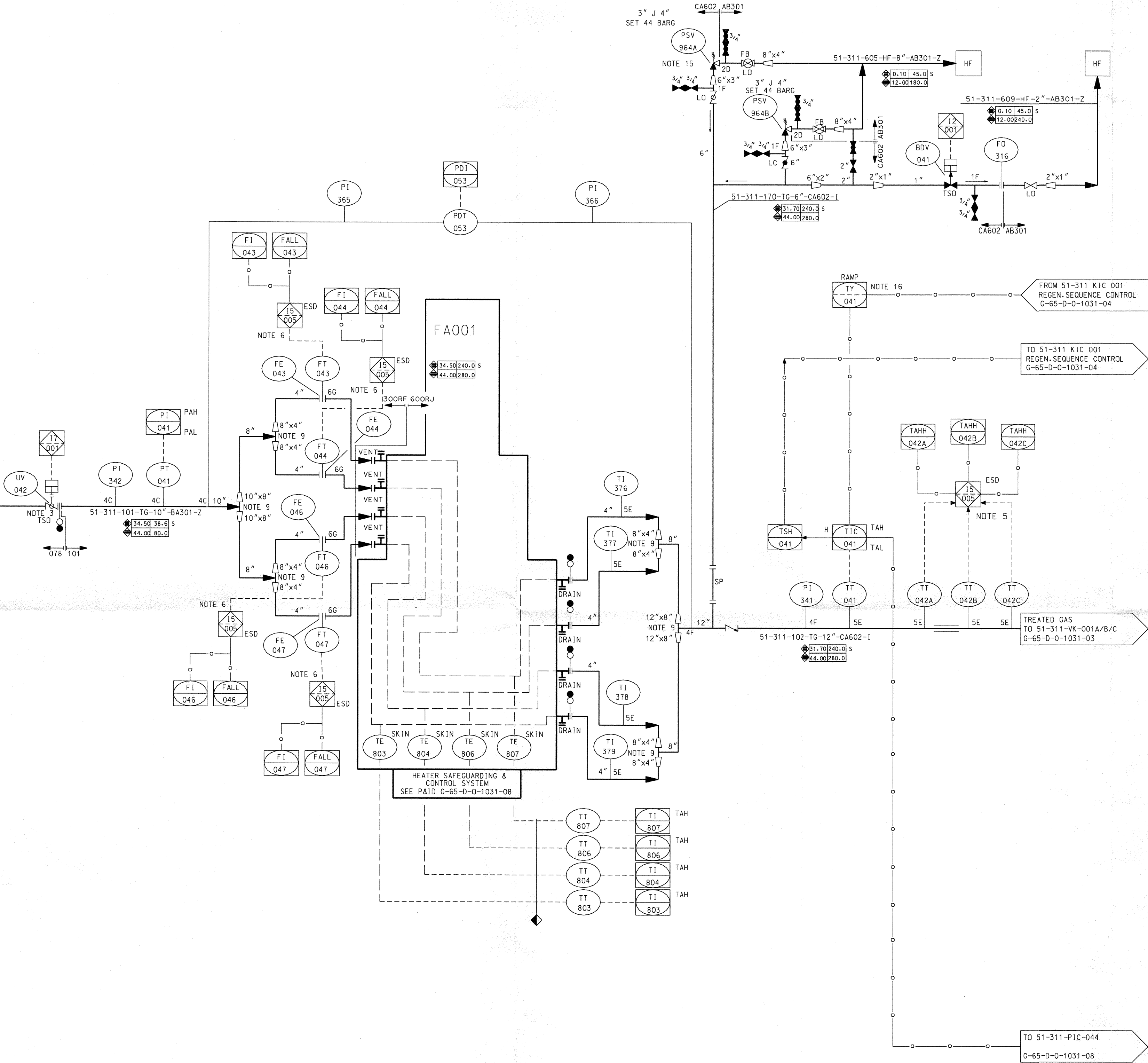
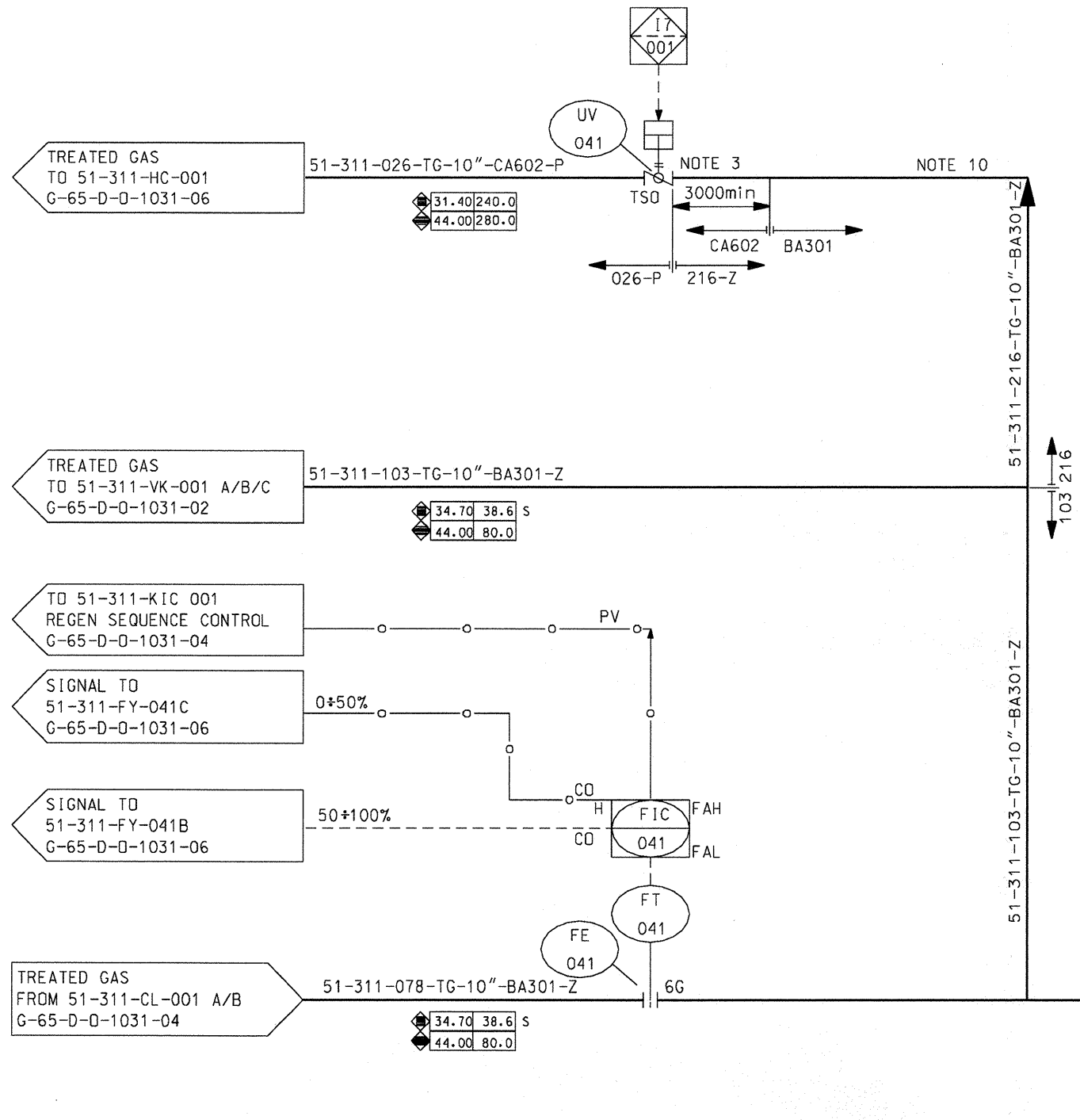
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- 1- DELETED.
- 2- DELETED.
- 3- DELETED.

EQUIPMENT LIST					
VK001A					

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A3	25/10/04	ISSUED FOR CONSTRUCTION REVISED WHERE INDICATED	CAREDOU	D. POLIDORO	G. LA MANTIA	A. MADDALONI	T. M. SAID
A2	16/04/04	ISSUED FOR CONSTRUCTION	CAREDOU	D. POLIDORO	G. LA MANTIA	A. MADDALONI	T. M. SAID
A1	23/10/03	ISSUED FOR CONSTRUCTION	CAREDOU	D. POLIDORO	G. LA MANTIA	A. MADDALONI	T. M. SAID
P2	30/05/03	ISSUED FOR DESIGN	R. CAREDOU	D. POLIDORO	G. LA MANTIA	A. MADDALONI	T. M. SAID
P1	18/02/03	ISSUED FOR REVIEW	MB	AL	GL		
Revision	DATE	DESCRIPTION	PREP.	CHECK.	APPR.	PMS CHECK.	AGBV APPR.
Agip Gas BV			WESTERN LIBYA GAS PROJECT MELLITAH PLANT GSLPAJ				AGBV DWG No. G-65-D-0-1031-02
Libyan Branch							AGBV JOB No. 0177/NC/02
Snamprogetti Lummus Gas Ltd							CTR DWG No. 51-311-GD-B-03811
Snamprogetti							Sh. of ...
ABB							CTR JOB NO. 310300
HYUNDAI							Revision ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐
PIPING & INSTRUMENT DIAGRAM UNIT 311 - GAS DEHYDRATION DRYER "A"							IN REPLACEMENT OF
							REPLACED BY



P1 25/02/03		ISSUED FOR REVIEW		CAREDDU	D-POLIGORO	D-LA MANTIA																																												
Revision	DATE	DESCRIPTION		PREP.	CHECK.	APPR.	PWS CHECK.	AGSV APPR.																																										
Agip Gas BV LIBYAN BRANCH				WESTERN LIBYA GAS PROJECT MELLITAH PLANT GSPLAJ		AGSV DWG No. G-65-D-0-1031-03 AGSV JOB No.0177NC/02																																												
 Snamprogetti				 ABB LOW VOLTAGE		 HYUNDAI INDUSTRIAL & CONSTRUCTION EQUIP. CO. LTD.		CTR DWG No. 51-311-GD-B-03812 Sp. _____ of _____																																										
TITLE: PIPING & INSTRUMENT DIAGRAM UNIT 311 - GAS DEHYDRATION DRYERS "B" AND "C"				CTR JOB NO. 310300 Revision <table data-bbox="2789 1999 2902 2009"><tr><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td><td>11</td><td>12</td><td>13</td><td>14</td><td>15</td><td>16</td><td>17</td><td>18</td><td>19</td><td>20</td><td>21</td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr></table> SCALE _____					1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21																					
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21																														



EQUIPMENT WITH FUEL FEED	51311FA001 REGENERATION GAS HEATER
DIMENSIONS	4500x3200 (MM X MM)
CAPACITY	(M3)
FLOW RATE	34100 (KG/H)
POTENTIAL	5034 (KW)
FUEL	L.P. FUEL GAS
HOURLY CONSUMPTION	415 (KG/H)
TRIM LINE	51-311-422-TG-1"-CA602-I
DESIGN TEMP./PRESS.	-29 +280/44 (C/BAR G)
WEIGHT	60000 (KG)
DATA SHEET No.	G-65-X-0-31FA-01

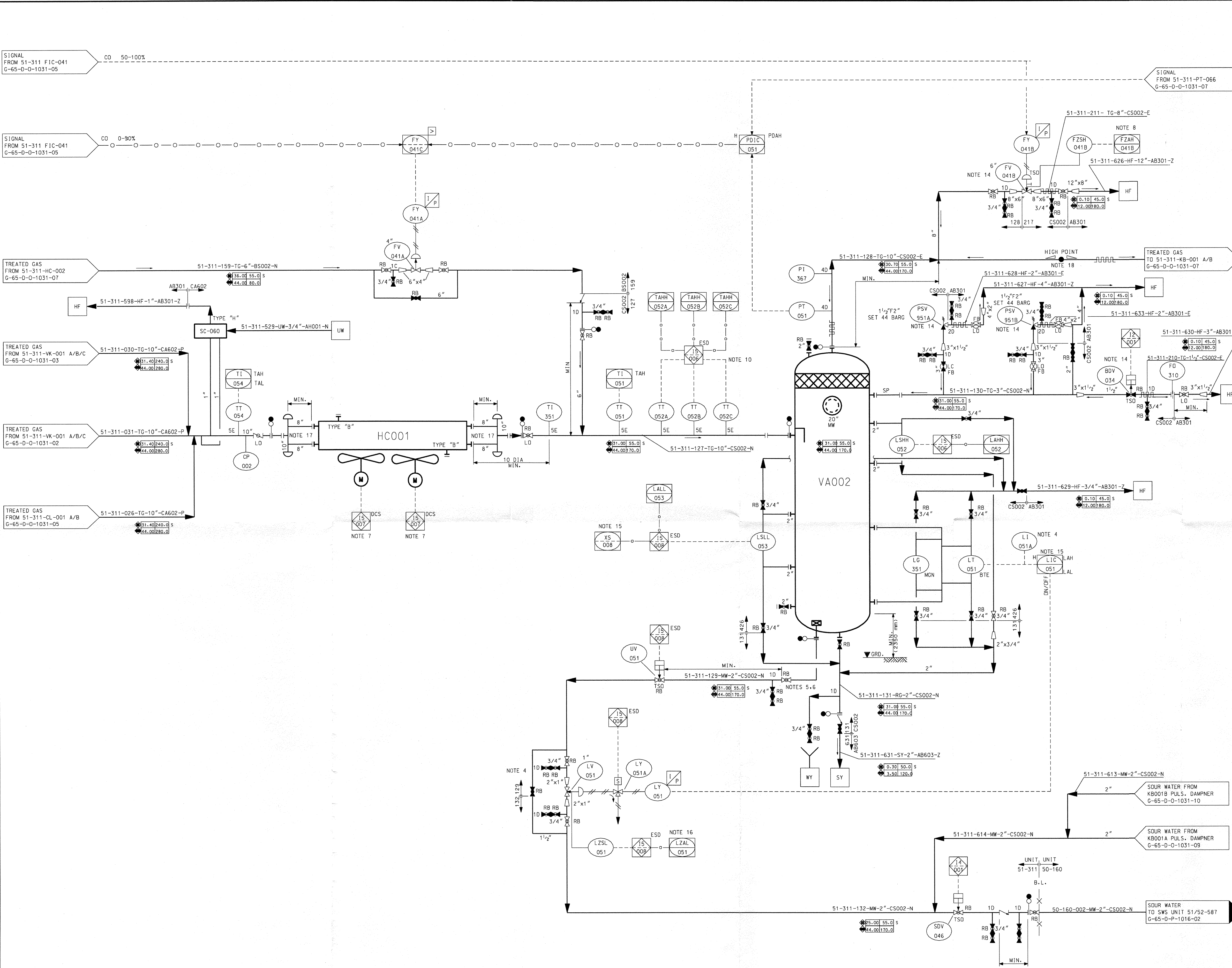
UTILITIES IDENTIFICATION					
CODE	DESCRIPTION	REF. DWG	CODE	DESCRIPTION	REF. DWG
DW	DRINKING WATER		CA	COMPRESS. AIR	
FW	FIRE-FIGHTING WATER		IA	INSTRUMENT AIR	
SW	SEA WATER		UA	UTILITIES AIR	
TW	DEMIN. WATER		UN	INERT GAS (NITROGEN)	
UW	DESALT. WATER		GD	DIESEL OIL	
BW	H.P. & L.P. BOILER FEED WATER		FG	H.P. & L.P. FUEL GAS	
CW	COOLING WATER		SY	CLOSED DRAIN	
HS	H.P. STEAM		WY	OPEN DRAIN	
MS	M.P. STEAM		RW	OILY STORM WATER	
LS	L.P. STEAM		HF	H.P. FLARE	G-65-D-0-1018-04
MC	M.P. STEAM COND.		LF	L.P. FLARE	
LC	L.P. STEAM COND.		CF	DRY/COLD FLARE	
LL	LOW LOW PRES. STEAM		AF	ACID BLOWDOWN	
SC	SUSPECT CONDENSATE		NW	NEUTRALIZATION WATER	
FF	FIRE FIGHTING FOAM				
JA	HYDROC. ACID SOLUTION				
JB	CAUSTIC SODA SOLUTION				

- NOTES
- FOR SYMBOLS, LEGEND, TYPICAL AND GENERAL NOTES SEE P&ID'S:
G-00-D-0-1000-01/02/03/04/05/06/07/08/09/10/11/12/13/14/15/16/17/18/19/25
 - FOR GENERAL SHUTDOWN LOGIC SEE P&ID'S: G-65-D-0-1000-20/21/22/23/24.
 - FOR PIPING ASSEMBLY (I.E. 4D) REFER TO DDC: G-65-S-P-0028-00
- 1- UNLESS OTHERWISE SPECIFIED, ALL INSTRUMENTS AND EQUIPMENTS SHOWN ON THIS DRAWING ARE PREFIXED WITH THE UNIT IDENTIFICATION NUMBER "51-311"
 - 2- DELETED
 - 3- HIGH PERFORMANCE BUTTERFLY VALVE FOR SPECIAL SERVICE: ANTIABRASION/ANTIERSION, DUST SERVICE WITH HIGH DIFFERENCE THERMAL EXPANSION
 - 4- DELETED
 - 5- TWO OUT OF THREE VOTING LOGIC.
 - 6- INHIBITION DURING COOLING AND STANDBY CYCLES.
 - 7- DELETED
 - 8- DELETED
 - 9- PROVIDE SYMMETRICAL INLET/OUTLET PIPING
 - 10- DRYERS BY PASS LINE (STAND BY PHASE)
 - 11- DELETED
 - 12- DELETED
 - 13- DELETED
 - 14- DELETED
 - 15- DELETED
 - 16- RAMP GENERATOR TO PERFORM HEATING AND PREHEATING TIME SCHEDULED CURVES.

- HOLD INDEX
- 1- DELETED.
 - 2- DELETED.
 - 3- DELETED.
 - 4- DELETED.

EQUIPMENT LIST					
FA001					

E1	30/08/07	AS BUILT	CAREDDU	D. POLIDORO	E. LA MANTIA	A. MADDALONI	T. M. SAID
A3	25/10/04	ISSUED FOR CONSTRUCTION REVISED WHERE INDICATED	CAREDDU	D. POLIDORO	E. LA MANTIA	A. MADDALONI	T. M. SAID
A2	16/04/04	ISSUED FOR CONSTRUCTION	CAREDDU	D. POLIDORO	E. LA MANTIA	A. MADDALONI	T. M. SAID
A1	23/10/03	ISSUED FOR CONSTRUCTION	CAREDDU	D. POLIDORO	E. LA MANTIA	A. MADDALONI	T. M. SAID
P2	30/05/03	ISSUED FOR DESIGN	CAREDDU	D. POLIDORO	E. LA MANTIA	A. MADDALONI	T. M. SAID
P1	18/02/03	ISSUED FOR REVIEW	MB	AL	OL		
Revision	DATE	DESCRIPTION	PREP.	CHECK.	APPR.	PMS CHECK.	AGB/ APRR.
AgipGas BV			WESTERN LIBYA GAS PROJECT MELLITAH PLANT CSPLAJ			AGBV DWG No. G-65-D-0-1031-05	
LIBYAN BRANCH						AGBV JOB No. 0177/NC02	
Snamprogetti Lummus Gas Ltd						CTR DWG No. 51-311-GD-B-03814	
Snamprogetti			HYUNDAI			CTR JOB NO. 310300	
PIPING & INSTRUMENT DIAGRAM UNIT 311 - GAS DEHYDRATION REGENERATION GAS HEATER						Revision: 1 2 3 4 5 6 7 8 9 10	
						SCALE	
						REPLACED BY	



TWO FLUIDS EQUIPMENT		PRESSURE EQUIPMENT	
51311HC001 REGENERATION GAS COOLER		51311VA002 REGEN. GAS COMPR. K.O. DRUM	
DIMENSIONS	12520x3940x2417 (MM X MM X MM)	DIAMETER	1050 (MM)
POTENTIAL	3.93 (MW)	LENGTH TL-TL	2700 (MM)
HOT FLUID/FLOW RATE	31000 (KG/H)	CAPACITY	2.6 (M3)
OPERATING TEMP./PRESS.	240/31.4 (C/BAR G)	OPERATING TEMP./PRESS.	55/31 (C/BAR G)
OUT OPER. TEMP.	55 (C)		
TRIM LINE		TRIM LINE	
DESIGN TEMP./PRESS.	-29 +280/44 (C/BAR G)	DESIGN TEMP./PRESS.	-29 +170/44 (C/BAR G)
WEIGHT	9560 (KG) x BUNDLE	WEIGHT	5460 (KG)
DATA SHEET No.	G-65-X-O-31HC-01	DATA SHEET No.	G-65-X-O-31VA-01

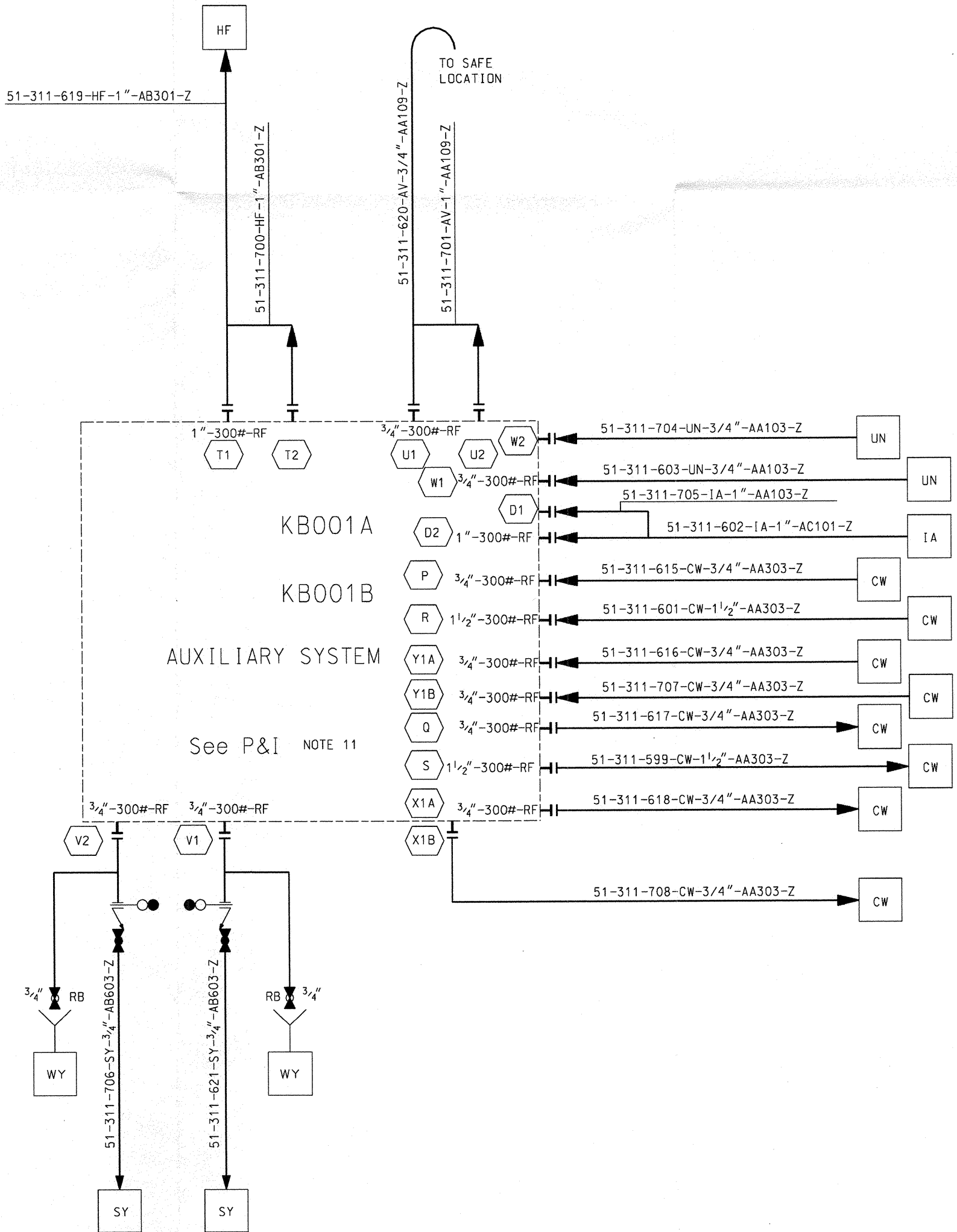
UTILITIES IDENTIFICATION					
CODE	DESCRIPTION	REF. DWG	CODE	DESCRIPTION	REF. DWG
DW	DRINKING WATER		CA	COMPRESS. AIR	
FW	FIRE-FIGHTING WATER		IA	INSTRUMENT AIR	
SW	SEA WATER		UA	UTILITIES AIR	
TW	DEMIN. WATER		UN	INERT GAS (NITROGEN)	
UW	DESALT. WATER		GO	DIESEL OIL	
BW	H.P. & L.P. BOILER FEED WATER		FG	H.P. & L.P. FUEL GAS	
CW	COOLING WATER		SY	CLOSED DRAIN	G-65-D-O-1018-06
HS	H.P. STEAM		WY	OPEN DRAIN	
MS	M.P. STEAM		RW	OILY STORM WATER	
LS	L.P. STEAM		HF	H.P. FLARE	G-65-D-O-1018-04
MC	M.P. STEAM COND.		LF	L.P. FLARE	
LC	L.P. STEAM COND.		CF	DRY/COLD FLARE	
LL	LOW LOW PRES. STEAM		AF	ACID BLOWDOWN	
SC	SUSPECT CONDENSATE		NW	NEUTRALIZATION WATER	
FF	FIREFIGHTING FOAM				
JA	HYDROC. ACID SOLUTION				
JB	CAUSTIC SODA SOLUTION				

- NOTES
- FOR SYMBOLS, LEGEND, TYPICAL AND GENERAL NOTES SEE P&ID'S: G-00-D-O-1000-01/02/03/04/05/06/07/08/09/10/11/12/13/14/15/16/17/18/19/25.
 - FOR GENERAL SHUTDOWN LOGIC SEE P&ID'S: G-65-D-O-1000-20/21/22/23/24.
 - FOR PIPING ASSEMBLY (I.E. 4D) REFER TO DOC:G-65-S-P-0028-00
- 1-UNLESS OTHERWISE SPECIFIED, ALL INSTRUMENTS AND EQUIPMENTS SHOWN ON THIS DRAWING ARE PREFIXED WITH THE UNIT IDENTIFICATION NUMBER "51-311"
- 2- DELETED.
- 3- DELETED.
- 4- LI VISIBLE FROM CONTROL VALVE BY PASS
- 5- LOCATE AT SAFE DISTANCE FROM THE NEAREST ASSUMED SOURCE OF LOSS OF CONTAINMENT.
- 6- TO BE OPERATED FROM GRADE.
- 7- FANS TO BE EQUALLY SHARED BETWEEN TWO INDEPENDENT BUS BARS.
- 8-ALARM TO BE ACTIVATED WHEN VALVE IS NOT CLOSED. (LIMIT SWITCH TO BE LOCATED IN VALVE CLOSE POSITION)
- 9- DELETED.
- 10-TWO OUT OF THREE VOTING LOGIC.
- 11-DELETED.
- 12-DELETED.
- 13-DELETED.
- 14-PROVIDE ELECTRIC TRACING ON VALVE BODY AND PART OF THE OUTLET LINE TO FLARE. EXTENT OF TRACING TO BE: 1000mm FOR PSV AND BDV 3000mm FOR FV
- 15- ACTIVATION OF XS-008 DRIVES LIC-051 TO MANUAL MODE AND OUTPUT NULL.
- 16- CLOSE COMMAND FAILURE ALARM.
- 17- PIPE TO BE SYMMETRICAL.
- 18- HIGH POINT TO BE LOCATED AS CLOSE AS POSSIBLE TO THE COMPRESSOR.

HOLD INDEX					
1- DELETED					
2- DELETED					
3- DELETED					
4- DELETED					

EQUIPMENT LIST					
HC001	VA002				

E1	30/08/07	AS BUILT	CAREDDU	D. POL. (DORO)	G. LA. MANTIA	A. MADDALONI	T. M. SAID
A3	25/10/04	ISSUED FOR CONSTRUCTION REVISED WHERE INDICATED	CAREDDU	D. POL. (DORO)	G. LA. MANTIA	A. MADDALONI	T. M. SAID
A2	16/04/04	ISSUED FOR CONSTRUCTION	CAREDDU	D. POL. (DORO)	G. LA. MANTIA	A. MADDALONI	T. M. SAID
A1	23/10/03	ISSUED FOR CONSTRUCTION	CAREDDU	D. POL. (DORO)	G. LA. MANTIA	A. MADDALONI	T. M. SAID
P2	30/06/03	ISSUED FOR DESIGN	CAREDDU	D. POL. (DORO)	G. LA. MANTIA	A. MADDALONI	T. M. SAID
P1	18/02/03	ISSUED FOR REVIEW					
Revision	DATE	DESCRIPTION	PREP.	CHECK.	APPR.	PWS CHECK.	AGBV APPR.
AgbGas BV LIBYAN BRANCH			WESTERN LIBYA GAS PROJECT MELLITAH PLANT GSP/LAJ			AGBV DWG No. G-65-D-O-1031-06	
Snamprogetti Lummus Gas Ltd Snamprogetti ABB HYUNDAI			AGBV JOB No.0177/NC/02			CTR DWG No. 51-311-GD-B-03816	
PIPING & INSTRUMENT DIAGRAM UNIT 311 - GAS DEHYDRATION REGENERATION GAS COOLING			CTR JOB NO. 310300			Revision 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100 101 102 103 104 105 106 107 108 109 110 111 112 113 114 115 116 117 118 119 120 121 122 123 124 125 126 127 128 129 130 131 132 133 134 135 136 137 138 139 140 141 142 143 144 145 146 147 148 149 150 151 152 153 154 155 156 157 158 159 160 161 162 163 164 165 166 167 168 169 170 171 172 173 174 175 176 177 178 179 180 181 182 183 184 185 186 187 188 189 190 191 192 193 194 195 196 197 198 199 200 201 202 203 204 205 206 207 208 209 210 211 212 213 214 215 216 217 218 219 220 221 222 223 224 225 226 227 228 229 230 231 232 233 234 235 236 237 238 239 240 241 242 243 244 245 246 247 248 249 250 251 252 253 254 255 256 257 258 259 260 261 262 263 264 265 266 267 268 269 270 271 272 273 274 275 276 277 278 279 280 281 282 283 284 285 286 287 288 289 290 291 292 293 294 295 296 297 298 299 300 301 302 303 304 305 306 307 308 309 310 311 312 313 314 315 316 317 318 319 320 321 322 323 324 325 326 327 328 329 330 331 332 333 334 335 336 337 338 339 340 341 342 343 344 345 346 347 348 349 350 351 352 353 354 355 356 357 358 359 360 361 362 363 364 365 366 367 368 369 370 371 372 373 374 375 376 377 378 379 380 381 382 383 384 385 386 387 388 389 390 391 392 393 394 395 396 397 398 399 400 401 402 403 404 405 406 407 408 409 410 411 412 413 414 415 416 417 418 419 420 421 422 423 424 425 426 427 428 429 430 431 432 433 434 435 436 437 438 439 440 441 442 443 444 445 446 447 448 449 450 451 452 453 454 455 456 457 458 459 460 461 462 463 464 465 466 467 468 469 470 471 472 473 474 475 476 477 478 479 480 481 482 483 484 485 486 487 488 489 490 491 492 493 494 495 496 497 498 499 500 501 502 503 504 505 506 507 508 509 510 511 512 513 514 515 516 517 518 519 520 521 522 523 524 525 526 527 528 529 530 531 532 533 534 535 536 537 538 539 540 541 542 543 544 545 546 547 548 549 550 551 552 553 554 555 556 557 558 559 560 561 562 563 564 565 566 567 568 569 570 571 572 573 574 575 576 577 578 579 580 581 582 583 584 585 586 587 588 589 590 591 592 593 594 595 596 597 598 599 600 601 602 603 604 605 606 607 608 609 610 611 612 613 614 615 616 617 618 619 620 621 622 623 624 625 626 627 628 629 630 631 632 633 634 635 636 637 638 639 640 641 642 643 644 645 646 647 648 649 650 651 652 653 654 655 656 657 658 659 660 661 662 663 664 665 666 667 668 669 670 671 672 673 674 675 676 677 678 679 680 681 682 683 684 685 686 687 688 689 690 691 692 693 694 695 696 697 698 699 700 701 702 703 704 705 706 707 708 709 710 711 712 713 714 715 716 717 718 719 720 721 722 723 724 725 726 727 728 729 730 731 732 733 734 735 736 737 738 739 740 741 742 743 744 745 746 747 748 749 750 751 752 753 754 755 756 757 758 759 760 761 762 763 764 765 766 767 768 769 770 771 772 773 774 775 776 777 778 779 780 781 782 783 784 785 786 787 788 789 790 791 792 793 794 795 796 797 798 799 800 801 802 803 804 805 806 807 808 809 810 811 812 813 814 815 816 817 818 819 820 821 822 823 824 825 826 827 828 829 830 831 832 833 834 835 836 837 838 839 840 841 842 843 844 845 846 847 848 849 850 851 852 853 854 855 856 857 858 859 860 861 862 863 864 865 866 867 868 869 870 871 872 873 874 875 876 877 878 879 880 881 882 883 884 885 886 887 888 889 890 891 892 893 894 895 896 897 898 899 900 901 902 903 904 905 906 907 908 909 910 911 912 913 914 915 916 917 918 919 920 921 922 923 924 925 926 927 928 929 930 931 932 933 934 935 936 937 938 939 940 941 942 943 944 945 946 947 948 949 950 951 952 953 954 955 956 957 958 959 960 961 962 963 964 965 966 967 968 969 970 971 972 973 974 975 976 977 978 979 980 981 982 983 984 985 986 987 988 989 990 991 992 993 994 995 996 997 998 999 1000 1001 1002 1003 1004 1005 1006 1007 1008 1009 1010 1011 1012 1013 1014 1015 1016 1017 1018 1019 1020 1021 1022 1023 1024 1025 1026 1027 1028 1029 1030 1031 1032 1033 1034 1035 1036 1037 1038 1039 1040 1041 1042 1043 1044 1045 1046 1047 1048 1049 1050 1051 1052 1053 1054 1055 1056 1057 1058 1059 1060 1061 1062 1063 1064 1065 1066 1067 1068 1069 1070 1071 1072 1073 1074 1075 1076 1077 1078 1079 1080 1081 1082 1083 1084 1085 1086 1087 1088 1089 1090 1091 1092 1093 1094 1095 1096 1097 1098 1099 1100 1101 1102 1103 1104 1105 1106 1107 1108 1109 1110 1111 1112 1113 1114 1115 1116 1117 1118 1119 1120 1121 1122 1123 1124 1125 1126 1127 1128 1129 1130 1131 1132 1133 1134 1135 1136 1137 1138 1139 1140 1141 1142 1143 1144 1145 1146 1147 1148 1149 1150 1151 1152 1153 1154 1155 1156 1157 1158 1159 1160 1161 1162 1163 1164 1165 1166 1167 1168 1169 1170 1171 1172 1173 1174 1175 1176 1177 1178 1179 1180 1181 1182 1183 1184 1185 1186 1187 1188 1189 1190 1191 1192 1193 1194 1195 1196 1197 1198 1199 1200 1201 1202 1203 1204 1205 1206 1207 1208 1209 1210 1211 1212 1213 1214 1215 1216 1217 1218 1219 1220 1221 1222 1223 1224 1225 1226 1227 1228 1229 1230 1231 1232 1233 1234 1235 1236 1237 1238 1239 1240 1241 1242 1243 1244 1245 1246 1247 1248 1249 1250 1251 1252 1253 1254 1255 1256 1257 1258 1259 1260 1261 1262 1263 1264 1265 1266 1267 1268 1269 1270 1271 1272 1273 1274 1275 1276 1277 1278 1279 1280 1281 1282 1283 1284 1285 1286 1287 1288 1289 1290 1291 1292 1293 1294 1295 1296 1297 1298 1299 1300 1301 1302 1303 1304 1305 1306 1307 1308 1309 1310 1311 1312 1313 1314 1315 1316 1317 1318 1319 1320 1321 1322 1323 1324 1325 1326 1327 1328 1329 1330 1331 1332 1333 1334 1335 1336 1337 1338 1339 1340 1341 1342 1343 1344 1345 1346 1347 1348 1349 1350 1351 1352 1353 1354 1355 1356 1357 1358 1359 1360 1361 1362 1363 1364 1365 1366 1367 1368 1369 1370 1371 1372 1373 1374 1375 1376 1377 1378 1379 1380 1381 1382 1383 1384 1385 1386 1387 1388 1389 1390 1391 1392 1393 1394 1395 1396 1397 1398 1399 1400 1401 1402 1403 1404 1405 1406 1407 1408 1409 1410 1411 1412 1413 1414 1415 1416 1417 1418 1419 1420 1421 1422 1423 1424 1425 1426 1427 1428 1429 1430 1431 1432 1433 1434 1435 1436 1437 1438 1439 1440 1441 1442 1443 1444 1445 1446 1447 1448 1449 1450 1451 1452 1453 1454 1455 1456 1457 1458 1459 1460 1461 1462 1463 1464 1465 1466 1467 1468 1469 1470 1471 1472 1473 1474 1475 1476 1477 1478 1479 1480 1481 1482 1483 1484 1485 1486 1487 1488 1489 1490 1491 1492 1493 1494 1495 1496 1497 1498 1499 1500 1501 1502 1503 1504 1505 1506 1507 1508 1509 1510 1511 1512 1513 1514 1515 1516 1517 1518 1519 1520 1521 1522 1523 1524 1525 1526 1527 1528 1529 1530 1531 1532 1533 1534 1535 1536 1537 1538 1539 1540 1541 1542 1543 1544 1545 1546 1547 1548 1549 1550 1551 1552 1553 1554 1555 1556 1557 1558 1559 1560 1561 1562 1563 1564 1565 1566 1567 1568 1569 1570 1571 1572 1573 1574 1575 1576 1577 1578 1579 1580 1581 1582 1583 1584 1585 1586 1587 1588 1589 1590 1591 1592 1593 1594 1595 1596 1597 1598 1599 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1800 1801 1802 1803 1804 1805 1806 1807 1808 1809 1810 1811 1812 1813 1814 1815 1816 1817 1818 1819 1820 1821 1822 1823 1824 1825 1826 1827 1828 1829 1830 1831 1832 1833 1834 1835 1836 1837 1838 1839 1840 1841 1842 1843 1844 1845 1846 1847 1848 1849 1850 1851 1852 1853 1854 1855 1856 1857 1858 1859 1860 1861 1862 1863 1864 1865 1866 1867 1868 1869 1870 1871 1872 1873 1874 1875 1876 1877 1878 1879 1880 1881 1882 1883 1884 1885 1886 1887 1888 1889 1890 1891 1892 1893 1894 1895 1896 1897 1898 1899 1900 1901 1902 1903 1904 1905 1906 1907 1908 1909 1910 1911 1912 1913 1914 1915 1916 1917 1918 1919 1920 1921 1922 1923 1924 1925 1926 1927 1928 1929 1930 1931 1932 1933 1934 1935 1936 1937 1938 1939 1940 1941 1942 1943 1944 1945 1946 1947 1948 1949 1950 1951 1952 1953 1954 1955 1956 1957 1958 1959 1960 1961 1962 1963 1964 1965 1966 1967 1968 1969 1970 1971 1972 1973 1974 1975 1976 1977 1978 1979 1980 1981 1982 1983 1984 1985 1986 1987 1988 1989 1990 1991 1992 1993 1994 1995 1996 1997 1998 1999 2000 2001 2002 2003 2004 2005 2006 2007 2008 2009 2010 2011 2012 2013 2014 2015 2016 2017 2018 2019 2020 2021 2022 2023 2024 2025 2026 2027 2028 2029 2030 2031 2032 2033 2034 2035 2036 2037 2038 2039 2040 2041 2042 2043 2044 2045 2046 2047 2048 2049 2050 2051 2052 2053 2054 2055 2056 2057 2058 2059 2060 2061 2062 2063 2064 2065 2066 2067 2068 2069 2070 2071 2072 2073 2074 2075 2076 2077 2078 2079 2080 2081 2082 2083 2084 2085 2086 2087 2088 2089 2090 2091 2092 2093 2094 2095 2096 2097 2098 2099 2100 2101 2102 2103 2104 2105 2106 2107 2108 2109 2110 2111 2112 2113 2114 2115 2116 2117 2118 2119 2120 2121 2122 2123 2124 2125 2126 2127 2128 2129 2130 2131 2132 2133 2134 2135 2136 2137 2138 2139 2140 2141 2142 2143 2144 2145 2146 2147 2148 2149 2150 2151 2152 2153 2154 2155 2156 2157 2158 2159 2160 2161 2162 2163 2164 2165 2166 2167 2168 2169 2170 2171 2172 2173 2174 2175 2176 2177 2178 2179 2180 2181 2182 2183 2184 2185 2186 2187 2188 2189 2190 2191 2192 2193 2194 2195 2196 2197 2198 2199 2200 2201 2202 2203 2204 2205 2206 2207 2208 2209 2210 2211 2212 2213 2214 2215 2216 2217 2218 2219 2220 2221 2222 2223 2224 2225 2226 2227 2228 2229 2230 2231 2232 2233 2234 2235 2236 2237 2238 2239 2240 2241 2242 2243 2244 2245 2246 2247 2248 2249 2250 2251 2252 2253 2254 2255 2256 2257 2258 2259 2260 2261 2262 2263 2264 2265 2266 2267 2268 2269 2270 2271 2272 2273 2274 2275 2276 2277 2278 2279 2280 2281 2282 2283 2284 2285 2286 2287 2288 2289 2290 2291 2292 2293 2294 2295 2296 2297 2298 2299 2300 2301 2302 2303 2304 2305 2306 2307 2308 2309 2310 2311 2312 2313 2314 2315 2316 2317 2318 2319 2320 2321 2322 2323 2324 2325 2326 2327 2328 2329 2330 2331 2332 2333 2334 2335 2336 2337 2338 2339 2340 2341 2342 2343 2344 2345 2346 2347 2348 2349 2350 2351 2352 2353 2354 2355 2356 2357 2358 2359 2360 2361 2362 2363 2364 2365 2366 2367 2368 2369 2370 2371 2372 2373 2374 2375 2376 2377 2378 2379 2380 2381 2382 2383 2384 2385 2386 2387 2388 2389 2390 2391 2392 2393 2394 2395 2396 2397 2398 2399 2400 2401 2402 2403 2404 2405 2406 2407 2408 2409 2410 2411 2412 2413 2414 2415 2416 2417 2418 2419 2420 2421 2422 2423 2424 2425 2426 2427 2428 2429 2430 2431 2432 2433 2434 2435 2436 2437 2438 2439 2440 2441 2442 2443 2444 2445 2446 2447 2448 2449 2450 2451 2452 2453 2454 2455 2456 2457 2458 2459 2460 2461 2462 2463 2464 2465 2466 2467 2468 2469 2470 2471 2472 2473 2474 2475 2476 2477 2478 2479 2480 2481 2482 2483 2484 2485 2486 2487 2488 2489 2490 2491 2492 2493 2494 2495 2496 2497 2498 2499 2500 2501 2502 2503 2504 2505 2506 2507 2508 2509 2510 2511 2512 2513 2514 2515 2516 2517 2518 2519 2520 2521 2522 2523 2524 2525 2526 2527 2528 2529 2530 2531 2532 2533 2534 2535 2536 2537 2538 2539 2540 2541 2	



UTILITIES IDENTIFICATION					
CODE	DESCRIPTION	REF. DWG	CODE	DESCRIPTION	REF. DWG
DW	DRINKING WATER		CA	COMPRESS. AIR	
FW	FIRE-FIGHTING WATER		IA	INSTRUMENT AIR	
SW	SEA WATER		UA	UTILITIES AIR	
TW	DEMIN. WATER		UN	INERT GAS (NITROGEN)	
UW	DESALT. WATER		GD	DIESEL OIL	
BW	H.P. & L.P. BOILER FEED WATER		FG	H.P. & L.P. FUEL GAS	
CW	CODLING WATER		SY	CLOSED DRAIN	
HS	H.P. STEAM		WY	OPEN DRAIN	
MS	M.P. STEAM		RW	OILY STORM WATER	
LS	L.P. STEAM		HF	H.P. FLARE	G-65-D-P-1018-04
MC	M.P. STEAM COND.		LF	L.P. FLARE	
LC	L.P. STEAM COND.		CF	DRY/COLD FLARE	
LL	LOW LOW PRES. STEAM		AF	ACID BLOWDOWN	
SC	SUSPECT CONDENSATE		NW	NEUTRALIZATION WATER	
FF	FIREFIGHTING FOAM				
JA	HYDROC. ACID SOLUTION				
JB	CAUSTIC SODA SOLUTION				

1- UNLESS OTHERWISE SPECIFIED, ALL INSTRUMENTS AND EQUIPMENTS SHOWN ON THIS DRAWING ARE PREFIXED WITH THE UNIT IDENTIFICATION NUMBER "51-311"

2- DELETED.

3- DELETED.

4- DELETED.

5- DELETED.

6- DELETED.

7- DELETED.

8- PRESSURIZING LINE

9- PROVIDE SYMMETRICAL PIPING ARRANGEMENT.

10- PROVIDE ELECTRICAL TRACING ON VALVE BODY AND PART OF OUTLET LINE TO FLARE. EXTENT OF TRACING TO BE 500mm FROM BDV.

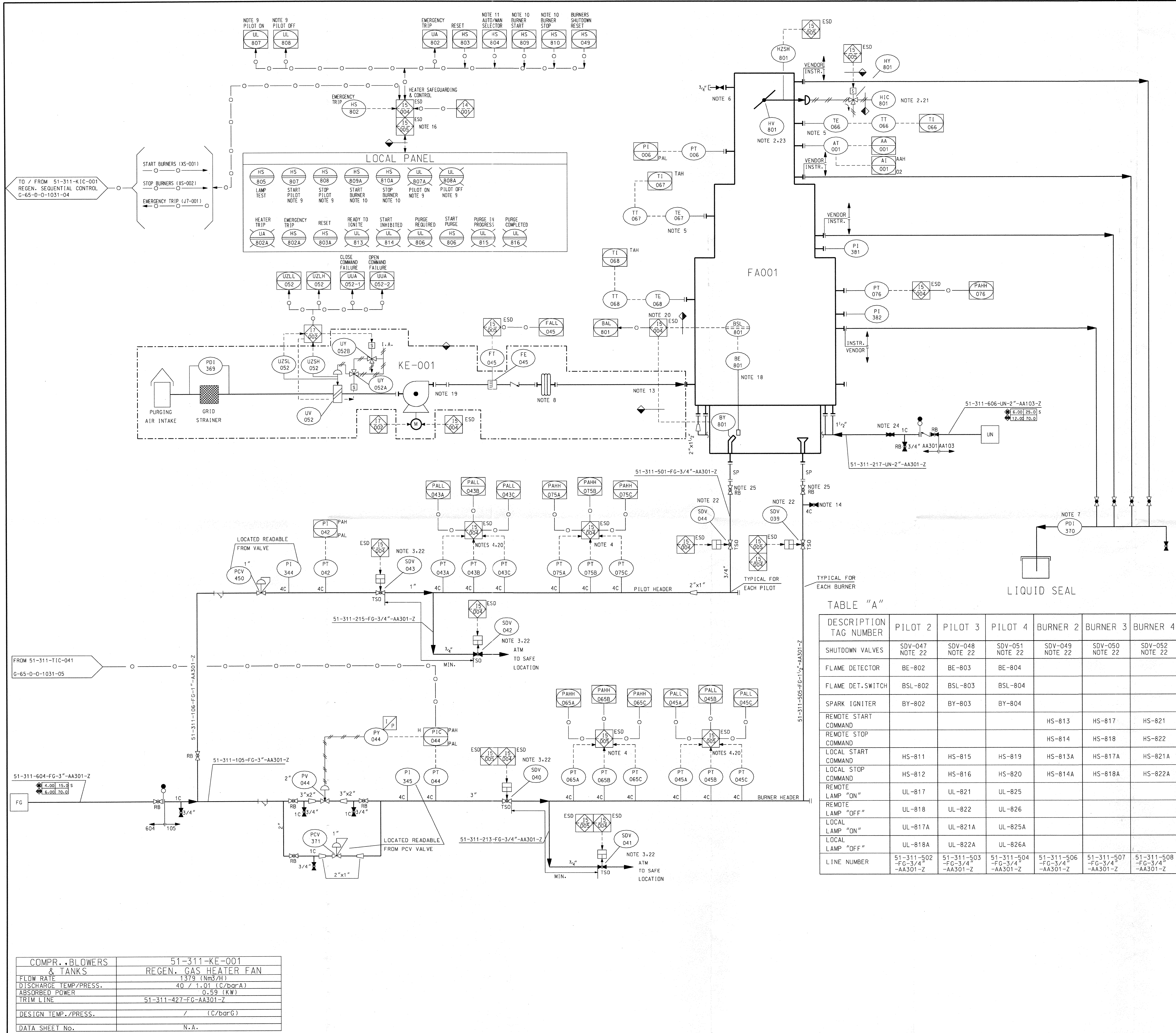
11- FOR COMPRESSOR AUXILIARY SYSTEM DETAIL REFER TO DWG. N°13120-51-D-L-0012-00/0018-00/0021-00/0015-00.

1- DELETED.
2- DELETED.
3- DELETED.
4- DELETED.

HC002				

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G-65-D-Q-1031-07-E1



UTILITIES IDENTIFICATION					
CODE	DESCRIPTION	REF. DWG	CODE	DESCRIPTION	REF. DWG
DW	DRINKING WATER		CA	COMPRESS. AIR	
FW	FIRE-FIGHTING WATER		IA	INSTRUMENT AIR	
SW	SEA WATER		UA	UTILITIES AIR	
TW	DEMIN. WATER		UN	INERT GAS (NITROGEN)	
UW	DESALT. WATER		GO	DIESEL OIL	
BW	H.P. & L.P. BOILER FEED WATER		FG	H.P. & L.P. FUEL GAS	G-65-D-0-1018-07
CW	COOLING WATER		SY	CLOSED DRAIN	
HS	H.P. STEAM		WY	OPEN DRAIN	
MS	M.P. STEAM		RW	OILY STORM WATER	
LS	L.P. STEAM		HF	H.P. FLARE	
MC	M.P. STEAM COND.		LF	L.P. FLARE	
LC	L.P. STEAM COND.		CF	DRY/COLD FLARE	
LL	LOW PRES. STEAM		AF	ACID BLOWDOWN	
SC	SUSPECT CONDENSATE		NW	NEUTRALIZATION WATER	
FF	FIREFIGHTING FOAM				
JA	HYDROC. ACID SOLUTION				
JB	CAUSTIC SODA SOLUTION				

- NOTES
- FOR SYMBOLS, LEGEND, TYPICAL AND GENERAL NOTES SEE P&ID'S:
G-00-D-0-1000-01/02/03/04/05/06/07/08/09/10/11/12/13/14/15/16/17/18/19/25.
 - FOR GENERAL SHUTDOWN LOGIC SEE P&ID'S: G-65-D-0-1000-20/21/22/23/24.
 - FOR PIPING ASSEMBLY (I.e. 4D) REFER TO DOC:G-65-S-P-0028-00
- 1- UNLESS OTHERWISE SPECIFIED, ALL INSTRUMENTS AND EQUIPMENTS SHOWN ON THIS DRAWING ARE PREFIXED WITH THE UNIT IDENTIFICATION NUMBER "51-311"
- 2- DAMPER OPERATED FROM GRADE AT SAFE LOCATION.
- 3- LOCATE FIRST BLOCK VALVE & BLEED VALVE AT SAFE LOCATION IN ORDER TO GUARANTEE NO FUEL GAS SUPPLY TO FURNACE EVEN IN THE EVENT OF AN EXPLOSION.
- 4- TWO OUT OF THREE VOTING LOGIC.
- 5- TEMPERATURE INDICATORS PROVIDED AT INLET/OUTLET OF CONVECTION SECTION.
- 6- SAMPLE POINT TO BE LOCATED AT AN ACCESSIBLE PLACE.
- 7- GROUPED AT GROUND LEVEL.
- 8- FLANGED EXPANSION JOINT.
- 9- PROVIDE ONE FOR EACH PILOT. SEE TABLE "A" FOR OTHER TAGS (PILOT 2, 3 & 4)
- 10- PROVIDE ONE FOR EACH BURNER. SEE TABLE "A" FOR OTHER TAGS (BURNERS 2, 3 & 4)
- 11- AUTO/MANUAL SWITCH TO SELECT THE BURNER LIGHT-ON BY LOCAL PUSH BUTTONS OR BY THE SEQUENCE 51-311-KIC-001.
- 12- DELETED.
- 13- PROVIDE SYMMETRICAL PIPING
- 14- CONNECTION FOR PORTABLE MANOMETER.
- 15- DELETED
- 16- HEATER SAFEGUARDING SYSTEM TO BE DESIGNED AS FOR NFPA 8502.
- 17- DELETED.
- 18- ULTRAVIOLET SCANNER WITH SELF DIAGNOSTIC (FAILURE). SUPPLIED BY HEATER VENDOR.
- 19- ANTIVIBRATION JOINT.
- 20- TRIP TO BE DELAYED ON HEATER START (0-20 SECONDS ADJUSTABLE).
- 21- PDI-370 TO BE READABLE FROM HIC-801 LOCATION
- 22- ALL HEATER SDVS SHALL NOT BE EQUIPPED WITH RESET PUSHBUTTON
- 23- PROVIDE MECHANICAL STOP AT 30% OF DAMPER OPENING
- 24- LOCATED AT SAFE LOCATION
- 25- LOCATED CLOSE TO INSPECTION DOOR

TABLE "A"						
DESCRIPTION TAG NUMBER	PILOT 2	PILOT 3	PILOT 4	BURNER 2	BURNER 3	BURNER 4
SHUTDOWN VALVES	SDV-047 NOTE 22	SDV-048 NOTE 22	SDV-051 NOTE 22	SDV-049 NOTE 22	SDV-050 NOTE 22	SDV-052 NOTE 22
FLAME DETECTOR	BE-802	BE-803	BE-804			
FLAME DET. SWITCH	BSL-802	BSL-803	BSL-804			
SPARK IGNITER	BY-802	BY-803	BY-804			
REMOTE START COMMAND				HS-813	HS-817	HS-821
REMOTE STOP COMMAND				HS-814	HS-818	HS-822
LOCAL START COMMAND	HS-811	HS-815	HS-819	HS-813A	HS-817A	HS-821A
LOCAL STOP COMMAND	HS-812	HS-816	HS-820	HS-814A	HS-818A	HS-822A
REMOTE LAMP "ON"	UL-817	UL-821	UL-825			
REMOTE LAMP "OFF"	UL-818	UL-822	UL-826			
LOCAL LAMP "ON"	UL-817A	UL-821A	UL-825A			
LOCAL LAMP "OFF"	UL-818A	UL-822A	UL-826A			
LINE NUMBER	51-311-502 -FG-3/4" -AA301-Z	51-311-503 -FG-3/4" -AA301-Z	51-311-504 -FG-3/4" -AA301-Z	51-311-506 -FG-3/4" -AA301-Z	51-311-507 -FG-3/4" -AA301-Z	51-311-508 -FG-3/4" -AA301-Z

COMPR. & BLOWERS	51-311-KE-001
& TANKS	REGEN. GAS HEATER FAN
FLOW RATE	1379 (Nm3/H)
DISCHARGE TEMP/PRESS.	40 / 1.01 (C/barA)
ABSORBED POWER	0.59 (KW)
TRIM LINE	51-311-427-FG-AA301-Z
DESIGN TEMP./PRESS.	/ (C/barG)
DATA SHEET No.	N.A.

EQUIPMENT LIST					
KE001	FA001				

E1	30/08/07	AS BUILT	CAREDDU	D. POLIDORO	G. LA MANTIA	A. MADDALONI	T. SAID
A3	25/10/04	ISSUED FOR CONSTRUCTION REVISED WHERE INDICATED	CAREDDU	D. POLIDORO	G. LA MANTIA	A. MADDALONI	T. SAID
A2	16/04/04	ISSUED FOR CONSTRUCTION	CAREDDU	D. POLIDORO	G. LA MANTIA	A. MADDALONI	T. SAID
A1	23/10/03	ISSUED FOR CONSTRUCTION	CAREDDU	D. POLIDORO	G. LA MANTIA	A. MADDALONI	T. SAID
P2	30/05/03	ISSUED FOR DESIGN	CAREDDU	D. POLIDORO	G. LA MANTIA	A. MADDALONI	T. SAID
P1	18/02/03	ISSUED FOR REVIEW	IRC	DP	GL		
Revision	DATE	DESCRIPTION	PREP.	CHECK.	APPR.	PWS CHECK.	AGBV APPR.

AgipGas BV
LIBYAN BRANCH

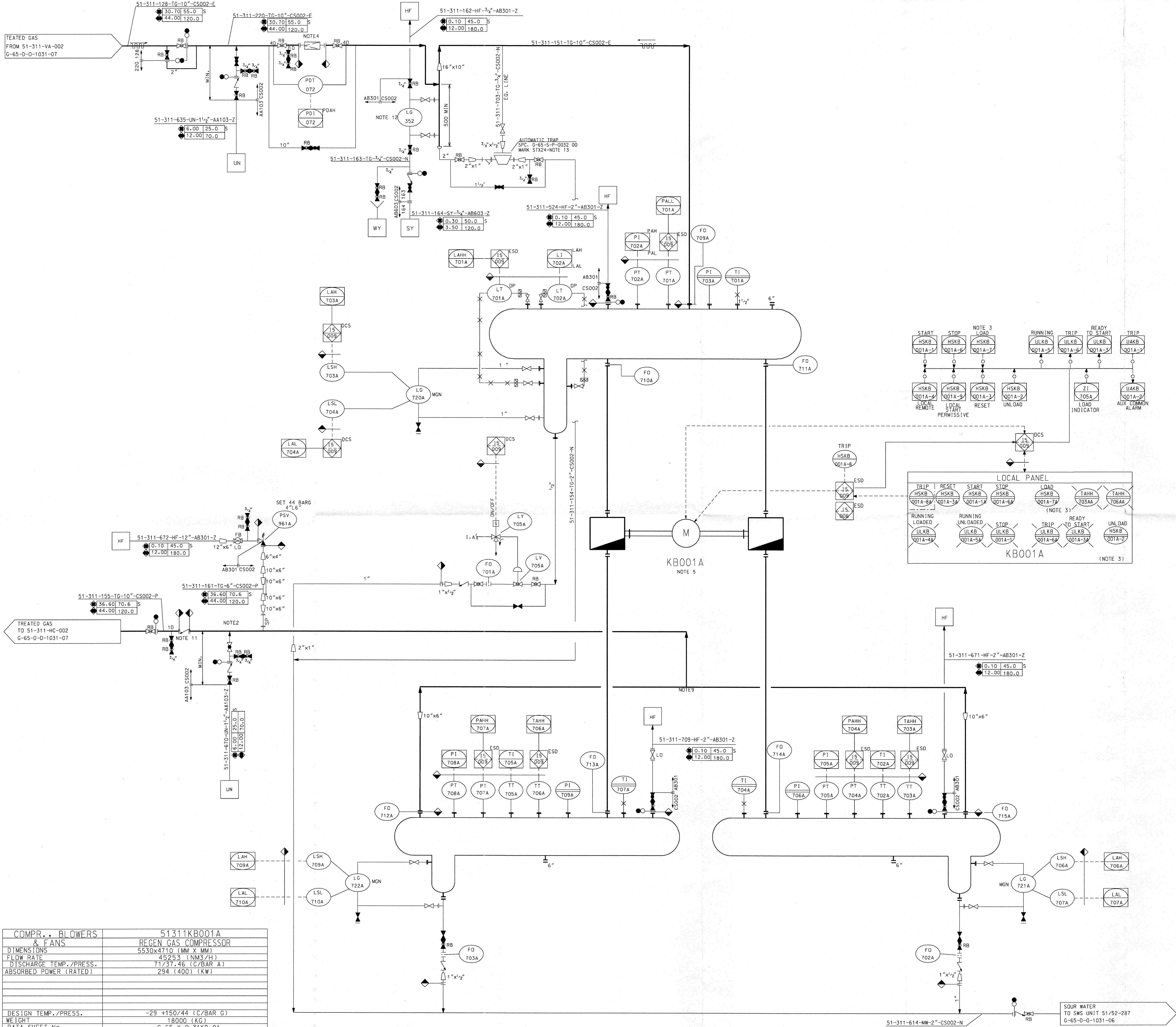
WESTERN LIBYA GAS PROJECT
MELLITAH PLANT
CSPLAJ

AGBV DWG No.
G-65-D-0-1031-08
AGBV JOB No.0177/NC02

Snamprogetti
LUMMUS Gas Ltd

HYUNDAI
CONSTRUCTION & OPERATION CO., LTD.

CTR DWG No.
51-311-GD-B-03815
Sh. of
CTR JOB NO. 310300
Revision
P1 P2 A1 A2 A3 E1
IN REPLACEMENT OF
REPLACED BY
G-65-D-0-1031-08-E1



COMPR. BLOWERS		51311KB001A
& FANS		RECEN GAS COMPRESSOR
DIMENSIONS		5530x4710 (MM X MM)
FLOW RATE		45253 (NM3/H)
DISCHARGE TEMP./PRESS.		71/37.46 (C/BAR A)
ABSORBED POWER (RATED)		294 (400) (KW)
DESIGN TEMP./PRESS.		-29 +150/44 (C/BAR G)
WEIGHT		18000 (KG)
DATA SHEET No.		G-65-X-D-31KB-01

UTILITIES IDENTIFICATION					
CODE	DESCRIPTION	REF. DWG	CODE	DESCRIPTION	REF. DWG
DW	DRINKING WATER		CA	COMPRESS. AIR	
FW	FIRE-FIGHTING WATER		IA	INSTRUMENT AIR	
SW	SEA WATER		UA	UTILITIES AIR	
TW	DEMIN. WATER		UN	INERT GAS (NITROGEN)	G-65-D-0-1018-03
UW	DESALT. WATER		GO	DIESEL OIL	
BW	H.P. & L.P. BOILER FEED WATER		FG	H.P. & L.P. FUEL GAS	
CW	COOLING WATER		SY	CLOSED DRAIN	
HS	H.P. STEAM		WY	OPEN DRAIN	
MS	M.P. STEAM		RW	OILY STORM WATER	
LS	L.P. STEAM		HF	H.P. FLARE	G-65-D-0-1018-04
MC	M.P. STEAM COND.		LF	L.P. FLARE	
LC	L.P. STEAM COND.		CF	DRY/COLD FLARE	
LL	LOW LOW PRES. STEAM		AF	ACID BLOWDOWN	
SC	SUSPECT CONDENSATE		NW	NEUTRALIZATION WATER	
FF	FIREFIGHTING FOAM				
JA	HYDROC. ACID SOLUTION				
JB	CAUSTIC SODA SOLUTION				

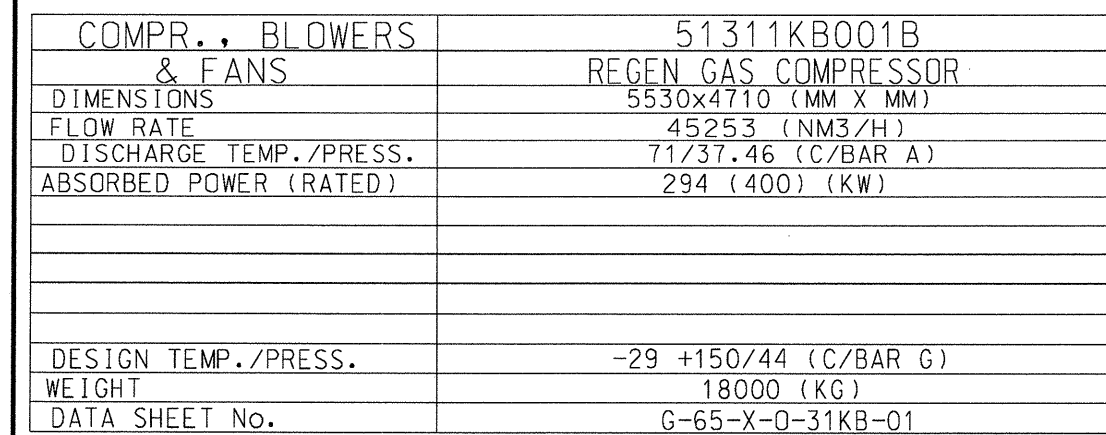
- NOTES
- FOR SYMBOLS, LEGEND, TYPICAL AND GENERAL NOTES SEE P&ID'S: G-00-D-0-1000-01/02/03/04/05/06/07/08/09/10/11/12/13/14/15/16/17/18/19/25.
 - FOR GENERAL SHUTDOWN LOGIC SEE P&ID'S: G-65-D-0-1000-20/21/22/23/24.
 - FOR PIPING ASSEMBLY (I.E. 4D) REFER TO DOC: G-65-S-P-0028-00

- 1- UNLESS OTHERWISE SPECIFIED, ALL INSTRUMENTS AND EQUIPMENTS SHOWN ON THIS DRAWING ARE PREFIXED WITH THE UNIT IDENTIFICATION NUMBER "51-311"
- 2- NON SLAM TYPE BY COMPRESSOR VENDOR.
- 3- FLOW CONTROL WILL BE BY STEPS 25-50-75-100 BY VALVE UNLOADING (MANUAL VALVE UNLOADING). PERMANENT RECIRCULATION LINE FOR AUTOMATIC CONTROL IS PROVIDED SEE P&ID G-65-D-0-1031-06.
- 4- COMPRESSOR SUCTION STRAINER 100 MESH (DEFINED BY COMPRESSOR VENDOR).
- 5- FOR COMPRESSOR AUXILIARY AND MONITORING DETAILS SEE VENDOR P&ID N°13120-51-D-L-0012-00/0018-00/0021-00/0015-00.
- 6- COMPRESSOR SHALL BE VENTED BEFORE CLOSING INLET MANUAL BLOCK VALVE.
- 7- DELETED.
- 8- DELETED.
- 9- PROVIDE SYMMETRICAL PIPING.
- 10- DELETED.
- 11- CHECK VALVE TO BE LOCATED AT MINIMUM DISTANCE FROM PULSATION DAMPNER OUTLET.
- 12- BEFORE COMPRESSOR START-UP OPERATOR SHALL VERIFY LEVEL GLASS OF THE DRAIN POT. MANUAL GLOBE VALVE SHALL BE IN VIEW OF LEVEL GLASS FOR POT DRAINAGE.
- 13- AUTOMATIC TRAP SUPPLIED WITH STRAINER.

- HOLD INDEX
- 1- DELETED.
- 2- DELETED.
- 3- DELETED.
- 4- DELETED.

EQUIPMENT LIST					
KB001A					

A1	30/08/07	AS BUILT	CAREDDU	D. POLI	D. LA MANTIA	A. MADALONI	T. M. SAID
A3	29/10/04	ISSUED FOR CONSTRUCTION REVISED WHERE INDICATED	CAREDDU	D. POLI	D. LA MANTIA	A. MADALONI	T. M. SAID
A2	16/04/04	ISSUED FOR CONSTRUCTION	CAREDDU	D. POLI	D. LA MANTIA	A. MADALONI	T. M. SAID
A1	23/10/03	ISSUED FOR CONSTRUCTION	CAREDDU	D. POLI	D. LA MANTIA	A. MADALONI	T. M. SAID
P2	30/05/03	ISSUED FOR DESIGN	CAREDDU	D. POLI	D. LA MANTIA	A. MADALONI	T. M. SAID
P1	18/02/03	ISSUED FOR REVIEW	RC	DP	GL		
Revision	DATE	DESCRIPTION	PREP.	CHECK	APPR.	PMS CHECK	AGBV APPR.
AgipGas BV			WESTERN LIBYA GAS PROJECT			AGBV DWG No. G-65-D-0-1031-09	
LIBYAN BRANCH			MELLITAH PLANT CSPLAJ			AGBV JOB No.0177/NC02	
Snamprogetti Lummus Gas Ltd			HYUNDAI			CTR DWG No. 51-311-GD-B-03817	
Snamprogetti			HYUNDAI ENGINEERING & CONSTRUCTION CO., LTD.			CTR JOB NO. 310300	
TITLE: PIPING & INSTRUMENT DIAGRAM UNIT 311 - GAS DEHYDRATION RE-COMPRESSORS "A"						Revision <u>PT</u> <u>PS</u> <u>AT</u> <u>AE</u> <u>AS</u> <u>ET</u>	
						SCALE	
						IN REPLACEMENT OF	
						REPLACED BY	

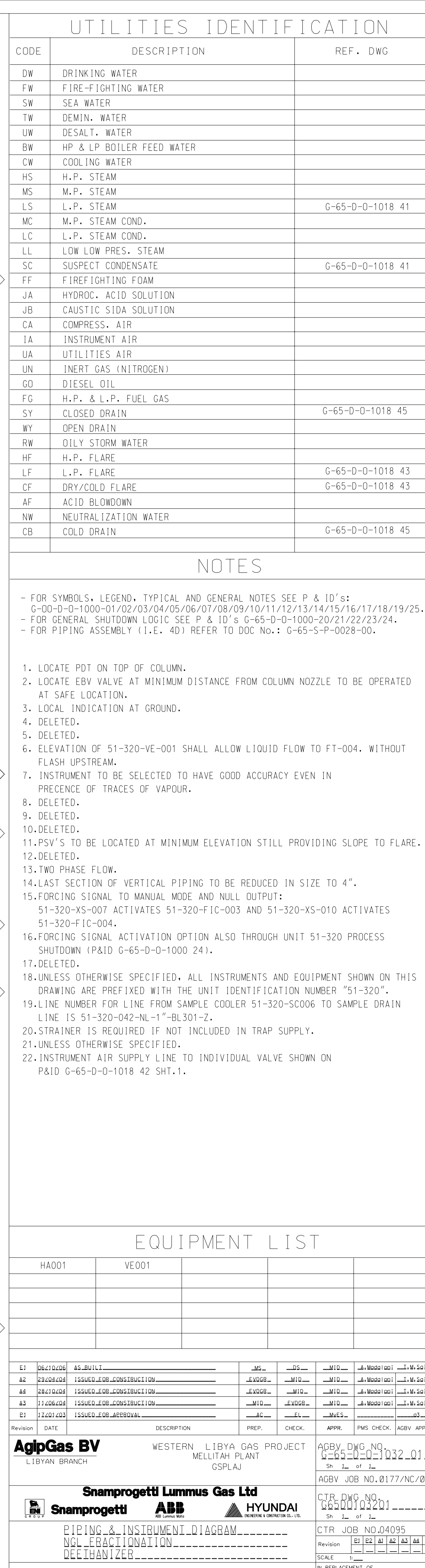


NOTES

- FOR SYMBOLS, LEGEND, TYPICAL AND GENERAL NOTES SEE P&ID'S:
G-00-D-0-1000-01/02/03/04/05/06/07/08/09/10/11/12/13/14/15/16/17/18/19/25.
- FOR GENERAL SHUTDOWN LOGIC SEE P&ID'S: G-65-D-0-1000-20/21/22/23/24.
- FOR PIPING ASSEMBLY (I.e. 4D) REFER TO DDC: G-65-S-P-0028-00

- 1- UNLESS OTHERWISE SPECIFIED, ALL INSTRUMENTS AND EQUIPMENTS SHOWN ON THIS DRAWING ARE PREFIXED WITH THE UNIT IDENTIFICATION NUMBER "51-311"
- 2- NON SLAM TYPE BY COMPRESSOR VENDOR.
- 3- FLOW CONTROL WILL BE BY STEPS 25-50-75-100 BY VALVE UNLOADING
(MANUAL VALVE UNLOADING),
PERMANENT RECIRCULATION LINE FOR AUTOMATIC CONTROL IS PROVIDED SEE P&ID
G-65-D-0-1031-06.
- 4- COMPRESSOR SUCTION STRAINER 100 MESH (DEFINED BY COMPRESSOR VENDOR).
- 5- FOR COMPRESSOR AUXILIARY AND MONITORING DETAILS SEE VENDOR P&ID
N°13120-S1-D-L-0012-00/0018-09/0021-00/0015-00.
- 6- COMPRESSOR SHALL BE VENTED BEFORE CLOSING INLET MANUAL BLOCK VALVE.
- 7- DELETED.
- 8- DELETED.
- 9- PROVIDE SYMMETRICAL PIPING.
- 10- DELETED.
- 11- CHECK VALVE TO BE LOCATED AT MINIMUM DISTANCE FROM PULSATION DAMPNER OUTLET.
- 12- BEFORE COMPRESSOR STAR-UP OPERATOR SHALL VERIFY LEVEL GLASS OF THE DRAIN POT.
MANUAL GLOBE VALVE SHALL BE IN VIEW OF LEVEL GLASS FOR POT DRAINAGE.
- 13- COMMENT OF THIS DRAWING WAS PREVIOUSLY SHOWN IN P&ID G-65-D-0-1031-09-A1.
- 14- AUTOMATIC TRAP SUPPLIED WITH STRANIER.

E1	30/08/07	AS BUILT		CHECKED	D.POL IDORC	M.A. MANTUA A.MADALLON	T.M. SAID
A3	29/10/04	ISSUED FOR CONSTRUCTION REVISED WHERE INDICATED		CAREDDO	D.POL IDORC	M.A. MANTUA A.MADALLON	T.M. SAID
A2	16/04/04	ISSUED FOR CONSTRUCTION		CAREDDO	D.POL IDORC	M.A. MANTUA A.MADALLON	T.M. SAID
Revision	DATE	DESCRIPTION	PREP.	CHECK	APPR.	FMS CHECK	ACCEV APPR.
AgipGas BV LIBYAN BRANCH			WESTERN LIBYA GAS PROJECT MELLITAH PLANT GSP/LAJ			AGWB DWG NO. G-65-D-0-1031-10 AGWB JOB NO.0177NCI02	
 Snamprogetti ANALISI E PROGETTAZIONE			 ABB ABB LOW VOLTAGE			 HYUNDAI ENGINEERING & CONSTRUCTION CO., LTD.	
Snamprogetti Lummus Gas Ltd						CTR DWG NO. 51-311-GD-B-03819 Sh.....of.....	
TITLE: PIPING & INSTRUMENT DIAGRAM UNIT 311 - GAS DEHYDRATION RE-COMPRESSORS "B" (NOTE13)						CTR JOB NO. 310300 Revision AS EXIST	
						SCALE:	
						IN REPLACEMENT OF REPLACED BY:	



UTILITIES IDENTIFICATION

CODE	DESCRIPTION	REF. DWG
DW	DRINKING WATER	
FW	FIRE-FIGHTING WATER	
SW	SEA WATER	
TW	DEMIN. WATER	
UW	DESALT. WATER	
BW	HP & LP BOILER FEED WATER	
CW	COOLING WATER	G-65-D-0-1018 44
HS	H.P. STEAM	
MS	M.P. STEAM	
LS	L.P. STEAM	
MC	M.P. STEAM COND.	
LC	L.P. STEAM COND.	
LL	LOW LOW PRES. STEAM	
SC	SUSPECT CONDENSATE	
FF	FIREFIGHTING FOAM	
JA	HYDROC. ACID SOLUTION	
JB	CAUSTIC SODA SOLUTION	
CA	COMPRESS. AIR	
IA	INSTRUMENT AIR	
UA	UTILITIES AIR	
UN	INERT GAS (NITROGEN)	
GO	DIESEL OIL	
FG	H.P. & L.P. FUEL GAS	
SY	CLOSED DRAIN	G-65-D-0-1018 45
WY	OPEN DRAIN	
RW	OILY STORM WATER	
HF	H.P. FLARE	G-65-D-0-1018 43
LF	L.P. FLARE	G-65-D-0-1018 43
CF	DRY/COLD FLARE	G-65-D-0-1018 43
AF	ACID BLOWDOWN	
NW	NEUTRALIZATION WATER	
CB	COLD DRAIN	G-65-D-0-1018 45

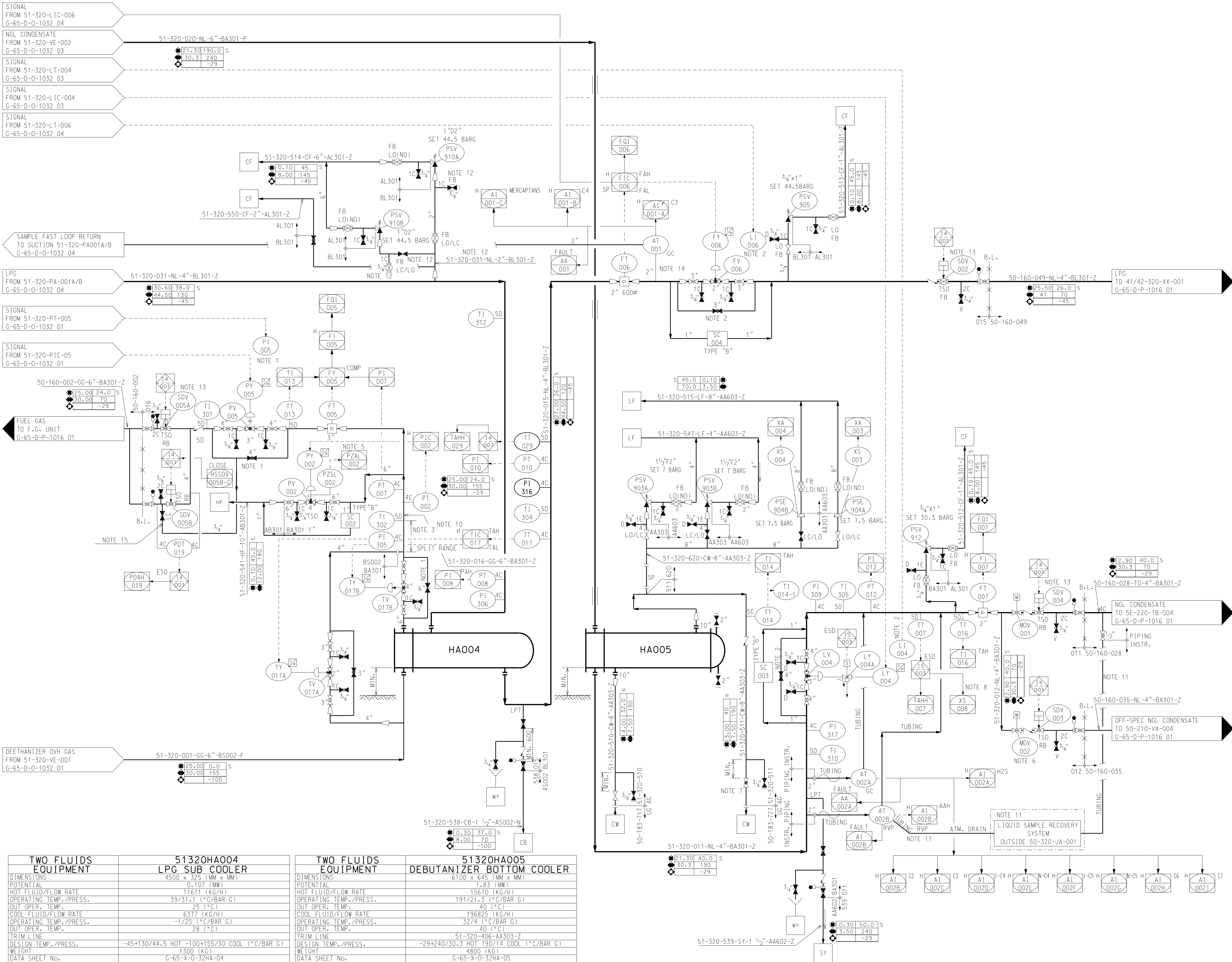
NOTES

- FOR SYMBOLS, LEGEND, TYPICAL AND GENERAL NOTES SEE P & ID'S:
G-00-D-0-1000-01/02/03/04/05/06/07/08/09/10/11/12/13/14/15/16/17/18/19/25.
- FOR GENERAL SHUTDOWN LOGIC SEE P & ID'S G-65-D-0-1000-20/21/22/23/24.
- FOR PIPING ASSEMBLY (I.E. 4D) REFER TO DOC No.: G-65-S-P-0028-00.
1. PI TO BE VISIBLE FROM GLOBE VALVE.
2. LI TO BE VISIBLE FROM GLOBE VALVE.
3. 10 PIPE DIAMETER MINIMUM.
4. DELETED.
5. PROVIDE ALARM WHEN VALVE POSITION IS NOT CLOSED.
6. DELETED.
7. TI-014 TO BE READABLE FROM GLOBE VALVE.
8. FORCING SIGNAL TO MANUAL MODE AND NULL % OUTPUT: 51-320-XS-008 ACTIVATES 51-520-LIC-004.
9. UNLESS OTHERWISE SPECIFIED, ALL INSTRUMENTS AND EQUIPMENT SHOWN ON THIS DRAWING ARE PREFIXED WITH THE UNIT IDENTIFICATION "51-320".
10. SPLIT RANGE CONTROLLER, FOR TEMPERATURE ABOVE SETPOINT TV-017A TO CLOSE AND TV-017B TO OPEN, FOR TEMPERATURE BELOW SET POINT TV-017A TO OPEN AND TV-017B TO CLOSE.
11. LIQUID SAMPLE RECOVERY SYSTEM AND RELEVANT RETURN LINE ARE COMMON FOR BOTH CONDENSATE TRAINS (51-320- & 52-320).
12. METAL SEATED BALL VALVES.
13. INSTRUMENT AIR SUPPLY LINE TO INDIVIDUAL VALVE SHOWN ON P&ID G-65-D-0-1018 42 SHT.1.
14. INSULATION IS REQUIRED FOR ANALYSER SAMPLE LINE.
15. TO BE USED DURING START-UP FOR PRESSURIZATION PURPOSE.

EQUIPMENT LIST

HA004	HA005				

E1	26/10/2024	AS-BUILT	JMS	JDS	WID	WSD	WSD	WSD	WSD	WSD
A1	29/09/2024	ISSUED FOR CONSTRUCTION	EWSD	WSD	WSD	WSD	WSD	WSD	WSD	WSD
A1	02/11/2024	ISSUED FOR CONSTRUCTION	EWSD	WSD	WSD	WSD	WSD	WSD	WSD	WSD
A1	28/10/2024	ISSUED FOR CONSTRUCTION	EWSD	WSD	WSD	WSD	WSD	WSD	WSD	WSD
E1	17/01/2025	ISSUED FOR APPROVAL	EWSD	WSD	WSD	WSD	WSD	WSD	WSD	WSD
Revision	DATE	DESCRIPTION	PREP.	CHECK	APPR.	FWSD	FWSD	FWSD	FWSD	FWSD
AgipGas BV			WESTERN LIBYA GAS PROJECT			AGBY DWG NO. G-65-D-0-1032-02				
LIBYAN BRANCH			MELLITAH PLANT			Sh 1 of 1				
Snamprogetti Lummus Gas Ltd			ABB			AGBY JOB NO. 0177/NC/02				
Snamprogetti			HYUNDAI			G-65-D-0-1032-02				
PIPING & INSTRUMENT DIAGRAM			NGL REACTION			CTR JOB NO. 04095				
PRODUCT COOLING			Revision			Sh 1 of 1				
			SCALE			IN REPLACEMENT OF				
						REPLACED BY				



TWO FLUIDS EQUIPMENT	51320HA004 LPG SUB COOLER
DIMENSIONS	4500 x 325 (MM x MM)
POTENTIAL	0.107 (MW)
HOT FLUID/FLOW RATE	11611 (KG/H)
OPERATING TEMP./PRESS.	39/31.1 (*C/BAR G)
OUT OPER. TEMP.	25 (*C)
COOL FLUID/FLOW RATE	6377 (KG/H)
OPERATING TEMP./PRESS.	-1/25 (*C/BAR G)
OUT OPER. TEMP.	28 (*C)
TRIM LINE	
DESIGN TEMP./PRESS.	-45+130/44.5 HOT -100+155/30 COOL (*C/BAR G)
WEIGHT	1300 (KG)
DATA SHEET No.	G-65-X-0-32HA-04

TWO FLUIDS EQUIPMENT	51320HA005 DEBUTANIZER BOTTOM COOLER
DIMENSIONS	6100 x 645 (MM x MM)
POTENTIAL	1.83 (MW)
HOT FLUID/FLOW RATE	15670 (KG/H)
OPERATING TEMP./PRESS.	191/21.3 (*C/BAR G)
OUT OPER. TEMP.	40 (*C)
COOL FLUID/FLOW RATE	196825 (KG/H)
OPERATING TEMP./PRESS.	32/4 (*C/BAR G)
OUT OPER. TEMP.	40 (*C)
TRIM LINE	51-320-406-AA303-Z
DESIGN TEMP./PRESS.	-29+240/30.3 HOT 190/14 COOL (*C/BAR G)
WEIGHT	4800 (KG)
DATA SHEET No.	G-65-X-0-32HA-05

UTILITIES IDENTIFICATION

CODE	DESCRIPTION	REF. DWG
DW	DRINKING WATER	
FW	FIRE-FIGHTING WATER	
SW	SEA WATER	
TW	DEMIN. WATER	
UW	DESALT. WATER	G-65-D-0-1018 44
BW	HP & LP BOILER FEED WATER	
CW	COOLING WATER	
HS	H.P. STEAM	
MS	M.P. STEAM	G-65-D-0-1018 41
LS	L.P. STEAM	
MC	M.P. STEAM COND.	G-65-D-0-1018 41
LC	L.P. STEAM COND.	
LL	LOW LOW PRES. STEAM	
SC	SUSPECT CONDENSATE	
FF	FIREFIGHTING FOAM	
JA	HYDROC. ACID SOLUTION	
JB	CAUSTIC SODA SOLUTION	
CA	COMPRESS. AIR	
IA	INSTRUMENT AIR	
UA	UTILITIES AIR	
UN	INERT GAS (NITROGEN)	
GO	DIESEL OIL	
FG	H.P. & L.P. FUEL GAS	
SY	CLOSED DRAIN	G-65-D-0-1018 45
WY	OPEN DRAIN	
RW	OILY STORM WATER	
HF	H.P. FLARE	G-65-D-0-1018 45
LF	L.P. FLARE	
CF	DRY/COLD FLARE	
AF	ACID BLOWDOWN	
NW	NEUTRALIZATION WATER	
CB	COLD DRAIN	

NOTES

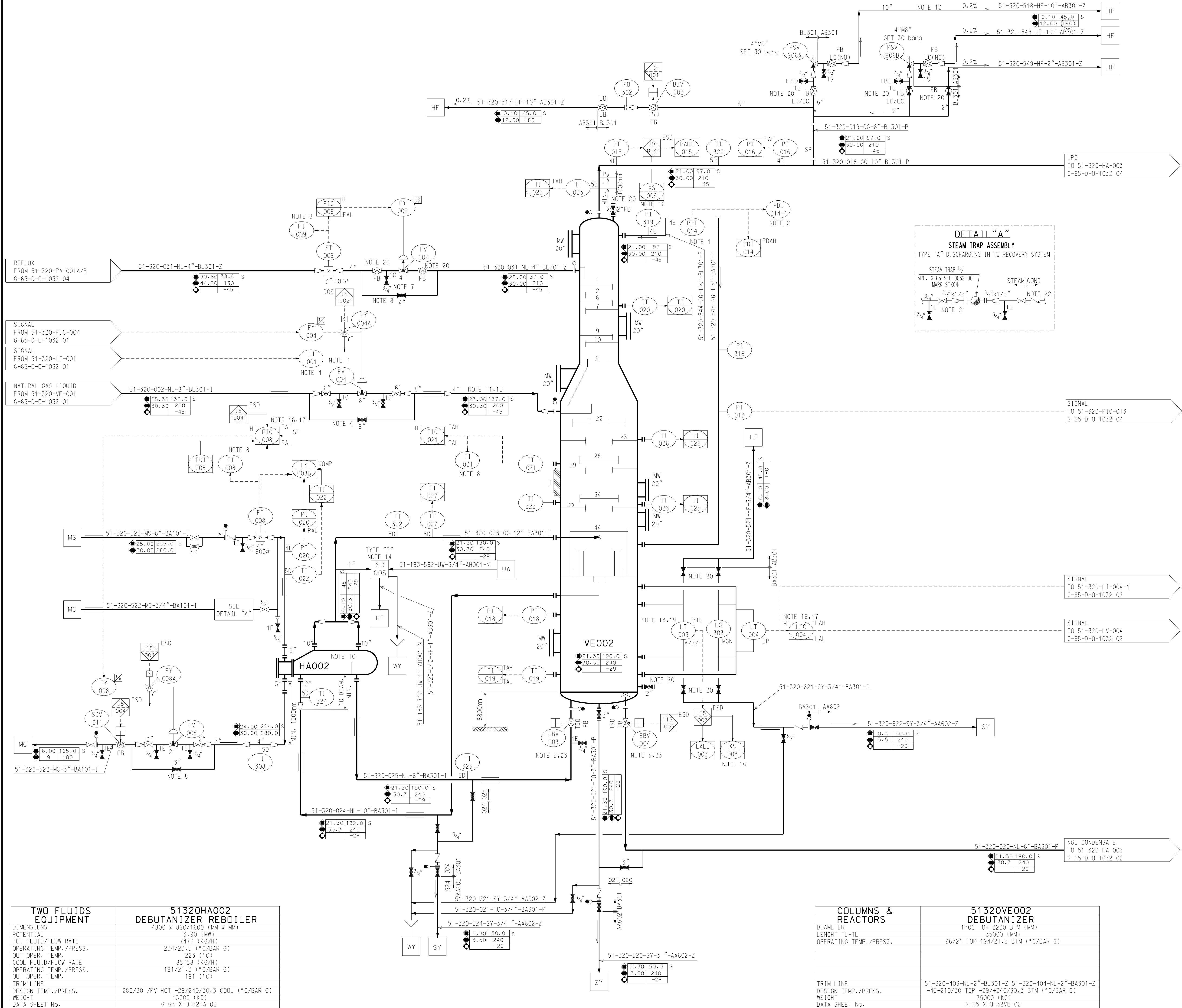
- FOR SYMBOLS, LEGEND, TYPICAL AND GENERAL NOTES SEE P & ID'S:
- G-00-D-0-1000-01/02/03/04/05/06/07/08/09/10/11/12/13/14/15/16/17/18/19/25.
- FOR GENERAL SHUTDOWN LOGIC SEE P & ID'S G-65-D-0-1000-20/21/22/23/24.
- FOR PIPING ASSEMBLY (I.E. 4D) REFER TO DOC No.: G-65-S-P-0028-00.

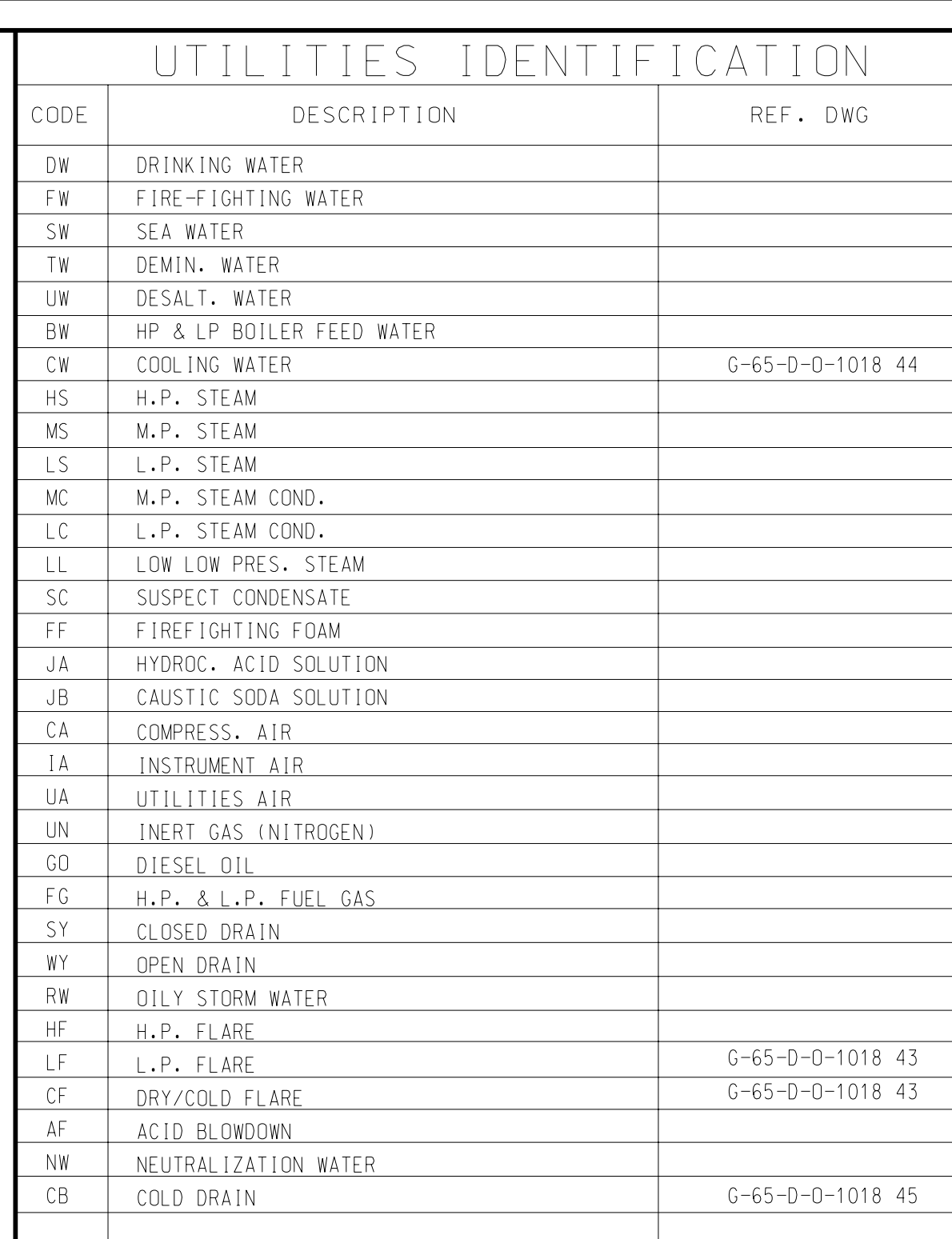
- PDT TO BE INSTALLED AT HIGH POINT.
- PDI TO BE INSTALLED AT GRADE.
- DELETED.
- READABLE FROM GLOBE VALVE.
- EBV LOCATED AT MINIMUM DISTANCE FROM COLUMN NOZZLE AND OPERATED AT SAFE LOCATION.
- DELETED.
- LOCATE FV AT GROUND.
- TI AND FI TO BE READABLE FROM GLOBE VALVE
- DELETED.
- ELEVATION OF REBOILER WEIR TO BE 300MM MIN. ABOVE 51-320-VE-002 HLL.
- TWO PHASE FLOW.
- PSV'S TO BE LOCATED AT MINIMUM ELEVATION STILL PROVIDING SLOPE TO FLARE.
- TWO OUT OF THREE VOTING SYSTEM
- WITH COOLER.
- LAST SECTION OF VERTICAL PIPING TO BE REDUCED IN SIZE TO 4".
- FORCING TO MANUAL MODE AND NULL' OUTPUT:
51-320-XS-008 ACTIVATES 51-320-LIC-004 AND 51-320-XS-009 ACTIVATES 51-320-FOIC-008.
- FORCING SIGNAL ACTIVATION OPTION THROUGH UNIT 51-320 PRESSURE SHUTDOWN (P&ID G-65-D-0-1000 24).
- UNLESS OTHERWISE SPECIFIED, ALL INSTRUMENTS AND EQUIPMENT SHOWN ON THIS DRAWING ARE PREFIXED WITH IDENTIFICATION NUMBER "51-320".
- SEPARATE NOZZLES FOR 51-320-LT-003A/B/C (THREE SETS) WILL BE PROVIDED DIRECTLY ON THE VESSEL.
- METAL SEATED BALL VALVES.
- STRAINER IS REQUIRED IF NOT INCLUDED IN TRAP SUPPLY.
- UNLESS OTHERWISE SPECIFIED.
- INSTRUMENT AIR SUPPLY LINE TO INDIVIDUAL VALVE SHOWN ON P&ID G-65-D-0 1018 42 SHT.1.

EQUIPMENT LIST

HA002	VE002			

E1	06/10/04	AS-BUILT	JMS	DS	WID	A.Was01/001	J.W.Sol01
A2	29/04/04	ISSUED_EGB_CONSTRUCTION	EX008	WID	WID	A.Was01/001	J.W.Sol01
A3	06/11/03	ISSUED_EGB_CONSTRUCTION	EX008	SC04	WID	A.Was01/001	J.W.Sol01
A4	28/10/04	ISSUED_EGB_CONSTRUCTION	EX008	WID	WID	A.Was01/001	J.W.Sol01
E1	17/01/03	ISSUED_EGB_APPROVAL	AC	EL	WES	A.Was01/001	J.W.Sol01
Revision	DATE	DESCRIPTION	PREP.	CHECK.	APPR.	PWS CHECK.	AGBY APPR.
AgipGas BV			WESTERN LIBYA GAS PROJECT			AGBY DWG NO. G-65-D-0-1032-03	
LIBYAN BRANCH			MELLITAH PLANT			Sh 1 of 1	
			GSLAJ			AGBY JOB NO.0177/NC/02	
Snamprogetti Lummus Gas Ltd						CTR DWG NO. G-65-D-0-1032-03	
Snamprogetti			ABB			Sh 1 of 1	
			HYUNDAI			CTR JOB NO.04095	
PIPELINE IDENTIFICATION						Revision E1 E2 A1 A2 A3 A4	
DEBUTANIZER						SCALE 1	
						IN REPLACEMENT OF	
						REPLACED BY	



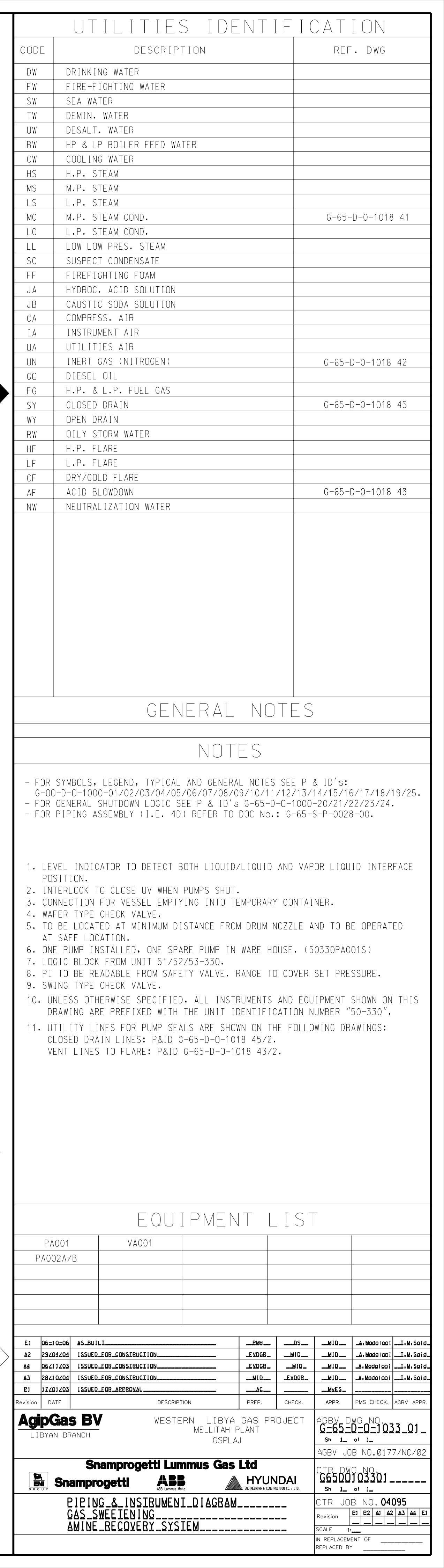


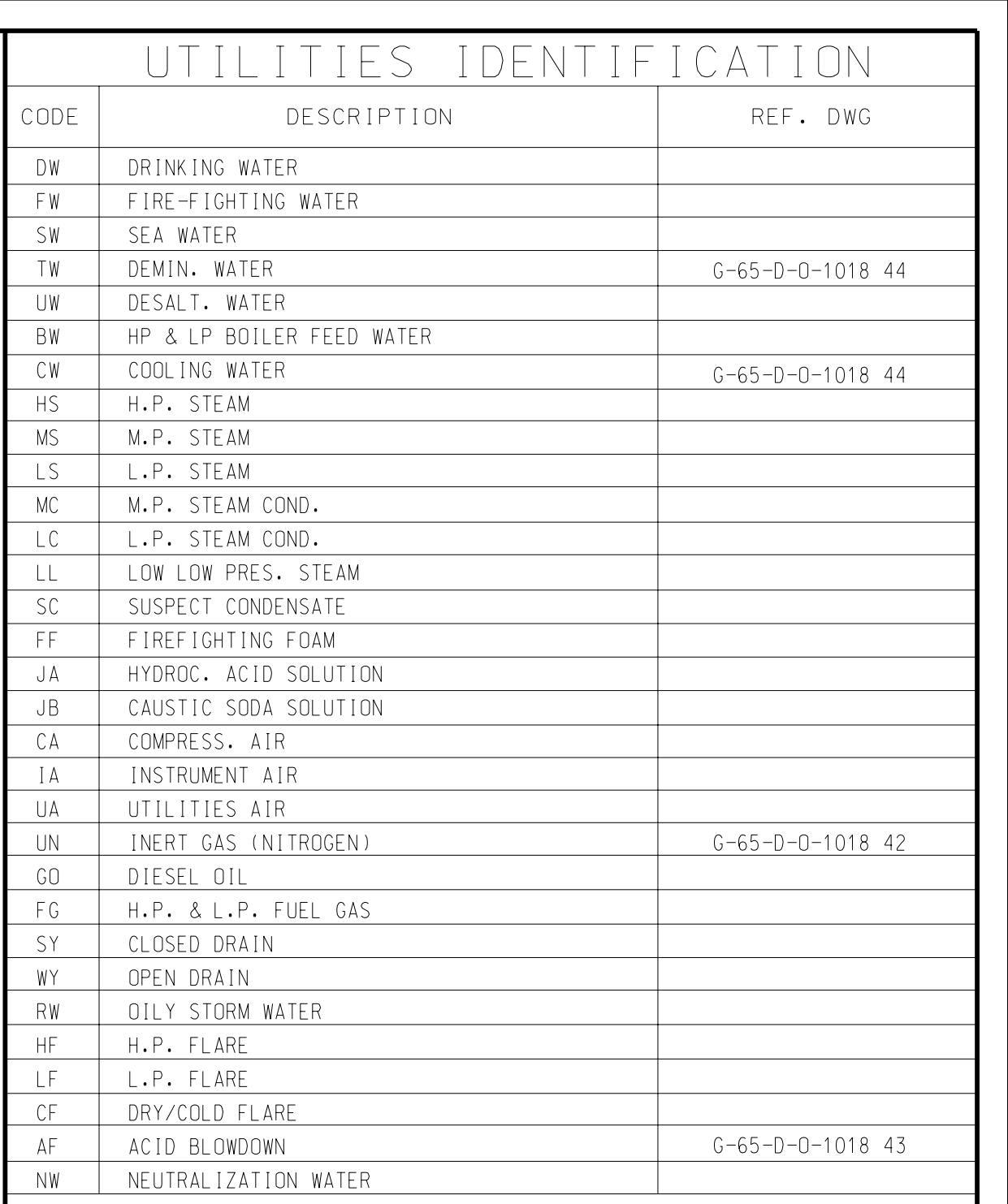
- FOR SYMBOLS, LEGEND, TYPICAL AND GENERAL NOTES SEE P & ID'S:
G-00-D-0-1000-01/02/03/04/05/06/07/08/09/10/11/12/13/14/15/16/17/18/19/25.
- FOR GENERAL SHUTDOWN LOGIC SEE P & ID'S G-65-D-0-1000-20/21/22/23/24.
- FOR PIPING ASSEMBLY (I.E. 4D) REFER TO DOC No.: G-65-S-P-0028-00.

1. DELETED.
2. PI TO BE READABLE FROM GLOBE VALVE.
3. LOCATE AT MINIMUM DISTANCE FROM DRUM NOZZLE
4. ALARM ACTIVATED WHEN VALVE IS NOT CLOSED.
5. CONNECTION ON LIQUID PHASE.
6. TO BE LOCATED AT MIN. DISTANCE FROM 51-320-VA001.
7. AUTOMATIC START OF SPARE PUMP WILL BE PROVIDED IN CASE OF MAIN PUMP FAILURE.
8. MAKE CONNECTION ON TOP OF HORIZONTAL PIPE OR SIDE OF VERTICAL PIPE.
BY PASS SHALL BE MINIMUM DISTANCE FROM REFLEX DRUM NOZZLE.
9. TI TO BE READABLE FROM VALVE.
10. UTILITY LINES FOR PUMPS SEALS ARE SHOWN ON THE FOLLOWING DRAWINGS:
COOLING WATER SUPPLY/RETURN LINES: P&ID G-65-D-0-1018 44 SHT.1
CLOSED DRAIN LINES: P&ID G-65-D-0-1018 45 SHT.1
VENT LINES TO FLARE: P&ID G-65-D-0-1018 43 SHT.1.
11. UNLESS OTHERWISE SPECIFIED, ALL INSTRUMENTS AND EQUIPMENT SHOWN ON THIS
DRAWING ARE PREFIXED WITH THE UNIT IDENTIFICATION NUMBER "51-320".
12. METAL SEATED BALL VALVES.
13. INSTRUMENT AIR SUPPLY LINE TO INDIVIDUAL VALVE SHOWN
ON P&ID G-65-D-0-1018- 42 SHT.1

A1	06/21/06	AS-BUILT		JMS	DS	MID	M-Jobid 001	I-W-50
A2	29/04/04	ISSUED FOR CONSTRUCTION		EVOGR	MID	MID	M-Jobid 001	I-W-50
A3	06/21/03	ISSUED FOR CONSTRUCTION		EVOGR	SCBA	MID	M-Jobid 001	I-W-50
A4	28/10/04	ISSUED FOR CONSTRUCTION		EVOGR	MID	MID	M-Jobid 001	I-W-50
E1	17/02/03	ISSUED FOR APPROVAL		AC	EL	MWES		
Revision	DATE	DESCRIPTION	PREP.	CHECK.	APPR.	PMS CHECK	AWBY APPR.	
Agip Gas BV LIBYAN BRANCH				WESTERN LIBYA GAS PROJECT MELLITAH PLANT GSP/LAJ		AGB DWG NO. G-65-D-0=1Q32_04 Sh. 1 of 1 AGB JOB NO. 0177/NC/0		
Snamprogetti Lummus Gas Ltd						CTR DWG NO. G65D001Q3204 Sh. 1 of 1		
	Snamprogetti	 FOR LUMMUS LTD	 ENGINEERING & CONSTRUCTION CO., LTD.					
PIPING & INSTRUMENT DIAGRAM NGL FRACTIONATION DEBUTANIZER REFLUX SYSTEM				CTR JOB NO. 04095 Revision _____ Scale _____ IN REPLACEMENT OF _____ REPLACED BY _____				

PRESSURE EQUIPMENT	51320VA001
DEBUTANIZER	REFLUX DRUM
DIAMETER	2200 (MM)
LENGTH TL-TL	4400 (MM)
CAPACITY	19.5 (M3)
OPERATING TEMP./PRESS.	38/20.5 (°C/BAR G)
TRIM LINE	51-320-405-GG-BL301-Z
DESIGN TEMP./PRESS.	-45/+210/30 (°C/BAR G)
WEIGHT	13250 (KG)
DATA SHEET NO.	G-65-X-0-32VA-01





- FOR SYMBOLS, LEGEND, TYPICAL AND GENERAL NOTES SEE P & ID's:
G-00-D-O-1000-01/02/03/04/05/06/07/08/09/10/11/12/13/14/15/16/17/18/19/25.
- FOR GENERAL SHUTDOWN LOGIC SEE P & ID's G-65-D-O-1000-20/21/22/23/24.
- FOR PIPING ASSEMBLY (I.E. 4D) REFER TO DOC NO.: G-65-S-P-0028-00.

- EQUIPMENT LIST

HA001A/B	PA003A/B	TA001	VA002	

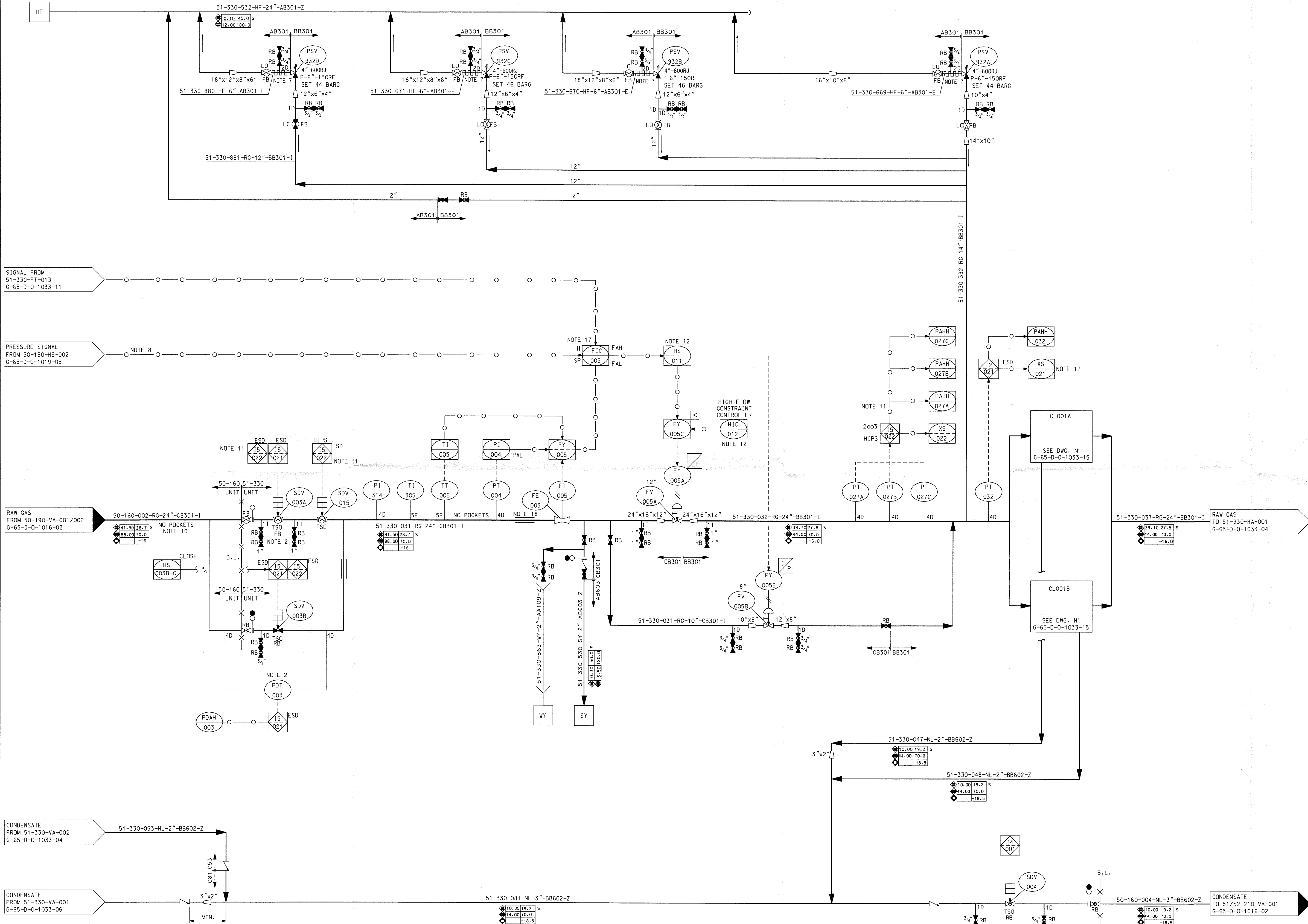
E1	06/23/02	A5-BUILT		-PMH-	D-5	M10	J-A-MO001	I-T-W-SIG1
E2	29/04/02	ISSUED_EOB_CONSTRUCTION		-EXVGR-	M10	M10	J-A-MO001	I-T-W-SIG1
A6	10/27/02	ISSUED_EOB_CONSTRUCTION		-XVGR-	M10	M10	J-A-MO001	I-T-W-SIG1
A8	28/12/02	ISSUED_EOB_CONSTRUCTION		-M10-	EVJ08B	M10	J-A-MO001	I-T-W-SIG1
R1	11/20/03	ISSUED_EOB_APPROVAL		-TST	E1-	MVES-		
Revision	DATE	DESCRIPTION	PREP.	CHECK.	APPRO.	PMS CHECK.	AGBY APPRO.	
AgipGas BV			WESTERN LIBYA GAS PROJECT			AGBY DWG NO.		
LIBYAN BRANCH			MELLITAT PLANT			C-065-D-0-C-1033-02-		
			(CSLAJ)			S _L of I _T		
						AGBY JOB NO. 0177/N/C/02		
 Snampromgetti Lummus Gas Ltd						CTR DWG NO.		
 Snampromgetti ABB			 HYUNDAI ENGINEERING & CONSTRUCTION CO., LTD.			C65D0102302		
						S _L of I _T		
PIPING & INSTRUMENT DIAGRAM						CTR JOB NO. 04095		
GAS SWEETENING						Revision		
AMINE PREP. & STORAGE SYSTEM						SCALE		
						N° REPLACEMENT OF REPLACED BY _____		

TWO FLUIDS EQUIPMENT	50330HA001A/B BFW COOLER
DIMENSIONS	3870 x 222 (MM x MM)
POTENTIAL	0.24 (MW)
HOT FLUID/FLOW RATE	2250 (KG/H)
OPERATING TEMP./PRESS.	139/61 (°C/BAR G)
OUT OPER. TEMP.	48 (°C)
COOL FLUID/FLOW RATE	20640 (KG/H)
OPERATING TEMP./PRESS.	32/5 (°C/BAR G)
OUT OPER. TEMP.	42 (°C)
TRIM LINE	
DESIGN TEMP./PRESS.	165/77 + FV HOT 145/8.5 + COOL (°C/BAR G)
WEIGHT	480 (KG)
DATA SHEET No.	G-65-X-0-33HA-07

PUMPS	50330PA003A/B
-	HP AMINE MAKE-UP PUMPS
DIMENSIONS	2145 x 910 x 690 (MM x MM x MM)
FLOW RATE	80 (M3/H)
FLUID VISC. (DIFF. PRESS.)	3.5 (BAR)
SPECIFIC WEIGHT	1.03 (KG/DM3)
NPSH AVAILABLE/REQ	7 (M)
DISCH. TEMP./PRESS.	50/3.5 (*C/BAR G)
ABSORBED POWER	12.5 (KW)
DESIGN TEMP./PRESS.	75/6 (*C/BAR G)
WEIGHT	673 (KG)
DATA SHEET No.	G-65-X-O-33PA-03

PRESSURE EQUIPMENT	50330VA002 AMINE DRAIN DRUM
DIAMETER	1500 (MM)
LENGTH TL-TL	3000 (MM)
CAPACITY	6-2 (M ³)
OPERATING TEMP./PRESS.	AMB/ATM (*C/BAR G)
TRIM LINE	50-330-403-AM-AA302-Z
DESIGN TEMP./PRESS.	75/3.5 + FULL VACUUM (*C/C/BAR G)
WEIGHT	2600 (KG)
DATA SHEET NO.	G-65-X-0-33VA-11

ATMOSPHERIC TANKS		50330TA001
HP FRESH AMINE STORAGE TANK		
DIAMETER	10500 (MM)	
DIMENSIONS	HEIGHT 12600 (MM)	
CAPACITY	1091 (M3)	
OPERATING TEMP./PRESS	AMB/ATM (°C/BAR G)	
TRIM LINE	50-330-402-AM-AA302-Z	
DESIGN TEMP./PRESS.	75/-50 + 350 (C/MM WC)	
WEIGHT	36000 (KG)	
DATA SHEET No.	G-65-X-O-33TA-02	



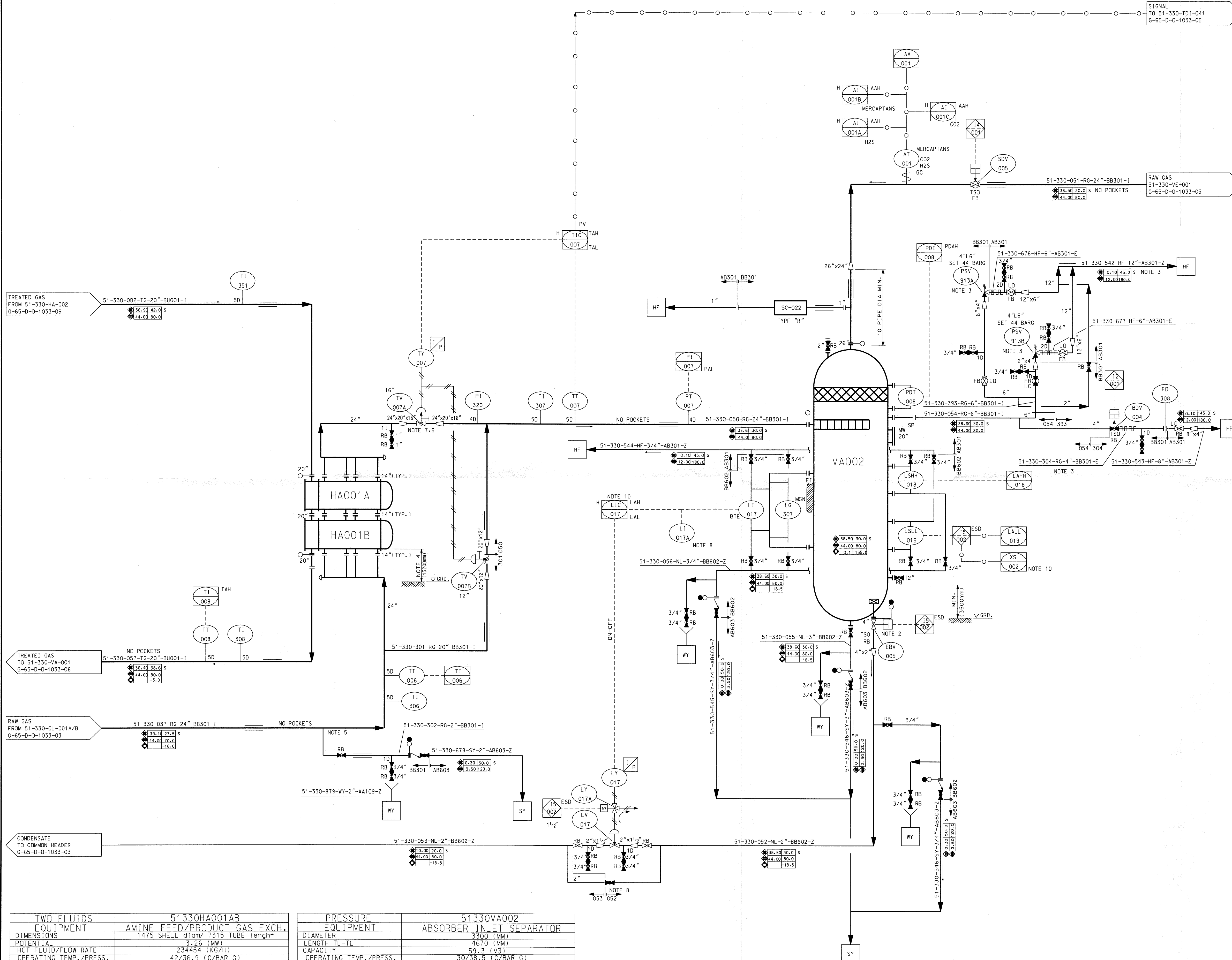
UTILITIES IDENTIFICATION					
CODE	DESCRIPTION	REF. DWG	CODE	DESCRIPTION	REF. DWG
DW	DRINKING WATER		CA	COMPRESS. AIR	
FW	FIRE-FIGHTING WATER		IA	INSTRUMENT AIR	
SW	SEA WATER		UA	UTILITIES AIR	
TW	DEMIN. WATER		UN	INERT GAS (NITROGEN)	G-65-D-0-1018-03
UW	DESALT. WATER		GO	DIESEL OIL	
BW	H.P. & L.P. BOILER FEED WATER		FG	H.P. & L.P. FUEL GAS	
CW	COOLING WATER		SY	CLOSED DRAIN	G-65-D-0-1018-06
HS	H.P. STEAM		WY	OPEN DRAIN	
MS	M.P. STEAM		RW	OILY STORM WATER	
LS	L.P. STEAM		HF	H.P. FLARE	G-65-D-0-1018-04
MC	M.P. STEAM COND.		LF	L.P. FLARE	
LC	L.P. STEAM COND.		CF	DRY/COLD FLARE	
LL	LOW LOW PRES. STEAM		AF	ACID BLOWDOWN	
SC	SUSPECT CONDENSATE		NW	NEUTRALIZATION WATER	
FF	FIREFIGHTING FOAM				
JA	HYDROC. ACID SOLUTION				
JB	CAUSTIC SODA SOLUTION				

- NOTES
- FOR SYMBOLS, LEGEND, TYPICAL AND GENERAL NOTES SEE P&ID'S:
G-00-D-0-1000-01/02/03/04/05/06/07/08/09/10/11/12/13/14/15/16/17/18/19/25.
- FOR GENERAL SHUTDOWN LOGIC SEE P&ID'S: G-65-D-0-1000-20/21/22/23/24.
- FOR PIPING ASSEMBLY (I.E. 4D) REFER TO DOC: G-65-S-P-0028-00
- 1- UNLESS OTHERWISE SPECIFIED, ALL INSTRUMENTS AND EQUIPMENTS SHOWN ON THIS DRAWING ARE PREFIXED WITH THE UNIT IDENTIFICATION NUMBER "51-330".
2- SHUT DOWN VALVE IS ALLOWED TO OPEN ONLY WHEN PRESSURE DROP IS LESS THAN 3 BAR.
3- DELETED.
4- DELETED.
5- DELETED.
6- DELETED.
7- PROVIDE ELECTRIC TRACING ON VALVE BODY AND PART OF THE OUTLET LINE TO FLARE. EXTENT OF TRACING TO BE 460mm FROM PSV-932A/B/C.
8- PRESSURE SIGNAL IS INACTIVE IF THE SELECTOR 50-190-HS-002 (P&ID N° G-65-D-0-1019-05) HAS BEEN POSITIONED TO SEND THE SIGNAL TO UNIT 52-330 OR TO UNIT 53-330.
9- DELETED.
10- REQUIREMENT OF INSULATION ON GAS LINE TO PREVENT EFFECT FROM AMBIENT TEMPERATURE.
11- HIPS SET AT 45 BARG. 2oo3 VOTING SYSTEM. FORESEEN FOR PROTECTION AGAINST FULL OPENING OF FLOW CONTROL VALVE DURING PLANT RESTART AFTER OFFSHORE PIPELINE PRESSURE SETTLING OR PACKING. OTHER OVER-PRESSURE CONTINGENCIES ARE COVERED BY PSV SET AT 44 BARG. HIPS RELIABILITY TO BE SAME LEVEL OF PSV.
12- DURING START UP, BEFORE TO SWITCH FIC-005 FROM FV-005B TO FV-005A AND IN ORDER TO AVOID FV-005A FAST OPENING, IT IS RECOMMENDED TO SET THE HIGH FLOW CONSTRAINT CONTROLLER HIC-012 IN ORDER TO MAINTAIN THE SAME FLOWRATE OF FV-005B THROUGH FV-005A.
13- DELETED.
14- DELETED.
15- DELETED.
16- DELETED.
17- ACTIVATION OF XS-021/022 DRIVER FIC-005 TO MANUAL MODE AND OUTPUT NULL.
18- TO BE LOCATED AT LOWEST POINT OF MAIN PIPE.

HOLD INDEX					
1-	DELETED.				
2-	DELETED.				
3-	DELETED.				
4-	DELETED.				
5-	DELETED.				
6-	DELETED.				

EQUIPMENT LIST					

E1	30/08/07	AS BUILT	CAREDDU	D.POL.DORD	LA MANTIA	A.MADALONI	T.M.SAID
A3	05/11/04	ISSUED FOR CONSTRUCTION REVISED WHERE INDICATED	CAREDDU	D.POL.DORD	LA MANTIA	A.MADALONI	T.M.SAID
A2	14/04/04	ISSUED FOR CONSTRUCTION	CAREDDU	D.POL.DORD	LA MANTIA	A.MADALONI	T.M.SAID
A1	22/10/03	ISSUED FOR CONSTRUCTION	CAREDDU	D.POL.DORD	LA MANTIA	A.MADALONI	T.M.SAID
P2	30/05/03	ISSUED FOR DESIGN	ALVESTAD	D.POL.DORD	LA MANTIA	A.MADALONI	T.M.SAID
P1	18/02/03	ISSUED FOR REVIEW	MD	AL	GL		
Revision	DATE	DESCRIPTION	PREP.	CHECK	APPR.	PMS CHECK	AGBV APPR.
Agip Gas BV LIBYAN BRANCH			WESTERN LIBYA GAS PROJECT MELLITAH PLANT GSP/LAJ			AGBV DWG No. G-65-D-0-1033-03	
Snamprogetti Lummus Gas Ltd Snamprogetti ABB HYUNDAI						AGBV JOB No.0177NC02	
PIPING & INSTRUMENT DIAGRAM UNIT 330 - GAS SWEETENING RAW GAS INLET MANIFOLD						CTR DWG No. 51-330-GD-B-03710 Sh. of	
						CTR JOB No. 310300	
						Revision PREP. APPR. E1	
						SCALE	
						IN REPLACEMENT OF	
						REPLACED BY	



TWO FLUIDS EQUIPMENT		51330HA001AB
AMINE FEED/PRODUCT GAS EXCH.		
DIMENSIONS	1475 SHELL diam/ 7315 TUBE length	
POTENTIAL	3.26 (MW)	
HOT FLUID/FLOW RATE	23454 (KG/H)	
OPERATING TEMP./PRESS.	42/36.9 (C/BAR G)	
OUT OPER. TEMP.	21 (C)	
COOL FLUID/FLOW RATE	332254 (KG/H)	
OPER. TEMP./PRESS.	12/39.1 (C/BAR G)	
OUT OPER. TEMP.	30 (C)	
TRIM LINE		
DESIGN TEMP./PRESS.	-29 +80/44(HOT) -29 +80/44(COOL) (C/BAR G)	
WEIGHT	73020 (KG)	
DATA SHEET No.	G-65-X-0-33HA-01	

PRESSURE EQUIPMENT		51330VA002
ABSORBER INLET SEPARATOR		
DIAMETER	3300 (MM)	
LENGTH TL-TL	4670 (MM)	
CAPACITY	59.3 (M3)	
OPERATING TEMP./PRESS.	30/36.5 (C/BAR G)	
TRIM LINE		
DESIGN TEMP./PRESS	S.O. -29 +80/44 (C/BAR G) 155°C @0.1 barg	
WEIGHT	67540 (KG)	
DATA SHEET No.	G-65-X-0-33VA-10	

UTILITIES IDENTIFICATION					
CODE	DESCRIPTION	REF. DWG	CODE	DESCRIPTION	REF. DWG
DW	DRINKING WATER		CA	COMPRESS. AIR	
FW	FIRE-FIGHTING WATER		IA	INSTRUMENT AIR	
SW	SEA WATER		UA	UTILITIES AIR	
TW	DEMIN. WATER		UN	INERT GAS (NITROGEN)	
UW	DESALT. WATER		GD	DIESEL OIL	
BW	H.P. & L.P. BOILER FEED WATER		FG	H.P. & L.P. FUEL GAS	
CW	COOLING WATER		SY	CLOSED DRAIN	G-65-D-0-1018-06
HS	H.P. STEAM		WY	OPEN DRAIN	
MS	M.P. STEAM		RW	OILY STORM WATER	
LS	L.P. STEAM		HF	H.P. FLARE	G-65-D-0-1018-04
MC	M.P. STEAM COND.		CF	DRY/COLD FLARE	
LC	L.P. STEAM COND.		AF	ACID BLOWDOWN	
LL	LOW LOW PRES. STEAM		NW	NEUTRALIZATION WATER	
SC	SUSPECT CONDENSATE				
FF	FIREFIGHTING FOAM				
JA	HYDROC. ACID SOLUTION				
JB	CAUSTIC SODA SOLUTION				

- NOTES
- FOR SYMBOLS, LEGEND, TYPICAL AND GENERAL NOTES SEE P&ID'S: G-00-D-0-1000-01/02/03/04/05/06/07/08/09/10/11/12/13/14/15/16/17/18/19/25.
 - FOR GENERAL SHUTDOWN LOGIC SEE P&ID'S: G-65-D-0-1000-20/21/22/23/24.
 - FOR PIPING ASSEMBLY (i.e. 4D) REFER TO DOC: G-65-S-P-0028-00
- 1- UNLESS OTHERWISE SPECIFIED, ALL INSTRUMENTS AND EQUIPMENTS SHOWN ON THIS DRAWING ARE PREFIXED WITH THE UNIT IDENTIFICATION NUMBER "51-330".
- 2- VALVE TO BE INSTALLED AT MIN. DISTANCE FROM DRUM NOZZLE.
- 3- PROVIDE ELECTRIC TRACING ON VALVE BODY AND PART OF THE OUTLET LINE TO FLARE. EXTENT OF TRACING TO BE 3000mm DOWNSTREAM PSV-913A/B AND 1500mm DOWNSTREAM BOV-004.
- 4- ELEV. ABOVE 51-330-VA-001 AND 51-330-VA-002 INLET NOZZLE.
- 5- TO BE LOCATED AT LOWEST POINT OF MAIN PIPE.
- 6- DELETED.
- 7- CONTROL VALVE SUITABLE FOR INLET MIXED PHASE.
- 8- LEVEL INDICATOR TO BE READABLE FROM GLOBE VALVE.
- 9- LOCATE CONTROL VALVE AT MINIMUM DISTANCE FROM EXCHANGER OUTLET NOZZLE.
- 10- ACTIVATION OF XS-002 DRIVES LIC-017 TO MANUAL MODE AND OUTPUT NULL.

- HOLD INDEX
- 1- DELETED.
 - 2- DELETED.
 - 3- DELETED.
 - 4- DELETED.

EQUIPMENT LIST			
HA001A/B	VA002		

E1	30/08/07	AS BUILT							
A3	06/11/08	ISSUED FOR CONSTRUCTION	REVISED WHERE INDICATED	CAREDDU	D.POL/DOPO	LA MANTIA	A.MAGDALONI	T.W.SAID	
A2	16/04/09	ISSUED FOR CONSTRUCTION		CAREDDU	D.POL/DOPO	LA MANTIA	A.MAGDALONI	T.W.SAID	
A1	31/10/03	ISSUED FOR CONSTRUCTION		CAREDDU	D.POL/DOPO	LA MANTIA	A.MAGDALONI	T.W.SAID	
P2	30/05/03	ISSUED FOR DESIGN		ALVESTAD	D.POL/DOPO	LA MANTIA	A.MAGDALONI	T.W.SAID	
P1	18/02/03	ISSUED FOR REVIEW		ME	AL	DL			
Revision	DATE	DESCRIPTION		PREP.	CHECK.	APPR.	PWS CHECK.	AGBV APPR.	

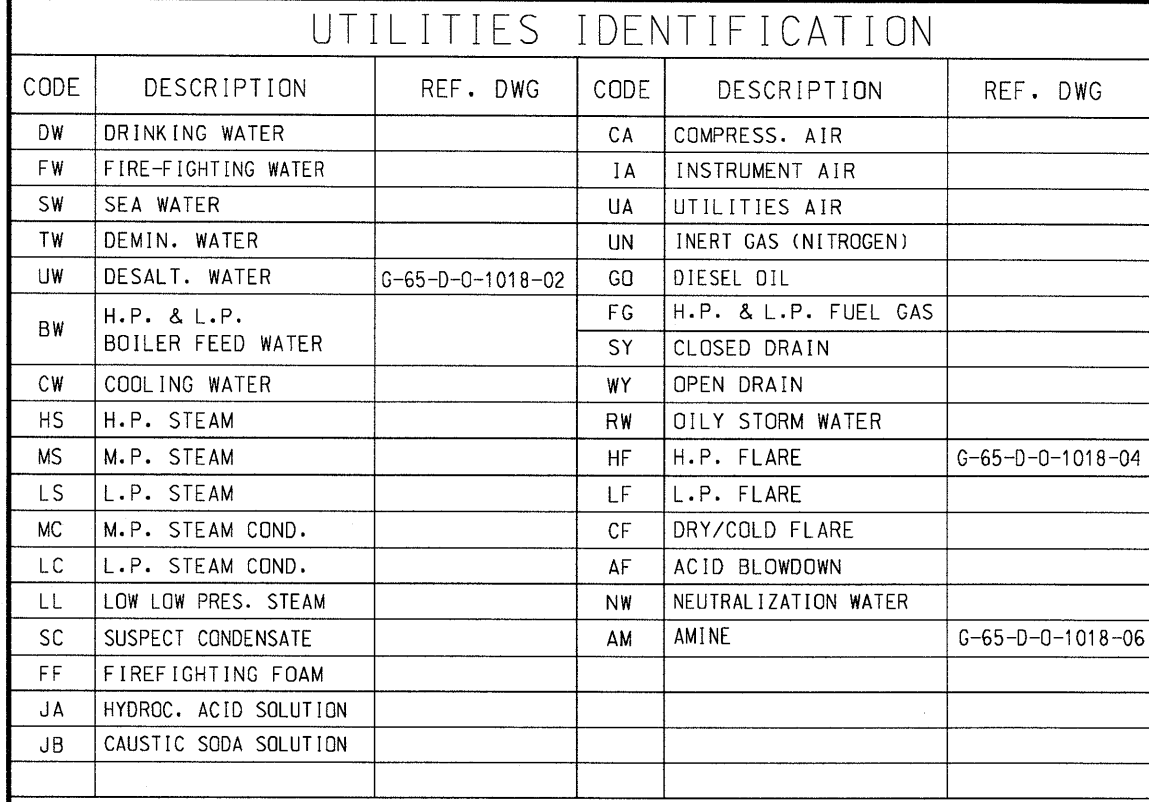
Agip Gas BV
LIBYAN BRANCH

WESTERN LIBYA GAS PROJECT
MELLITAH PLANT
GSPLAJ

Snamprogetti Lummus Gas Ltd
Snamprogetti **ABB** **HYUNDAI**

PIPING & INSTRUMENT DIAGRAM
UNIT 330 - GAS SWEETENING
RAW GAS PREHEATING

AGBV DWG No.
G-65-D-0-1033-04
AGBV JOB No.0177/NC02
CTR DWG No.
51-330-GD-B-03711
CTR JOB NO. **310300**
Revision
SCALE
IN REPLACEMENT OF
REPLACED BY

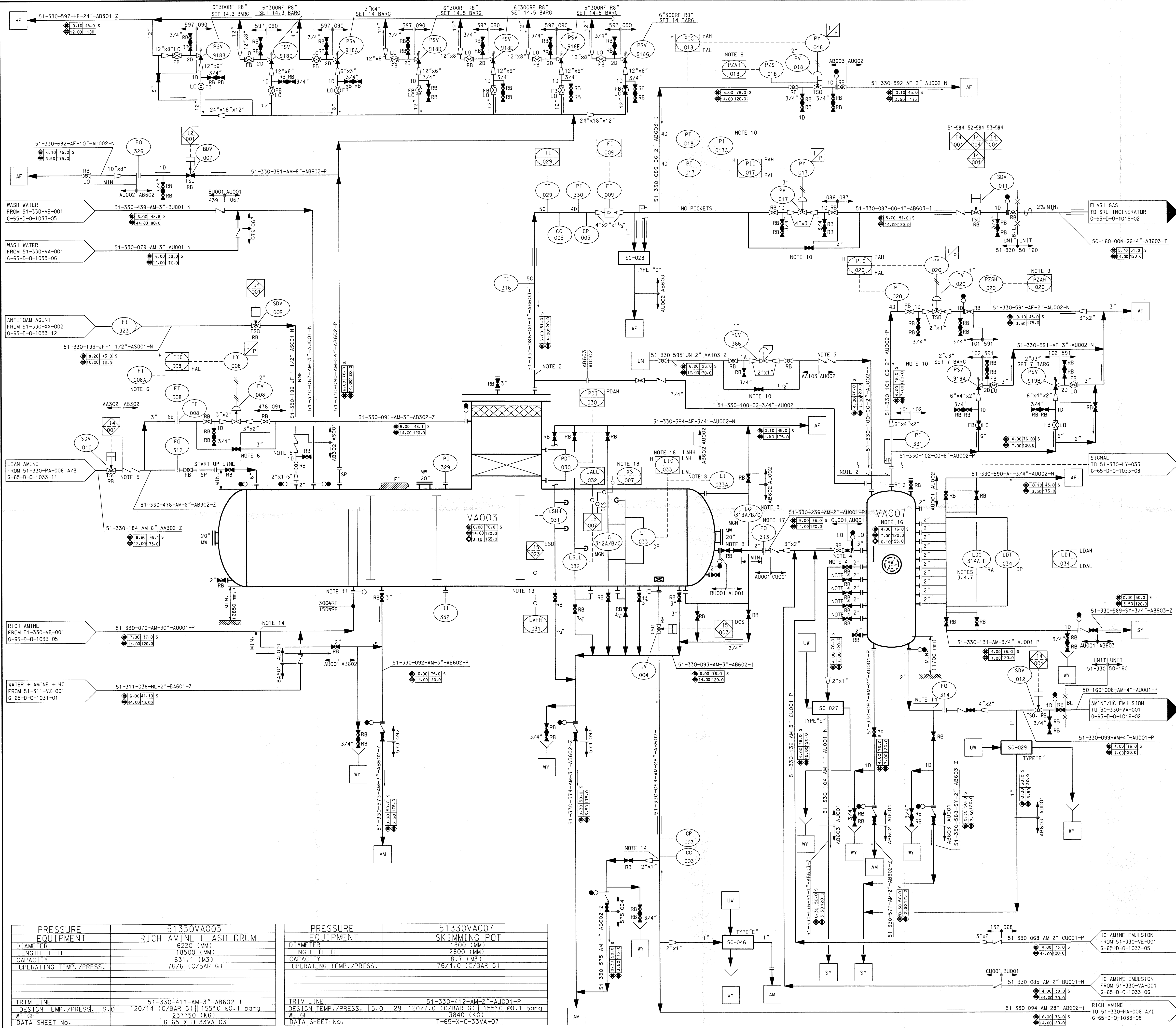


- 1- UNLESS OTHERWISE SPECIFIED, ALL INSTRUMENTS AND EQUIPMENTS SHOWN ON THIS DRAWING ARE PREFIXED WITH THE UNIT IDENTIFICATION NUMBER "51-330".
- 2- LOCAL LI AND MANUAL VALVE TO BE POSITIONED NEAR TO THE SKIMMING POT AND TO BE VISIBLE FROM SKIMMING POT AMINE/WATER LOCAL LEVEL INDICATOR.
- 3- PRESSURE DIFFERENTIAL TRANSMITTER AND PRESSURE DIFFERENTIAL GAUGE TO BE INSTALLED AT HIGH POINT.
- 4- LI-020A TO BE READABLE FROM LV-020 BYPASS. LI-025B TO BE READABLE FROM LV-025 HANDWHEEL.
- 5- HYDROCARBON SKIMMING MANUAL BY OPERATOR.
- 6- LOCAL FI TO BE READABLE FROM CV BY-PASS.
- 7- REINFORCED ANCHORING VALVE AT GRADE.
- 8- PROVIDE ELECTRIC TRACING ON VALVE BODY AND PART OF THE OUTLET LINE TO FLARE. EXTENT OF TRACING TO BE 3000 mm DOWNSTREAM BDV & PSV VALVES.
- 9- TWO CHECK VALVES OF DIFFERENT TYPE.
- 10- HIGH POINT TO PREVENT AMINE SPILL BACK.
- 11- PSV'S TO BE LOCATED AT MINIMUM POSSIBLE ELEVATION STILL PROVIDING SLOPE TO FLARE HEADER.
- 12- 51-330-413-AM-3"-BU001-P TOP
51-330-414-AM-2"-BB001-I BTM.
- 13- DELETED.
- 14- 2 OF 3 VOTING SYSTEM TO CUT OFF THE FEED TO THE RICH AMINE FLASH DRUM.
- 15- DELETED.
- 16- DELETED.
- 17- DUAL PLATE CHECK VALVE.
- 18- ALTERNATIVE DESIGN CONDITION: 155 °C @ 0.1 bar g. (DURING STEAM OUT).
- 19- PROVIDE SYMMETRICAL DISTRIBUTION.
- 20- FOR ALL CARBON STEEL LINE ON RICH AMINE SERVICE THE FOLLOWING RECOMMENDATIONS MUST BE APPLIED:
REINFORCED PIPING SCHEDULE TO BE SELECTED.
LONG RADIUS ELBOWS TO BE PREFERRED.
90° ANGLE RADIUS TO BE AVOIDED.
TAP MINI SIZE : 2"
- 21- TO BE LOCATED AT LOWEST POINT OF MAIN PIPE.
- 22- FLOW RATE SHALL BE LESS THAN 10m/h.²
- 23- ACTIVATION OF XS-003 DRIVES LIC-025 TO MANUAL MODE AND OUTPUT NULL.
- 24- ACTIVATION OF XS-004 DRIVES LIC-020 AND FIC-006 TO MANUAL MODE AND OUTPUT NULL.
- 25- TRIP DELAYED ON PUMP START (0-20 SECONDS).
- 26- CLOSE COMMAND FAILURE ALARM.
- 27- PUMP PROVIDED WITH AUTO START FACILITIES.
- 28- VALVE PROVIDED WITH MECHANICAL STOP AT 120°. OF CALCULATED CV.
- 29- INPUT SIGNAL SELECTOR.

EQUIPMENT LIST				
PA003A/B	VE001			

PUMPS	51330PA003A/B
	TRD GAS WATER WASH PUMPS
DIMENSIONS	1840X910X1010 (MM X MM X MM)
FLOW RATE	114.3 (M ³ /H)
PREVALENCE (DIFF. PRESS.)	1.9 (BAR)
SPECIFIC WEIGHT	0.986 (KG/DM ³)
NPSH AVAILABLE/REQ.	>7 (M)
DISCH. TEMP./PRESS.	49/43 (C/BAR G)
ABSORBED POWER @ RATED FLOW	8.7 (KW) (ESTIMATED)
TRIM LINE	51-330-427-AM-3/4"-CU001-N
DESIGN TEMP./PRESS.	80/50 (C/BAR G)
WEIGHT	867 (KG)
DATA SHEET No.	G-65-X-O-33PA-06

COLUMNS & REACTORS	51330VE001 HP ABSORBER
DIAMETER	5300 (MM)
LENGTH TL-TL	38390 (MM)
OPERATING TEMP./PRESS.	50/37.5 TOP 77/38.5 BTM (C/BAR G)
TRIM LINE	NOTE 12
DESIGN TEMP./PRESS. NOTE 18	-29 +80/44 TOP -22 +120/44 BTM (C/BAR G)
WEIGHT	566400 (KG)
DATA SHEET No.	G-65-X-0-33VE-01



UTILITIES IDENTIFICATION					
CODE	DESCRIPTION	REF. DWG	CODE	DESCRIPTION	REF. DWG
DW	DRINKING WATER		CA	COMPRESS. AIR	
FW	FIRE-FIGHTING WATER		IA	INSTRUMENT AIR	
SW	SEA WATER		UA	UTILITIES AIR	
TW	DEMIN. WATER		UN	INERT GAS (NITROGEN)	G-65-D-0-1018-03
UW	DESALT. WATER	G-65-D-0-1018-02	GO	DIESEL OIL	
BW	H.P. & L.P. BOILER FEED WATER		FG	H.P. & L.P. FUEL GAS	
CW	COOLING WATER		SY	CLOSED DRAIN	G-65-D-0-1018-06
HS	H.P. STEAM		RW	OILY STORM WATER	
MS	M.P. STEAM		HF	H.P. FLARE	G-65-D-0-1018-04
LS	L.P. STEAM		LF	L.P. FLARE	
MC	M.P. STEAM COND.		CF	DRY/COLD FLARE	
LC	L.P. STEAM COND.		AF	ACID BLOWDOWN	G-65-D-0-1018-04
LL	LOW LOW PRES. STEAM		NW	NEUTRALIZATION WATER	
SC	SUSPECT CONDENSATE		AM	AMINE	G-65-D-0-1018-06
FF	FIRE FIGHTING FOAM				
JA	HYDROC. ACID SOLUTION				
JB	CAUSTIC SODA SOLUTION				

NOTES					
- FOR SYMBOLS, LEGEND, TYPICAL AND GENERAL NOTES SEE P&ID'S: G-00-D-0-1000-01/02/03/04/05/06/07/08/09/10/11/12/13/14/15/16/17/18/19/25.					
- FOR GENERAL SHUTDOWN LOGIC SEE P&ID'S: G-65-D-0-1000-20/21/22/23/24.					
- FOR PIPING ASSEMBLY (i.e. 4D) REFER TO DOC: G-65-S-P-0028-00					

1-UNLESS OTHERWISE SPECIFIED, ALL INSTRUMENTS AND EQUIPMENTS SHOWN ON THIS DRAWING ARE PREFIXED WITH THE UNIT IDENTIFICATION NUMBER "51-330".

2- SPOOL PIECE FOR MAINTENANCE.

3- LEVEL GLASS AND MANUAL VALVES TO BE LOCATED NEAR SKIMMING POT AND VISIBLE FROM SKIMMING POT LEVEL INDICATOR.

4- LOCAL LEVEL INDICATION AND MANUAL VALVES TO BE ACCESSIBLE FROM OPERATION POINT.

5- TWO CHECK VALVES IN SERIES OF DIFFERENT TYPE. (ONE SHALL BE DUAL PLATE).

6- FLOW INDICATOR TO BE READABLE FROM GLOBE VALVE.

7- LOG-314 TO BE SELECTED IN ORDER TO MEASURE BOTH VAPOR/LIQUID AND LIQUID/LIQUID INTERFACE POSITION IN THE WHOLE RANGE BETWEEN BTL AND HHL. LOCATION OF LEVEL GAUGES CONNECTION ON SKIMMING POT BY INSTRUMENT DEPARTMENT.

8- LOCAL LI TO BE VISIBLE FROM LV-033 HANDWHEEL (SEE P&ID 1033-08).

9- ALARM TO BE ACTIVATED WHEN VALVE IS NOT CLOSED. (LIMIT SWITCH TO BE LOCATED IN VALVE CLOSE POSITION)

10-LOCAL PJ TO BE VISIBLE FROM CV BY-PASS.

11-300# FLANGE.

12-DELETED.

13-DELETED.

14- TO BE LOCATED AT LOWEST POINT OF MAIN PIPE.

15- DELETED.

16-SKIMMING POT TO BE OPERATED MANUALLY.

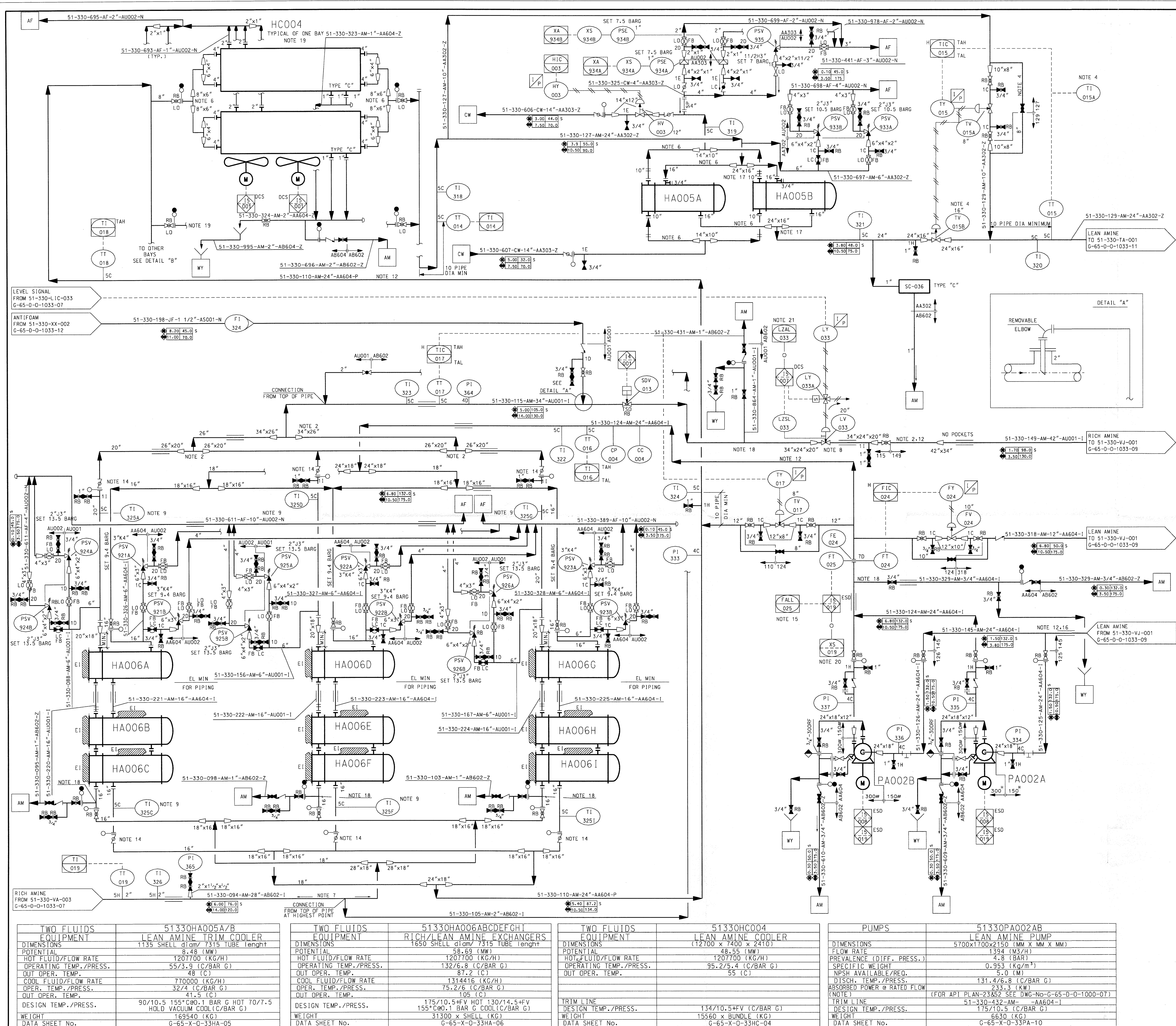
17-FLOW RATE SHALL BE LESS THAN 10m3/h.

18-ACTIVATION OF XS-007 DRIVES LIC-033 TO MANUAL MODE AND OUTPUT NULL.

19-15-023 WILL ACTIVATE XS-003/004/005.

HOLD INDEX					
1-	DELETED.				
2-	DELETED.				
3-	DELETED.				
4-	DELETED.				

EQUIPMENT LIST					
VA003					
VA007					



UTILITIES IDENTIFICATION

CODE	DESCRIPTION	REF. DWG	CODE	DESCRIPTION	REF. DWG
DW	DRINKING WATER		CA	COMPRESS. AIR	
FW	FIRE-FIGHTING WATER		IA	INSTRUMENT AIR	
SW	SEA WATER		UA	UTILITIES AIR	
TW	DEMIN. WATER		UN	INERT GAS (NITROGEN)	
UW	DESALT. WATER		GD	DIESEL OIL	
BW	H.P. & L.P. BOILER FEED WATER		FG	H.P. & L.P. FUEL GAS	
CW	COOLING WATER	G-65-D-0-1018-02	SY	CLOSED DRAIN	
HS	H.P. STEAM		WY	OPEN DRAIN	
MS	M.P. STEAM		RW	OILY STORM WATER	
LS	L.P. STEAM		HF	H.P. FLARE	
MC	M.P. STEAM COND.		LF	L.P. FLARE	
LC	L.P. STEAM COND.		CF	DRY/COLD FLARE	
LL	LOW LOW PRESS. STEAM		AF	ACID BLOWDOWN	G-65-D-0-1018-04
SC	SUSPECT CONDENSATE		NW	NEUTRALIZATION WATER	
FF	FIRE FIGHTING FOAM		AM	AMINE	G-65-D-0-1018-06
JA	HYDRO. ACID SOLUTION				
UB	CAUSTIC SODA SOLUTION				

NOTES

- FOR SYMBOLS, LEGEND, TYPICAL AND GENERAL NOTES SEE PAID'S:
G-00-D-0-1000-01/02/03/04/05/06/07/08/09/10/11/12/13/14/15/16/17/18/19/25.
- FOR GENERAL SHUTDOWN LOGIC SEE PAID'S: G-65-D-0-1000-20/21/22/23/24.
- FOR PIPING ASSEMBLY (I.E. 4D) REFER TO DOC: G-65-S-P-0028-00

1- UNLESS OTHERWISE SPECIFIED, ALL INSTRUMENTS AND EQUIPMENTS SHOWN ON THIS DRAWING ARE PREFIXED WITH THE UNIT IDENTIFICATION NUMBER "51-330".

2- TWO PHASE FLOW

3- DELETED.

4- LOCAL TEMPERATURE INSTRUMENT TO BE VISIBLE FROM TV-015B HANDWHEEL AND FROM TV-015A BYPASS

5- DELETED.

6- PROVIDE SYMMETRICAL PIPING ARRANGEMENT.

7- FOR ALL CARBON STEEL LINE ON RICH AMINE SERVICE THE FOLLOWING RECOMMENDATIONS MUST BE APPLIED:
REINFORCED PIPING SCHEDULE TO BE SELECTED.
LONG RADIUS ELBOWS TO BE PREFERRED.
90° ANGLE RADIUS TO BE AVOIDED.
TAP MINI SIZE : 2".

8- LI 033A (SEE PAID 1033-07) TO BE VISIBLE FROM LV-033 HANDWHEEL. VALVE TO BE POSITIONED AT GROUND WITH STRONG ANCHORING FOR TWO PHASE FLOW.
RISER PIPE TO REGENERATOR INLET NOZZLE WILL START JUST AFTER VALVE.

9- LOCAL TEMPERATURE INSTRUMENT TO BE INSTALLED VISIBLE FROM BUTTERFLY VALVE

10- DELETED.

11- DELETED.

12- CONSIDER FULL VACUUM CONDITIONS FOR THE LINE

13- DELETED.

14- BUTTERFLY VALVE FOR FLOW DISTRIBUTION ADJUSTMENT.

15- TRIP TO BE DELAYED ON PUMP START (0-20:sec. ADJUSTABLE)

16- AVOID HIGH POINTS (VAPOR POCKETS) - LEAN AMINE LINE FROM 51-330-VJ-001 TO BE RUN BELOW 51-330-VJ-001 BOTTOM TANGENT LINE.

17- DELETED.

18- TO BE LOCATED AT LOWEST POINT OF MAIN PIPE.

19- FOR AIR COOLER LAYOUT SEE DETAIL "B" NUMBER OF BAYS IS 8.

20- ACTIVATION OF XS-019 DRIVES LIC-035/PAID 1033-09 TO MANUAL MODE AND OUTPUT NULL.

21- CLOSE COMMAND FAILURE ALARM.

DETAIL "A"

DETAIL "B"

EQUIPMENT LIST

EQUIPMENT	HA005A/B	HA006A-I	HA006A/B	HA006C	HA006D	HA006E	HA006F	HA006G	HA006H	HA006I
HA005A/B	HC004		PA002A/B							
HA006A-I										

REVISION	DATE	DESCRIPTION	PREP.	CHECK.	APPR.	PM'S CHECK.	AGB APPROV.
E1	30/08/07	AS BUILT					
A3	25/11/04	ISSUED FOR CONSTRUCTION REVISED WHERE INDICATED	CAREDDU	D.POL/DORO	LA MANTIA	A.MADDAONI	T.M.SAID
A2	16/04/04	ISSUED FOR CONSTRUCTION	CAREDDU	D.POL/DORO	LA MANTIA	A.MADDAONI	T.M.SAID
A1	31/01/03	ISSUED FOR CONSTRUCTION	CAREDDU	D.POL/DORO	LA MANTIA	A.MADDAONI	T.M.SAID
P2	30/05/03	ISSUED FOR DESIGN	ALVESTAD	D.POL/DORO	LA MANTIA	A.MADDAONI	T.M.SAID
P1	18/02/03	ISSUED FOR REVIEW	MB	AL	GL		

AgipGas BV
LIBYAN BRANCH

WESTERN LIBYA GAS PROJECT
MELLITAH PLANT
CSPLAJ

AGB DWG No.
G-65-D-0-1033-08

AGB JOB No.0177/NO2

Snamprogetti Lummus Gas Ltd
Snamprogetti
HYUNDAI

CTB DWG No.
51-330-GD-B-03715

CTB JOB No.
310300

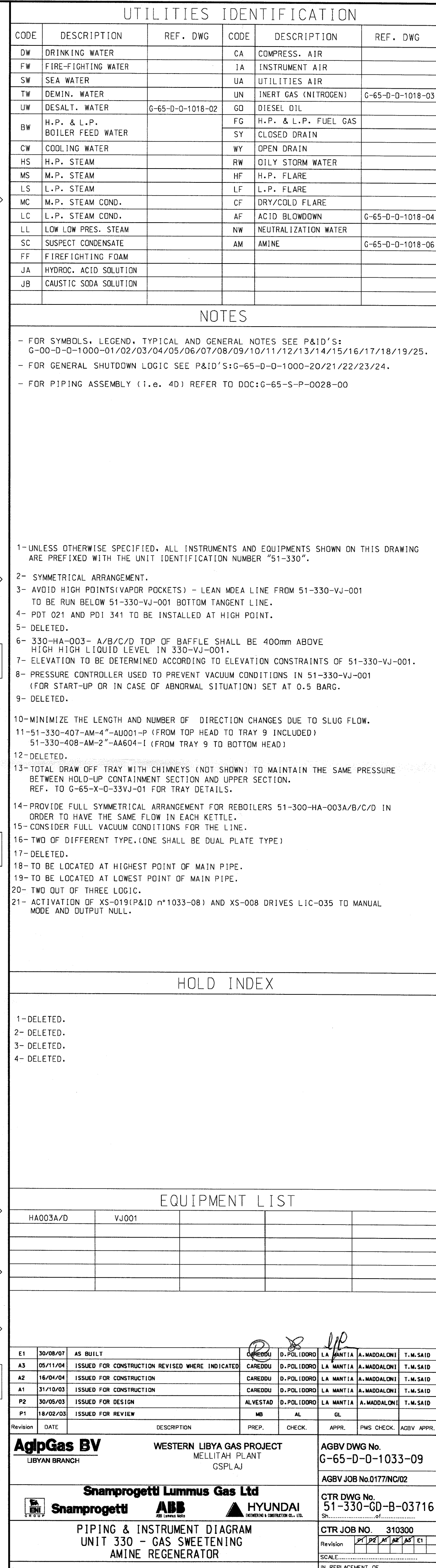
PIPING & INSTRUMENT DIAGRAM
UNIT 330 - GAS SWEETENING
RICH LEAN AMINE EXCHANGERS

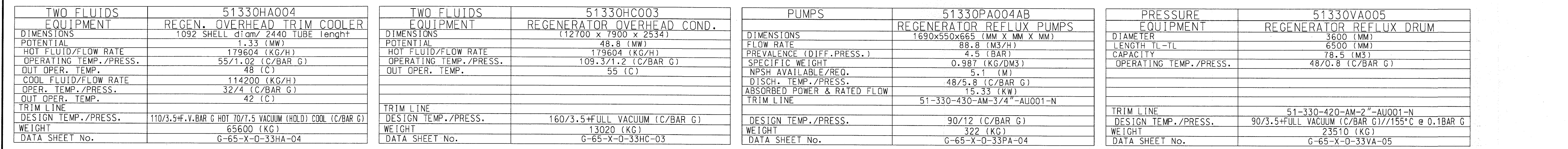
SCALE
1:1

BY REPLACEMENT OF
REPLACED BY

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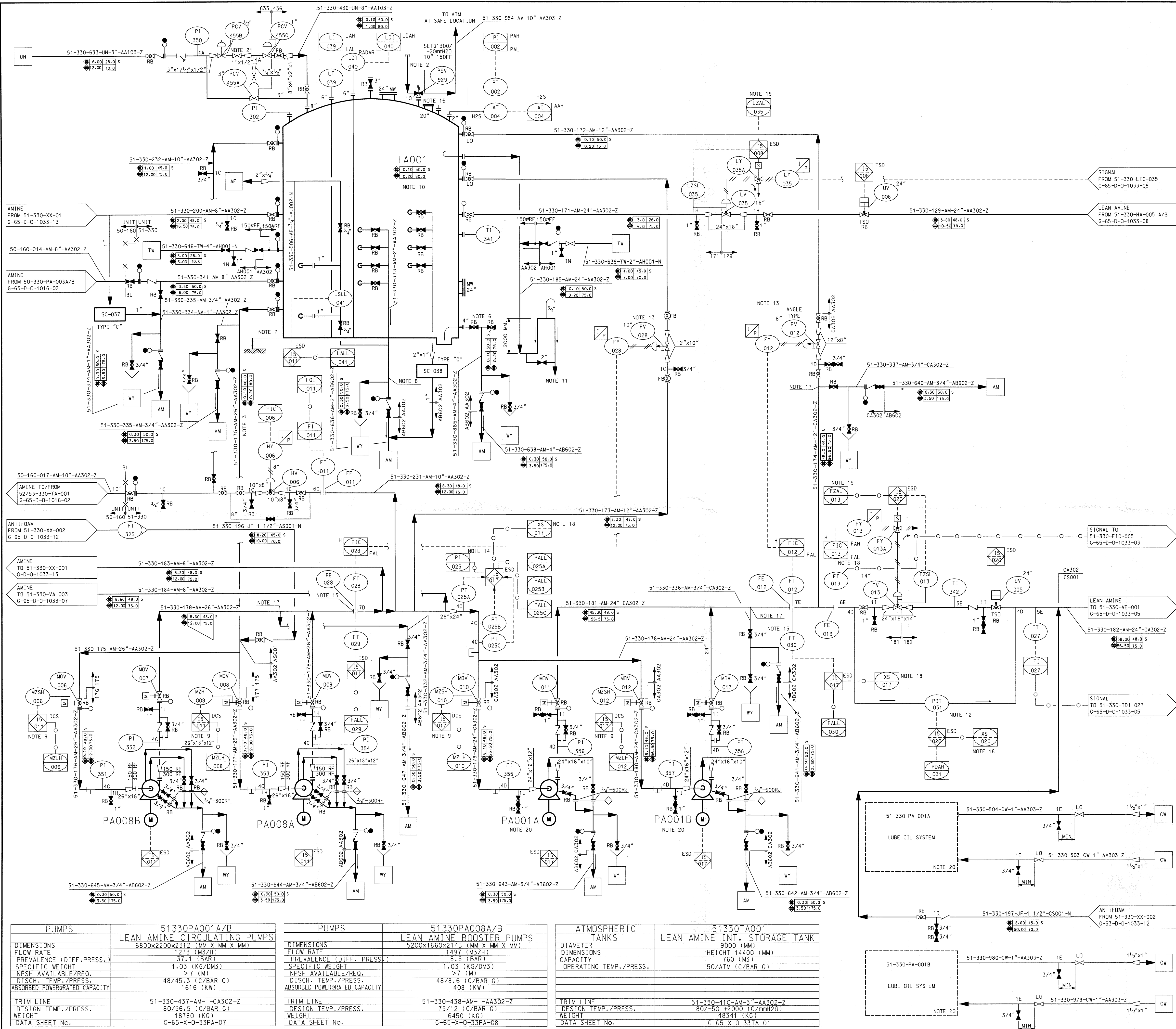
G-65-D-0-1033-08-E1



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1-	DELETED.
2-	DELETED.
3-	DELETED.
4-	DELETED.

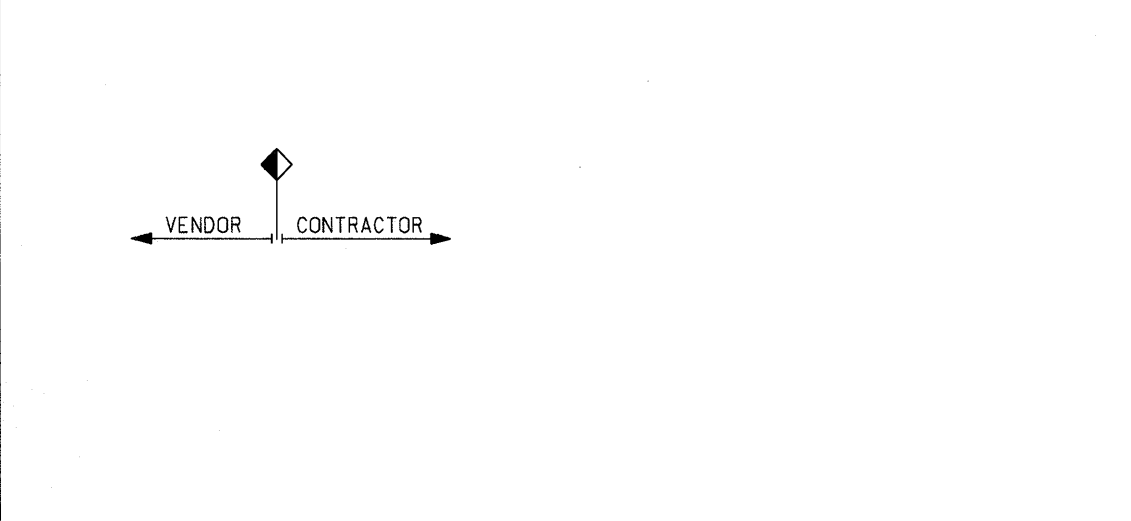
EQUIPMENT LIST								
HA004		HCO03	PA004A/B	VAO05				
E1	30/08/07	AS BUILT		(P) EDRU	D. POL. (DOPO)	L.A. MANTIA	A.-MAGDALONI	T.M.SAID
A3	26/10/04	ISSUED FOR CONSTRUCTION REVISED WHERE INDICATED		CAREDDU	D. POL. (DOPO)	L.A. MANTIA	A.-MAGDALONI	T.M.SAID
A2	16/04/04	ISSUED FOR CONSTRUCTION		CAREDDU	D. POL. (DOPO)	L.A. MANTIA	A.-MAGDALONI	T.M.SAID
A1	31/10/03	ISSUED FOR CONSTRUCTION		CAREDDU	D. POL. (DOPO)	L.A. MANTIA	A.-MAGDALONI	T.M.SAID
P3	30/05/03	ISSUED FOR DESIGN		AL VESTAD	D. POL. (DOPO)	L.A. MANTIA	A.-MAGDALONI	T.M.SAID
P1	18/02/03	ISSUED FOR REVIEW		MO	AL	GL		
Revision	DATE	DESCRIPTION		PREP.	CHECK.	APPR.	PMS CHECK.	AGBV APPR.
AgipGas BV LIBYAN BRANCH				WESTERN LIBYA GAS PROJECT MELLITAH PLANT GSP/LAJ			AGBV DWG No. G-65-D-0-1033-10 AGBV JOB No. 0177/MC02	
Snamprogetti Lummus Gas Ltd ABB ABB LOWPRESS DIVISION				HYUNDAI ENGINEERING & CONSTRUCTION CO., LTD.				
TITLE: PIPING & INSTRUMENT DIAGRAM UNIT 330 - GAS SWEETENING AMINE REGENERATOR REFLEX SECTION				CTR DWG No. 51-330-GD-B-03717 Sh..... of Revision PT PR AS AT ET SCALE IN REPLACEMENT OF				



UTILITIES IDENTIFICATION					
CODE	DESCRIPTION	REF. DWG	CODE	DESCRIPTION	REF. DWG
DW	DRINKING WATER		CA	COMPRESS. AIR	
FW	FIRE-FIGHTING WATER		IA	INSTRUMENT AIR	
SW	SEA WATER		UA	UTILITIES AIR	
TW	DEMIN. WATER	G-65-D-0-1018-02	UN	INERT GAS (NITROGEN)	G-65-D-0-1018-03
UW	DESALT. WATER		GO	DIESEL OIL	
BW	H.P. & L.P. BOILER FEED WATER		FG	H.P. & L.P. FUEL GAS	
CW	COOLING WATER	G-65-D-0-1018-02	SY	CLOSED DRAIN	
HS	H.P. STEAM		WY	OPEN DRAIN	
MS	M.P. STEAM		RW	OILY STORM WATER	
LS	L.P. STEAM		HF	H.P. FLARE	
MC	M.P. STEAM COND.		CF	DRY/COLD FLARE	
LC	L.P. STEAM COND.		AF	ACTO BLOWDOWN	G-65-D-0-1018-04
LL	LOW LOW PRES. STEAM		NW	NEUTRALIZATION WATER	
SC	SUSPECT CONDENSATE		AM	AMINE	G-65-D-0-1018-06
FF	FIRE FIGHTING FOAM				
JA	HYDROC. ACID SOLUTION				
JB	CAUSTIC SODA SOLUTION				

- NOTES
- FOR SYMBOLS, LEGEND, TYPICAL AND GENERAL NOTES SEE P&ID'S: G-00-D-0-1000-01/02/03/04/05/06/07/08/09/10/11/12/13/14/15/16/17/18/19/25.
 - FOR GENERAL SHUTDOWN LOGIC SEE P&ID'S: G-65-D-0-1000-20/21/22/23/24.
 - FOR PIPING ASSEMBLY (i.e. 4D) REFER TO DOC: G-65-S-P-0028-00
- 1-UNLESS OTHERWISE SPECIFIED, ALL INSTRUMENTS AND EQUIPMENTS SHOWN ON THIS DRAWING ARE PREFIXED WITH THE UNIT IDENTIFICATION NUMBER "51-330".
- 2- TANK BREATHER VALVE SPECIFIED AND SUPPLIED BY TANK MANUFACTURER.
- 3- LINE FROM 51-330-TA-001 TO 51-330-PA 008 A/B TO BE ROUTED TO MINIMIZE PRESSURE DROP.
- 4- DELETED
- 5- DELETED
- 6- PUMP OUT CONNECTION.
- 7- MINIMUM ELEVATION ABOVE GROUND.
- 8- HYDROCARBON SKIMMING BY OPERATOR.
- 9- PUMP RESET/START TO BE INHIBITED AND LOCAL "REMOTE" SELECTOR TO BE LOCKED IN "REMOTE" POSITION WHEN MOV IS NOT FULLY OPEN.
- 10- TANK PROVIDED WITH EXTERNAL BOUNDARY (30 cm)
- 11- PIPE TO BE LEFT AT GRADE INSIDE BUNDED AREA.
- 12- UV-005 TO TRIP CLOSED WHEN PDT-031 DETECTS HIGHER PRESSURE DOWNSTREAM OF FV-013A.
- UV-005 OPEN ONLY WHEN PRESSURE UPSTREAM OF FV-013A IS HIGHER THAN DOWNSTREAM.
- 13- SELECTED CV FOR CONTROL VALVE SHALL BE CLOSER TO CALCULATED CV.
- 14- 2 OUT OF 3 LOGIC.
- 15- PUMP TRIP TO BE DELAYED ON START (0-20 SEC ADJUSTABLE.)
- 16- EXPLOSION HATCH.
- 17- TO BE LOCATED AT LOWEST POINT OF MAIN PIPE.
- 18- ACTIVATION OF XS-017/020 DRIVES FIC-013 TO MANUAL MODE AND OUTPUT NULL.
- 19- CLOSE COMMAND FAILURE ALARM.
- 20- FOR DETAIL OF LUBE OIL & SEALING SYSTEM FOR PUMP 51-330-PA-001A/B REFER TO DWG. 13349-51-DL-0621-00
- 21- VALVE (SIZE 1/2" NPT) SUPPLIED BY INSTRUMENT DEPT.

- HOLD INDEX
- 1- DELETED.
 - 2- DELETED.
 - 3- DELETED.



EQUIPMENT LIST					
PA001A/B	TA001				
PA008A/B					

E1	30/08/07	AS BUILT	CANEDDU	D-POL/DOPO	LA MANTIA	A-MADALONI	T.M.SAID
A3	05/11/04	ISSUED FOR CONSTRUCTION REVISED WHERE INDICATED	CANEDDU	D-POL/DOPO	LA MANTIA	A-MADALONI	T.M.SAID
A2	16/04/04	ISSUED FOR CONSTRUCTION	CANEDDU	D-POL/DOPO	LA MANTIA	A-MADALONI	T.M.SAID
A1	31/03/03	ISSUED FOR CONSTRUCTION	CANEDDU	D-POL/DOPO	LA MANTIA	A-MADALONI	T.M.SAID
P2	30/05/03	ISSUED FOR DESIGN	ALVESTAD	D-POL/DOPO	LA MANTIA	A-MADALONI	T.M.SAID
P1	18/02/03	ISSUED FOR REVIEW	MB	AL	OL		

Revision DATE DESCRIPTION PREP. CHECK. APPR. PMS CHECK. AGBY APPR.

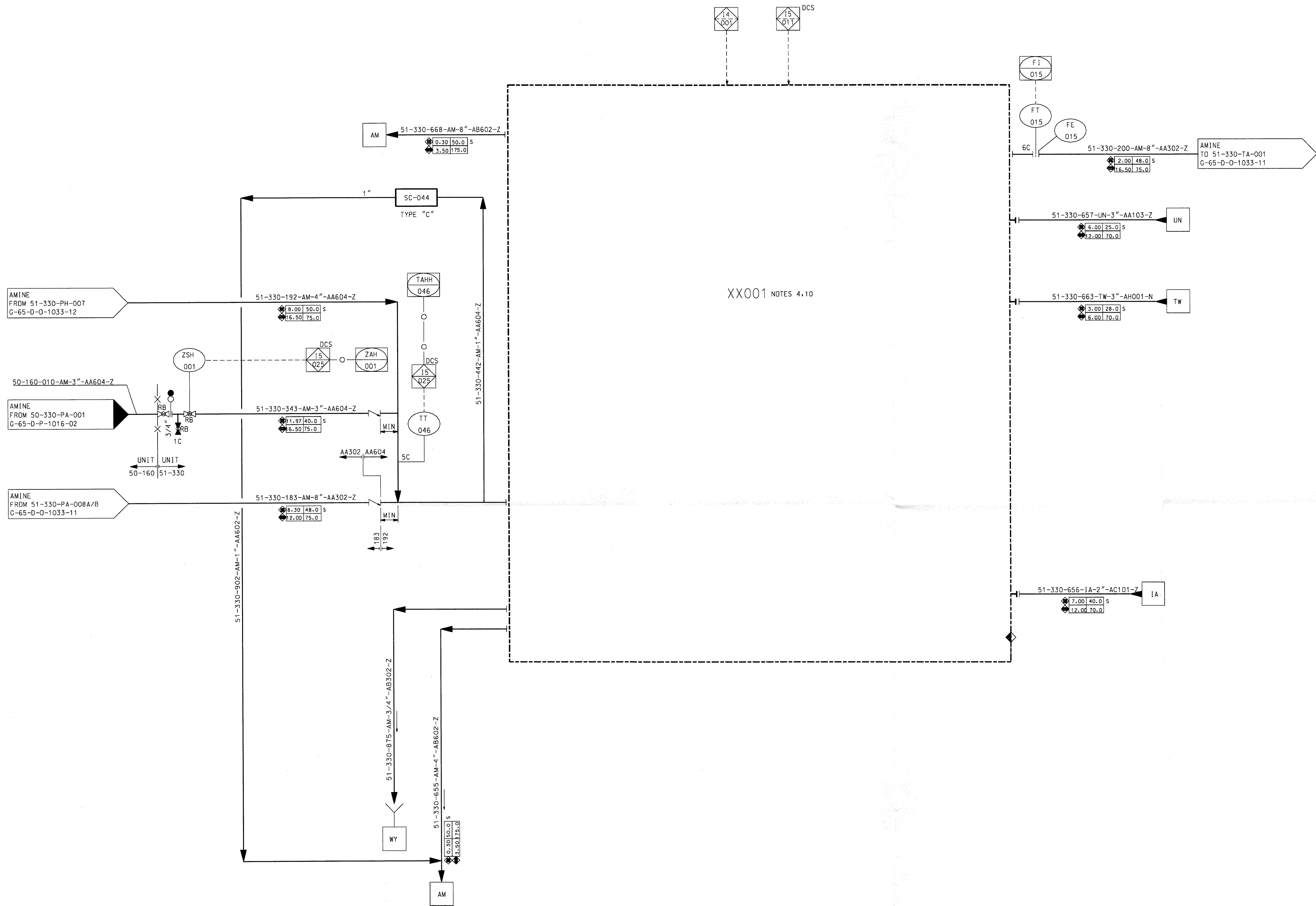
AgpGas BV LIBYAN BRANCH

WESTERN LIBYA GAS PROJECT MELIUTAH PLANT (GSPJAJ)

Snamprogetti Lummus Gas Ltd **ABB** **HYUNDAI**

PIPING & INSTRUMENT DIAGRAM
UNIT 330 - GAS SWEETENING
AMINE PUMP & STORAGE

AGBY DWG No. **G-65-D-0-1033-11**
AGBY JOB No. **0177N002**
CTR DWG No. **51-330-GD-B-03718**
CTR JOB NO. **310300**
Revision **1** **2** **3** **4** **5** **6** **7** **8** **9** **10**
SCALE _____
N. REPLACEMENT OF _____
REPLACED BY _____
G-65-D-0-1033-11-E1



XX001 NOTES 4.10

UTILITIES IDENTIFICATION					
CODE	DESCRIPTION	REF. DWG	CODE	DESCRIPTION	REF. DWG
DW	DRINKING WATER		CA	COMPRESS. AIR	
FW	FIRE-FIGHTING WATER		IA	INSTRUMENT AIR	G-65-D-0-1018-03
SW	SEA WATER		UA	UTILITIES AIR	G-65-D-0-1018-03
TW	DEMIN. WATER	G-65-D-0-1018-02	UN	INERT GAS (NITROGEN)	G-65-D-0-1018-03
UW	DESALT. WATER		GO	DIESEL OIL	
BW	H.P. & L.P. BOILER FEED WATER		FG	H.P. & L.P. FUEL GAS	
CW	COOLING WATER	G-65-D-0-1018-02	SY	CLOSED DRAIN	
HS	H.P. STEAM		WY	OPEN DRAIN	
MS	M.P. STEAM		RW	DILY STORM WATER	
LS	L.P. STEAM		HF	H.P. FLARE	
MC	M.P. STEAM COND.		LF	L.P. FLARE	
LC	L.P. STEAM COND.		CF	DRY/COLD FLARE	
LL	LOW LOW PRES. STEAM		AF	ACID BLOWDOWN	G-65-D-0-1018-04
SC	SUSPECT CONDENSATE		NW	NEUTRALIZATION WATER	
FF	FIREFIGHTING FOAM		CB	COLD DRAIN	
JA	HYDROC. ACID SOLUTION		AM	AMINE	G-65-D-0-1018-06
JB	CAUSTIC SODA SOLUTION				

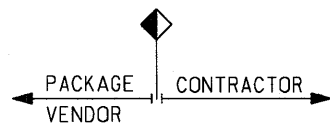
NOTES

- FOR SYMBOLS, LEGEND, TYPICAL AND GENERAL NOTES SEE P&ID'S: G-00-D-0-1000-01/02/03/04/05/06/07/08/09/10/11/12/13/14/15/16/17/18/19/25.
- FOR GENERAL SHUTDOWN LOGIC SEE P&ID'S: G-65-D-0-1000-20/21/22/23/24.
- FOR PIPING ASSEMBLY (I.E. 4D) REFER TO DOC: G-65-S-P-0028-00

- 1- UNLESS OTHERWISE SPECIFIED, ALL INSTRUMENTS AND EQUIPMENTS SHOWN ON THIS DRAWING ARE PREFIXED WITH THE UNIT IDENTIFICATION NUMBER "51-330"
- 2- DELETED
- 3- DELETED
- 4- THE CONTROL OF THE UNIT WILL BE PERFORMED THROUGH DCS IN CONTROL ROOM. AN OPERATOR INTERFACE PANEL WILL BE PROVIDED BY MFR FOR LOCAL INSTALLATION.
- 5- DELETED
- 6- DELETED
- 7- DELETED
- 8- DELETED
- 9- DELETED
- 10- FOR PACKAGE DETAIL SEE VENDOR DRAWING 13595-65-D-N-0010-00 SH 1&2

HOLD INDEX

- 1- DELETED
- 2- DELETED
- 3- DELETED.

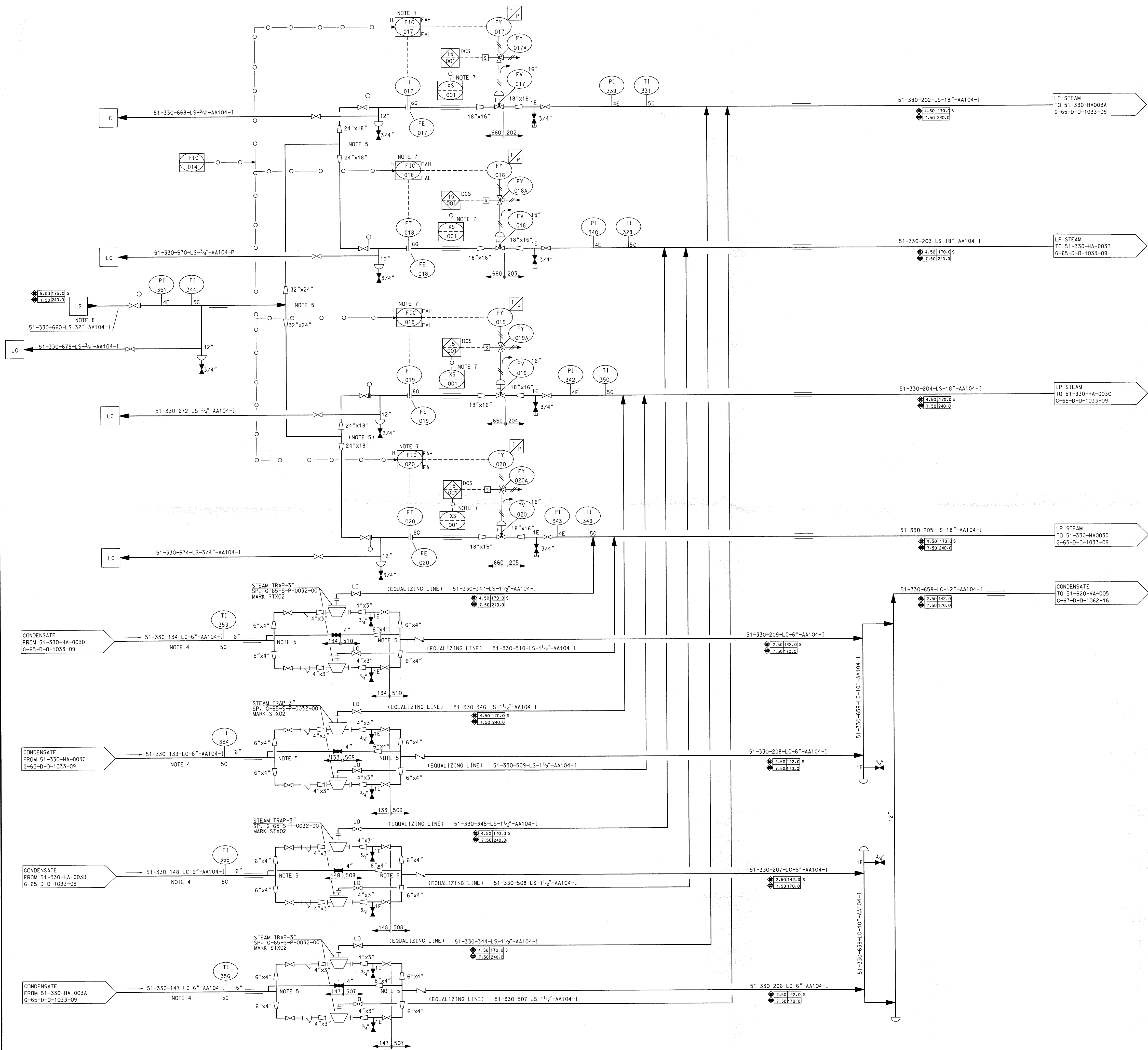


EQUIPMENT LIST

XX001					

EQUIPMENT SERVICE		51-330-XX-001 LEAN AMINE FILTRATION PACKAGE
DESIGN FLOWRATE	206 (M3/H)	
OPERATING TEMP./PRESS.	48/8.6 (C/BARG)	
DESIGN TEMP./PRESS.	75/16.5 (C/BARG)	
WEIGHT	N.A. (KG)	
DATA SHEET No.	G-65-X-D-33XX-01	

E1	30/08/07	AS BUILT	CAREDDU	D.POLIDORO	LA MANTIA	A.MADDALONI	T.M.SAID
A3	29/10/04	ISSUED FOR CONSTRUCTION REVISED WHERE INDICATED	CAREDDU	D.POLIDORO	LA MANTIA	A.MADDALONI	T.M.SAID
A2	16/04/04	ISSUED FOR CONSTRUCTION	CAREDDU	D.POLIDORO	LA MANTIA	A.MADDALONI	T.M.SAID
A1	31/10/03	ISSUED FOR CONSTRUCTION	CAREDDU	D.POLIDORO	LA MANTIA	A.MADDALONI	T.M.SAID
P2	30/05/03	ISSUED FOR DESIGN	ALVESTAD	D.POLIDORO	LA MANTIA	A.MADDALONI	T.M.SAID
P1	18/02/03	ISSUED FOR REVIEW	MD	AL	GL		
Revision	DATE	DESCRIPTION	PREP.	CHECK.	APPR.	PMS CHECK.	AGBV APPR.
AgipGas BV LIBYAN BRANCH			WESTERN LIBYA GAS PROJECT MELLITAH PLANT GSP/LAJ			AGBV DWG No. G-65-D-0-1033-13	
Snamprogetti Lummus Gas Ltd Snamprogetti ABB HYUNDAI						AGBV JOB No.0177/NC02	
TITLE: PIPING & INSTRUMENT DIAGRAM UNIT 330 GAS SWEETENING LEAN AMINE FILTRATION						CTR DWG No. 51-330-GD-B-03720	
						CTR JOB NO. 310300	
						Revision	
						IN REPLACEMENT OF REPLACED BY	
						G-65-D-0-1033-13-E1	



UTILITIES IDENTIFICATION					
CODE	DESCRIPTION	REF. DWG	CODE	DESCRIPTION	REF. DWG
DW	DRINKING WATER		CA	COMPRESS. AIR	
FW	FIRE-FIGHTING WATER		IA	INSTRUMENT AIR	
SW	SEA WATER		UA	UTILITIES AIR	
TW	DEMIN. WATER		UN	INERT GAS (NITROGEN)	
UW	DESALT. WATER		GD	DIESEL OIL	
BW	H.P. & L.P. BOILER FEED WATER		FG	H.P. & L.P. FUEL GAS	
CW	COOLING WATER		SY	CLOSED DRAIN	
HS	H.P. STEAM		WY	OPEN DRAIN	
MS	M.P. STEAM		RW	OILY STORM WATER	
LS	L.P. STEAM		HF	H.P. FLARE	
MC	M.P. STEAM COND.		LF	L.P. FLARE	
LC	L.P. STEAM COND.		CF	DRY/COLD FLARE	
LL	LOW LOW PRES. STEAM		AF	ACID BLOWDOWN	
SC	SUSPECT CONDENSATE		NW	NEUTRALIZATION WATER	
FF	FIRE FIGHTING FOAM				
JA	HYDROC. ACID SOLUTION				
JB	CAUSTIC SODA SOLUTION				

NOTES

1- FOR SYMBOLS, LEGEND, TYPICAL AND GENERAL NOTES SEE P&ID'S:
G-00-D-0-1000-01/02/03/04/05/06/07/08/09/10/11/12/13/14/15/16/17/18/19/25.

2- FOR GENERAL SHUTDOWN LOGIC SEE P&ID'S: G-65-D-0-1000-20/21/22/23/24.

3- FOR PIPING ASSEMBLY (i.e. 4D) REFER TO DOC: G-65-S-P-0028-00

- 1- UNLESS OTHERWISE SPECIFIED, ALL INSTRUMENTS AND EQUIPMENTS SHOWN ON THIS DRAWING ARE PREFIXED WITH THE UNIT IDENTIFICATION NUMBER "51-330".
- 2- DELETED.
- 3- DELETED.
- 4- NO HIGH POCKET ABOVE REBOILER BOTTOM T.L.
- 5- SYMMETRICAL PIPING.
- 6- DELETED.
- 7- ACTIVATION OF XS-001 DRIVES FIC-017/018/019/020 TO MANUAL MODE AND OUTPUT NULL.
- 8- STEAM IN SATURATED CONDITION.
- 9- DELETED.

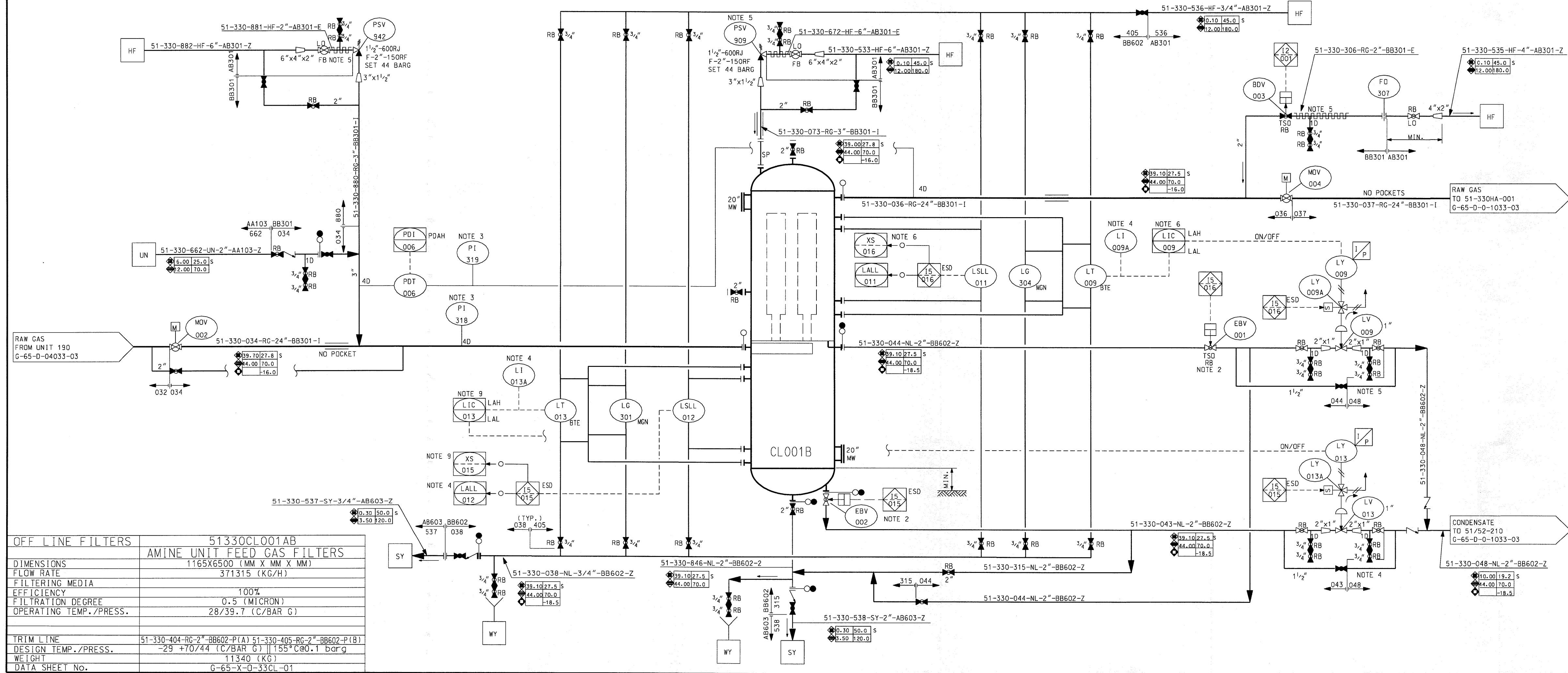
HOLD INDEX

1- DELETED.

EQUIPMENT LIST

NO.	DESCRIPTION	QTY	UNIT

E1 30/06/07 AS BUILT		D. POLI (DPO)		L.A. MANTIA (A. MADDALONI)		T.M. SAID	
A3 29/10/04 ISSUED FOR CONSTRUCTION REVISED WHERE INDICATED		D. POLI (DPO)		L.A. MANTIA (A. MADDALONI)		T.M. SAID	
A2 16/04/04 ISSUED FOR CONSTRUCTION		D. POLI (DPO)		L.A. MANTIA (A. MADDALONI)		T.M. SAID	
A1 31/10/03 ISSUED FOR CONSTRUCTION		D. POLI (DPO)		L.A. MANTIA (A. MADDALONI)		T.M. SAID	
P2 30/09/03 ISSUED FOR DESIGN		ALVESTAD		D. POLI (DPO)		L.A. MANTIA (A. MADDALONI)	
P1 18/02/03 ISSUED FOR REVIEW		MB		AL		GL	
Revision	DATE	DESCRIPTION	PREP.	CHECK.	APPR.	PMS CHECK.	AGBV APPR.
Agip Gas BV LIBYAN BRANCH			WESTERN LIBYA GAS PROJECT MELLITAH PLANT GSP LAJ			AGBV DWG No. G-65-D-0-1033-14 AGBV JOB No. 0177/NC02	
Snamprogetti Lummus Gas Ltd Snamprogetti			HYUNDAI			CTR DWG No. 51-330-CD-B-03721 CTR JOB NO. 310300	
TITLE: PIPING & INSTRUMENT DIAGRAM UNIT 330 - GAS SWEETENING STEAM & CONDENSATE HEADER			SCALE: 1:1			REPLACEMENT OF: REPLACED BY:	

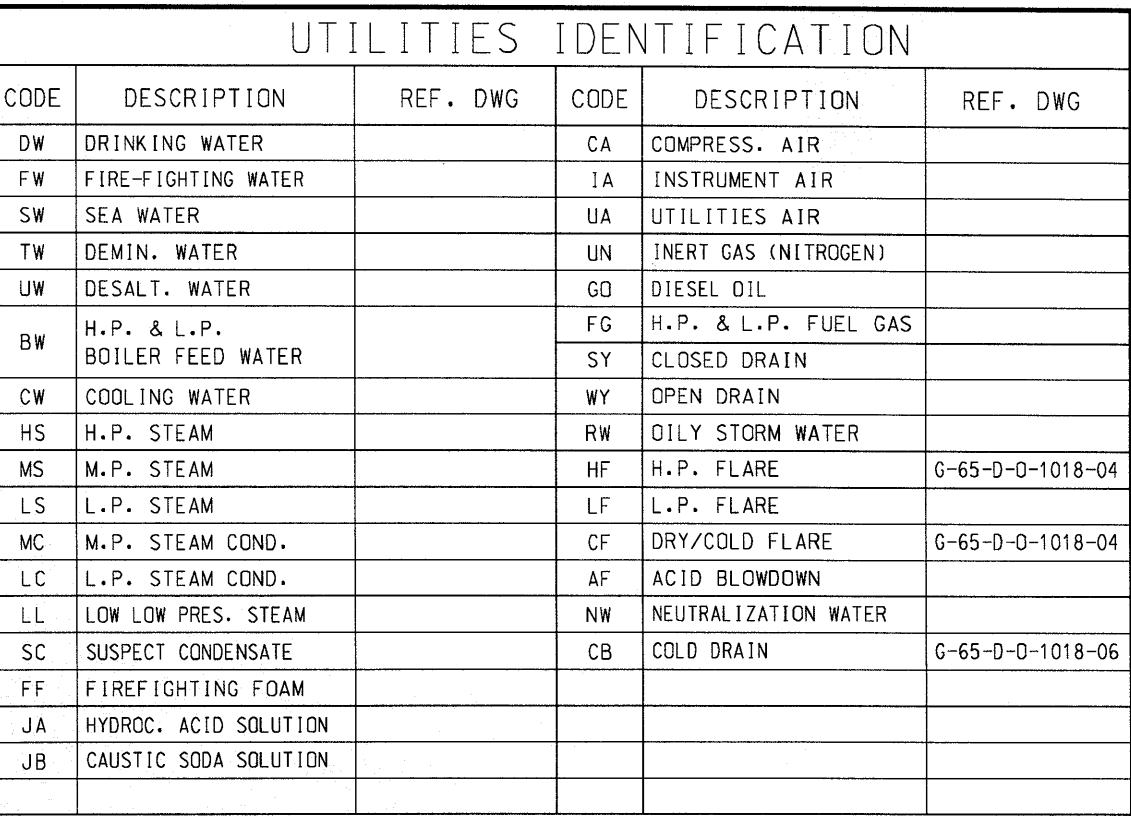


NOTES	
-	FOR SYMBOLS, LEGEND, TYPICAL AND GENERAL NOTES SEE P&ID'S: 0-00-0-0-1000-01/02/03/04/05/06/07/08/09/10/11/12/13/14/15/16/17/18/19/25.
-	FOR GENERAL SHUTDOWN LOGIC SEE P&ID'S:0-65-0-0-1000-20/21/22/23/24.
-	FOR PIPING ASSEMBLY (I.e., 4D) REFER TO DOC:G-65-S-P-0028-00

- 1- UNLESS OTHERWISE SPECIFIED, ALL INSTRUMENTS AND EQUIPMENTS SHOWN ON THIS DRAWING ARE PREFIXED WITH THE UNIT IDENTIFICATION NUMBER "51-330".
- 2- TO BE INSTALLED AT MIN. DISTANCE FROM EQUIPMENT NOZZLE.
- 3- PI TO BE VISIBLE FROM FILTER OPENING FACILITIES.
- 4- LEVEL INDICATOR TO BE READABLE FROM GLOBE VALVE.
- 5- PROVIDE ELECTRIC TRACING ON VALVE BODY AND PART OF THE OUTLET LINE TO FLARE. EXTENT OF TRACING TO BE 2100mm. FROM PSV-908/909, FROM PSV 941/942 AND 800mm FROM BDV-002/003.
- 6- ACTIVATION OF XS-016 DRIVES LIC-009 TO MANUAL MODE AND OUTPUT NULL.
- 7- ACTIVATION OF XS-013 DRIVES LIC-015 TO MANUAL MODE AND OUTPUT NULL.
- 8- ACTIVATION OF XS-014 DRIVES LIC-010 TO MANUAL MODE AND OUTPUT NULL.
- 9- ACTIVATION OF XS-015 DRIVES LIC-013 TO MANUAL MODE AND OUTPUT NULL.
- 10- CONTENT OF THIS DRAWING WAS PREVIOUSLY SHOWN IN P&ID N° G-65-D-G-1033-03.

EQUIPMENT LIST					
CL001A/B					

E1	30/08/07	AS BUILT		(P)	D.POL/DORG	LA MANT TA	A.MADALONI	T.M.SAIO
A3	29/10/04	ISSUED FOR CONSTRUCTION REVISED WHERE INDICATED	CAREDDU	D.POL/DORG	LA MANT TA	A.MADALONI	T.M.SAIO	
A2	16/04/04	ISSUED FOR CONSTRUCTION	CAREDDU	D.POL/DORG	LA MANT TA	A.MADALONI	T.M.SAIO	
Revision	DATE	DESCRIPTION	PREP.	CHECK.	APPR.	FMS CHECK.	AGDV APPR.	
Agip Gas BV LIBYAN BRANCH			WESTERN LIBYA GAS PROJECT MELLITA PLANT GSP/LAJ			AGBV DWG No. G-65-D-0-1033-15 AGBV JOB NO.0177N002		
 Snamprogetti   HYUNDAI ENGINEERING CORPORATION CO., LTD.			CTR DWG No. V1-330-GD-B-03722 Sh. _____ of _____					
TITLE: PIPING & INSTRUMENT DIAGRAM UNIT 330 - GAS SWEETENING RAW GAS INLET FILTERS (NOTE 10)			CTR JOB NO. 310300 Revision <u> 1 </u> of <u> 1 </u> SCALE: _____ IN REPLACEMENT OF: _____ REPLACED BY: _____					



- 1- UNLESS OTHERWISE SPECIFIED, ALL INSTRUMENTS AND EQUIPMENTS SHOWN ON THIS DRAWING ARE PREFIXED WITH THE UNIT IDENTIFICATION NUMBER "51-340".
- 2- ELEVATION ABOVE HA-002 INLET NOZZLE.
- 3- ALARM TO BE ACTIVATED WHEN VALVE IS NOT CLOSED. (LIMIT SWITCH TO BE LOCATED IN VALVE CLOSE POSITION)
- 4- HIGH AND LOW VALUES ALARMS ARE FOR WOBBE INDEX AND GHV.
- 5- PROVIDE SYMMETRICAL PIPING DISTRIBUTION.
- 6- T1 TO BE INSTALLED AT MIN. DISTANCE FROM EXCHANGER NOZZLE.
- 7- SHUTDOWN VALVE IS ALLOWED TO OPEN ONLY WHEN THE DIFFERENTIAL PRESSURE DROP IS LESS THAN 3 bar.
- 8- PI-302 TO BE READABLE FROM PV-010 HANDWHEEL.
- 9- ALARM ACTIVATED WHEN VALVE IS FULLY OPEN.

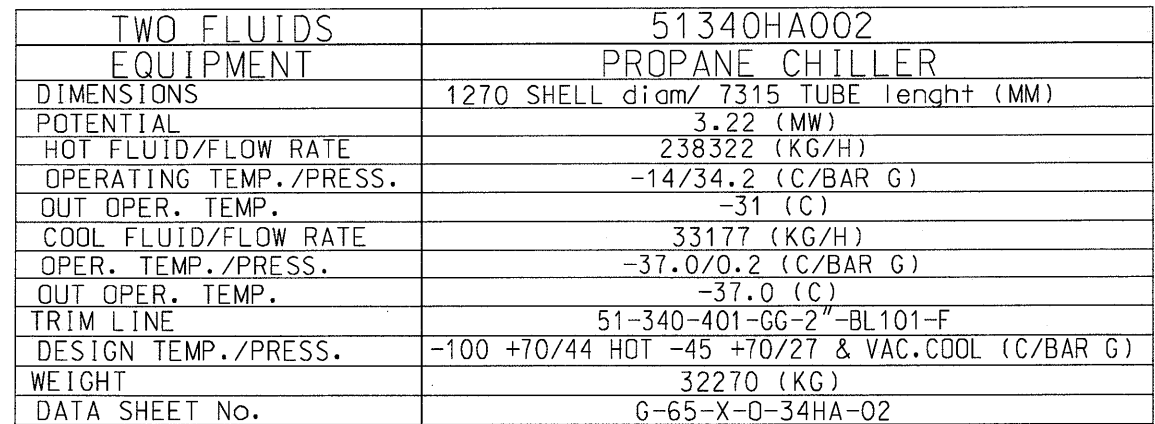
1-DELETED.
2- DELETED.
3- DELETED.

HA001A + F				

A1	10/08/07	AS BUILT		POL	D.LORDO	G.LA MANTIA	A.MADALONI	T.M.SAID
E1		ISSUED FOR CONSTRUCTION REVISED WHERE INDICATED	CAREDDU	D.POL	D.LORDO	G.LA MANTIA	A.MADALONI	T.M.SAID
A2	16/04/04	ISSUED FOR CONSTRUCTION	CAREDDU	D.POL	D.LORDO	G.LA MANTIA	A.MADALONI	T.M.SAID
F1	17/10/03	ISSUED FOR CONSTRUCTION	CAREDDU	D.POL	D.LORDO	G.LA MANTIA	A.MADALONI	T.M.SAID
P2	30/05/03	ISSUED FOR DESIGN	ALVESTAD	D.POL	D.LORDO	G.LA MANTIA	A.MADALONI	T.M.SAID
R1	18/02/03	ISSUED FOR REVIEW	MD	AL	GL			
Revision	DATE	DESCRIPTION	PREP.	CHECK.	APPR.	PMS CHECK.	ACSV	APPR

AgipGas BV LIBYAN BRANCH	WESTERN LIBYA GAS PROJECT MELLITAH PLANT GSP/LAJ	AGBV DWG No. G-65-D-0-1034-01						
 Snamprogetti	Snamprogetti Lummus Gas Ltd  ABB AN ALSTOM COMPANY  HYUNDAI PIPELINE & CONSTRUCTION CO., LTD.	AGBV JOB No 0177/NC/02 CTR DWG No. 51-340-CD-B-03860 Scale of						
PIPING & INSTRUMENT DIAGRAM UNIT 340 - GAS HC DEWPOINT CONTROL DRY GAS COOLING		CTR JOB NO. 3103000 Revision: <table border="1" data-bbox="2813 1955 2905 1965"> <tr> <td>1</td> <td>2</td> <td>3</td> <td>4</td> <td>5</td> <td>6</td> </tr> </table> SCALE: IN REPLACEMENT OF: REPLACED BY:	1	2	3	4	5	6
1	2	3	4	5	6			

TWO FLUIDS EQUIPMENT		51340HA001AB0DEF
RAW/GAS TREATED GAS EXCH.		
DIMENSIONS	1170 SHELL diam/ 9150 TUBE length (MM)	
POTENTIAL	9.06 (MW)	
HOT FLUID/FLOW RATE	238322 (KG/H)	
OPERATING TEMP./PRESS.	38.6/34.7 (C/BAR G)	
Cool FLUID/FLOW RATE	-14 (C)	
COOL FLUID/FLOW RATE	215876 (KG/H)	
OPER. TEMP./PRESS.	-31/32.5 (C/BAR G)	
OUT OPER. TEMP.	34.6 (C)	
TRIM LINE		
DESIGN TEMP./PRESS.	-45/+70/44 HOT -45/+70/44 COOL (C/BAR G)	
WEIGHT	27930 EACH (KG)	
DATA SHEET NO.	6-65-X-0-34HA-01	



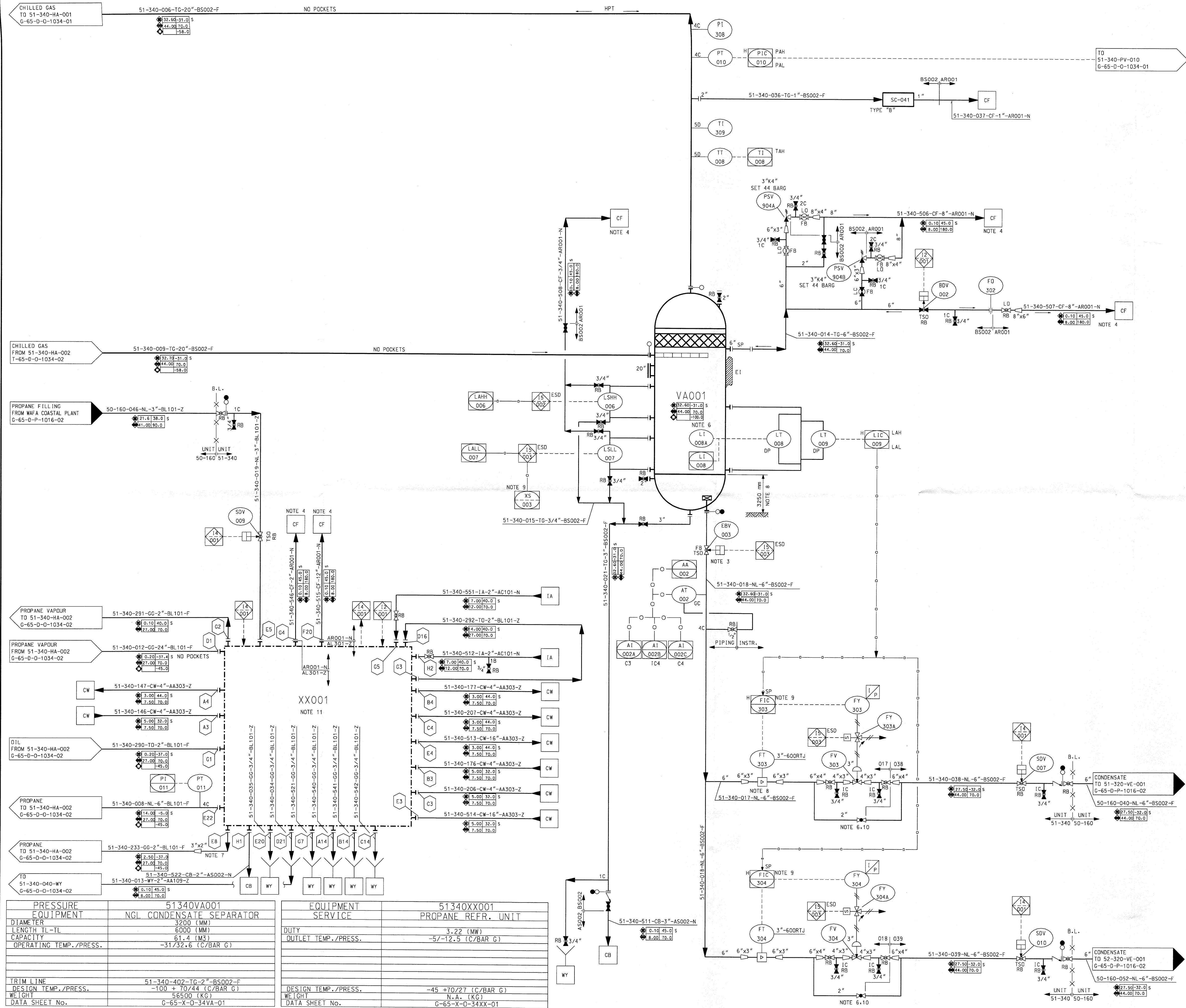
NOTES

- FOR SYMBOLS, LEGEND, TYPICAL AND GENERAL NOTES SEE P&ID'S:
G-00-D-0-1000-01/02/03/04/05/06/07/08/09/10/11/12/13/14/15/16/17/18/19/25.
- FOR GENERAL SHUTDOWN LOGIC SEE P&ID'S:G-65-D-0-1000-20/21/22/23/24.
- FOR PIPING ASSEMBLY (I.e. 4D) REFER TO DDC:G-65-S-P-0028-00

- 1-UNLESS OTHERWISE SPECIFIED, ALL INSTRUMENTS AND EQUIPMENTS SHOWN ON THIS DRAWING ARE PREFIXED WITH THE UNIT IDENTIFICATION NUMBER "51-340"
- 2- DELETED.
- 3- LV VALVE TO BE AT MINIMUM DISTANCE FROM 51-340-HA-002 AND AT THE SAME ELEVATION.
- 4- EBV VALVE TO BE INSTALLED AT MIN DISTANCE FROM HA-002-NOZZLE.
- 5- ELEVATION ABOVE VA-001 INLET NOZZLE.
- 6- DELETED.
- 7- DELETED.
- 8- CF HEADER IN SS INSIDE UNIT 340 DUE TO LOW DEPRESSURIZATION TEMPERATURE (BELOW -45°C)
SS HEADER TO BE CONNECTED ON KCS HEADER AFTER RELIEF HEATING UP.
- 9- LI TO BE VISIBLE FROM GLOBE VALVE.
- 10-PROVIDE SYMMETRICAL PIPING.
- 11-TO BE INSTALLED AT MINIMUM DISTANCE FROM HA-002 OUTLET NOZZLE.
- 12-CONSIDER FULL VACUUM CONDITION FOR THE LINE.

EQUIPMENT LIST				
HA002				

E1	30/08/07	AS BUILT		CHEUNG	D-POL.DORNOG-LLA MANTIA A-MADAGALI	T-W SAID								
A1	30/08/07	ISSUED FOR CONSTRUCTION REVISED WHERE INDICATED		CAREDDU	D-POL.DORNOG-LLA MANTIA A-MADAGALI	T-W SAID								
A2	16/04/04	ISSUED FOR CONSTRUCTION		CAREDDU	D-POL.DORNOG-LLA MANTIA A-MADAGALI	T-W SAID								
A1	17/10/03	ISSUED FOR CONSTRUCTION		CAREDDU	D-POL.DORNOG-LLA MANTIA A-MADAGALI	T-SAID								
P2	30/05/03	ISSUED FOR DESIGN		ALVESTAD	D-POL.DORNOG-LLA MANTIA A-MADAGALI	T-SAID								
P1	17/02/03	ISSUED FOR REVIEW		MD	AL	GL								
Revision	DATE	DESCRIPTION	PREP.	CHECK.	APPR.	PWS CHECK	AGBY APPROV.							
AgipGas BV LIBYAN BRANCH				WESTERN LIBYA GAS PROJECT MELLITAT PLANT GSPLAJ		AGBY DWG No. G-65-D-O-1034-02 AGBY JOB NO.D177NC/O2								
Snamprogetti Lummus Gas Ltd  Snamprogetti  ABB  HYUNDAI ABB LOW VOLTAGE DIVISION ENGINEERING & CONSTRUCTION INC. - U.S.A.				CTR DWG No. 51-340-GD-B-03861 Sp..... of.....										
PIPING & INSTRUMENT DIAGRAM UNIT 340 - GAS HC DEWPOINT CONTROL DRY GAS COOLING				CTR JOB NO. 310300 Revision <table><tr><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>ET</td></tr></table> IN REPLACEMENT OF REPLACED BY				1	2	3	4	5	6	ET
1	2	3	4	5	6	ET								



UTILITIES IDENTIFICATION					
CODE	DESCRIPTION	REF. DWG	CODE	DESCRIPTION	REF. DWG
DW	DRINKING WATER		CA	COMPRESS. AIR	
FW	FIRE-FIGHTING WATER		IA	INSTRUMENT AIR	G-65-D-0-1018-03
SW	SEA WATER		UA	UTILITIES AIR	
TW	DEMIN. WATER		UN	INERT GAS (NITROGEN)	
UW	DESALT. WATER		GD	DIESEL OIL	
BW	H.P. & L.P. BOILER FEED WATER		FG	H.P. & L.P. FUEL GAS	
CW	COOLING WATER	G-65-D-0-1018-02	SY	CLOSED DRAIN	
HS	H.P. STEAM		WY	OPEN DRAIN	
MS	M.P. STEAM		RW	DILY STORM WATER	
LS	L.P. STEAM		HF	H.P. FLARE	
MC	M.P. STEAM COND.		LF	L.P. FLARE	
LC	L.P. STEAM COND.		CF	DRY/COLD FLARE	G-65-D-0-1018-04
LL	LOW LOW PRES. STEAM		AF	ACID BLOWDOWN	
SC	SUSPECT CONDENSATE		NW	NEUTRALIZATION WATER	
FF	FIREFIGHTING FOAM		CB	COLD DRAIN	G-65-D-0-1018-06
JA	HYDROC. ACID SOLUTION				
JB	CAUSTIC SODA SOLUTION				

- NOTES
- FOR SYMBOLS, LEGEND, TYPICAL AND GENERAL NOTES SEE P&ID'S:
G-00-D-0-1000-01/02/03/04/05/06/07/08/09/10/11/12/13/14/15/16/17/18/19/25.
- FOR GENERAL SHUTDOWN LOGIC SEE P&ID'S: G-65-D-0-1000-20/21/22/23/24.
- FOR PIPING ASSEMBLY (i.e. 40) REFER TO DDC:G-65-S-P-0028-00
- 1- UNLESS OTHERWISE SPECIFIED, ALL INSTRUMENTS AND EQUIPMENTS SHOWN ON THIS DRAWING ARE PREFIXED WITH THE UNIT IDENTIFICATION NUMBER "51-340"
- 2 - DELETED.
- 3 - EBV VALVE TO BE INSTALLED AT MINIMUM DISTANCE FROM VESSEL NOZZLE.
- 4 - CF HEADER IN SS INSIDE UNIT 340 DUE TO LOW DEPRESSURIZATION TEMPERATURE (BELOW -45 C)
SS HEADER TO BE CONNECTED ON KCS HEADER AFTER RELIEF HEATING UP.
- 5 - DELETED.
- 6 - LI TO BE VISIBLE FROM GLOBE VALVE.
- 7 - HOT GAS BY-PASS LINE.
- 8 - FLOW ELEMENT TO HAVE MIN. PRESSURE DROP IN ORDER TO AVOID ANY CHANGE OF PHASE. (DP max 125 mm H2O). ELEVATION OF 340-VA001 B.T.L. TO BE MINIMUM 1800mm ABOVE FLOW ELEMENT.
- 9 - ACTIVATION OF XS-003 DRIVES FIC-303/304 TO MANUAL MODE AND OUTPUT NULL.
- 10 - CV OF BY-PASS VALVE NOT TO EXCEED CV OF SELECTED CONTROL VALVE.
- 11 - FOR PACKAGE DETAIL SEE THE FOLLOWING P&ID'S:
12744-65-D-N-1034-01/02/03/04/05/06/07/08/09/10/11/12/13/14.

- HOLD INDEX
- 1- DELETED.
2- DELETED.
3- DELETED.
4- DELETED.
5- DELETED.

EQUIPMENT LIST					
NO.	DESCRIPTION	UNIT	QTY	REMARKS	DATE
VA001	NGL CONDENSATE SEPARATOR		1		
XX001	PROPANE REFR. UNIT		1		

E1	30/08/07	AS BUILT	Q1000	D. POLIDORO	G. LA MANTIA	M. MADDALONI	T.M. SAID
A3	18/10/04	ISSUED FOR CONSTRUCTION	REVISÉ	CAREDDU	D. POLIDORO	G. LA MANTIA	M. MADDALONI
A2	16/04/04	ISSUED FOR CONSTRUCTION	REVISÉ	CAREDDU	D. POLIDORO	G. LA MANTIA	M. MADDALONI
A1	17/10/03	ISSUED FOR CONSTRUCTION	REVISÉ	CAREDDU	D. POLIDORO	G. LA MANTIA	M. MADDALONI
P2	30/06/03	ISSUED FOR DESIGN	REVISÉ	ALVESTAD	D. POLIDORO	G. LA MANTIA	M. MADDALONI
P1	18/02/02	ISSUED FOR DESIGN	REVISÉ	ALVESTAD	D. POLIDORO	G. LA MANTIA	M. MADDALONI

REVISION	DATE	DESCRIPTION	PREP.	CHECK.	APPR.	PWS CHECK.	AGBV APPR.

AgipGas BV LIBYAN BRANCH

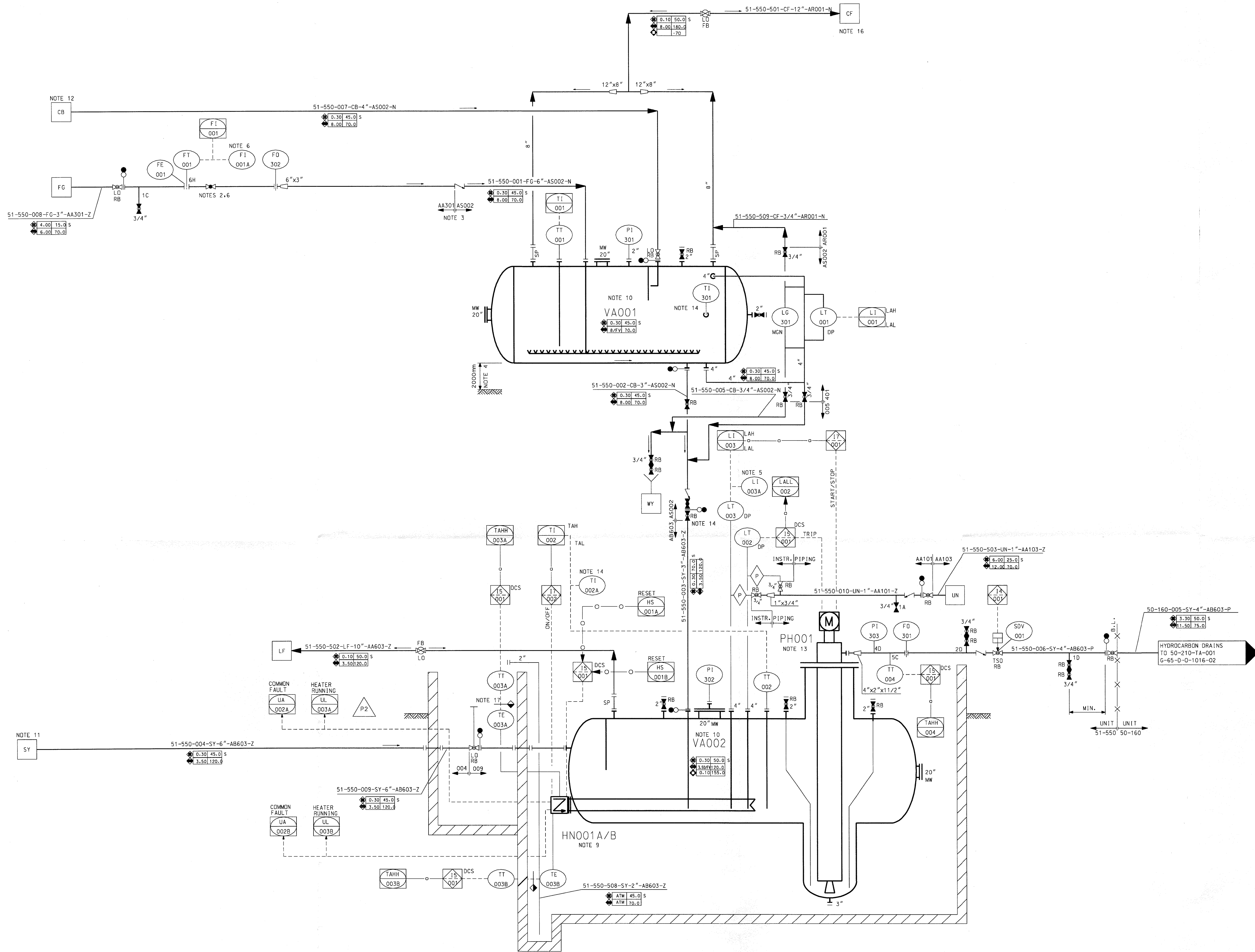
WESTERN LIBYA GAS PROJECT MELLITAH PLANT CSPLAJ

Snamprogetti Lummus Gas Ltd

HYUNDAI

PIPING & INSTRUMENT DIAGRAM
UNIT 340 - GAS HC DEWPOINT CONTROL
CONDENSATE SEPARATOR

AGBV DWG No. G-65-D-0-1034-03
AGBV JOB No. 0177/NCJ02
CTR DWG No. 51-340-GD-B-03862
CTR JOB NO. 310300
Revision: 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25
SCALE: _____
IN REPLACEMENT OF: _____
REPLACED BY: _____



UTILITIES IDENTIFICATION

CODE	DESCRIPTION	REF. DWG	CODE	DESCRIPTION	REF. DWG
DW	DRINKING WATER		CA	COMPRESS. AIR	
FW	FIRE-FIGHTING WATER		IA	INSTRUMENT AIR	
SW	SEA WATER		UA	UTILITIES AIR	
TW	DEMIN. WATER		UN	INERT GAS (NITROGEN)	G-65-D-0-1018-03
UW	DESALT. WATER		GO	DIESEL OIL	
BW	H.P. & L.P. BOILER FEED WATER		FG	H.P. & L.P. FUEL GAS	G-65-D-0-1018-07
CW	COOLING WATER		SY	CLOSED DRAIN	G-65-D-0-1018-06
HS	H.P. STEAM		RW	DILY STORM WATER	
MS	M.P. STEAM		WF	OPEN DRAIN	
LS	L.P. STEAM		LF	L.P. FLARE	G-65-D-0-1018-04
MC	M.P. STEAM COND.		CF	DRY/COLD FLARE	G-65-D-0-1018-04
LC	L.P. STEAM COND.		AF	ACID BLOWDOWN	
LL	LOW LOW PRES. STEAM		NW	NEUTRALIZATION WATER	
FF	FIREFIGHTING FOAM		AM	AMINE	
JA	HYDROC. ACID SOLUTION				
JB	CAUSTIC SODA SOLUTION				

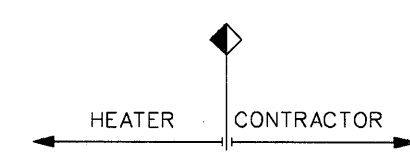
NOTES

- 1- FOR SYMBOLS, LEGEND, TYPICAL AND GENERAL NOTES SEE P&ID'S:
G-00-D-0-1000-01/02/03/04/05/06/07/08/09/10/11/12/13/14/15/16/17/18/19/25.
- 2- FOR GENERAL SHUTDOWN LOGIC SEE P&ID'S: G-65-D-0-1000-20/21/22/23/24.
- 3- FOR PIPING ASSEMBLY (I.e. 4D) REFER TO DOC: G-65-S-P-0028-00

- 1- UNLESS OTHERWISE SPECIFIED, ALL INSTRUMENTS AND EQUIPMENTS SHOWN ON THIS DRAWING ARE PREFIXED WITH THE UNIT IDENTIFICATION NUMBER "51-550"
- 2- VALVE TO BE LOCATED MINIMUM 1 METER ABOVE 51-550-VA-001 HIGH LEVEL.
- 3- CLASS BREAK TO BE LOCATED MINIMUM 1 METER FROM 51-550-VA-001.
- 4- MINIMUM ELEVATION TO ALLOW FREE DRAINING TO 51-550-VA-002.
- 5- LI TO BE READABLE FROM THE LOCAL START/STOP PUSH BUTTON.
- 6- FI-001A TO BE READABLE FROM GLOBE VALVE.
- 7- DELETED.
- 8- DELETED.
- 9- PIT TO BE SUITABLE FOR HN001A/B MAINTENANCE.
- 10- OPERATING PROCEDURE IN ORDER TO ENSURE THAT THE DRUM IS SAFELY OPERATED FOR MAINTENANCE CONDITION (IN/OUT CONNECTION BLOCKED IN VENT CONNECTION OPEN, ETC), TO BE PROVIDED.
- 11- WET DRAINS FROM GAS TRAIN AREA.
- 12- COLD DRAINS FROM GAS TRAIN AREA.
- 13- ONE COMMON SPARE FOR 51/52/53-550-PH001 IN WAREHOUSE.
- 14- TEMPERATURE WILL BE READABLE FROM BLOCK VALVE.
- 15- DELETED.
- 16- TO BE CONNECTED TO CF HEADER IN UNIT 340 AREA.
- 17- TO BE SUPPLIED BY HEATER VENDOR.

HOLD INDEX

- 1- DELETED.
- 2- DELETED.
- 3- DELETED.



EQUIPMENT LIST

HN001A/B	PH001	VA001	VA002

Revision	DATE	DESCRIPTION	PREP.	CHECK.	APPR.	PM'S CHECK.	AGBV APPR.
E1	30/08/07	AS BUILT					
A3	18/10/04	ISSUED FOR CONSTRUCTION	CAREDDU	D. POLI	DORO D. LA MANTIA	A. MADDALONI	T. M. SAID
A2	16/04/04	ISSUED FOR CONSTRUCTION	CAREDDU	D. POLI	DORO D. LA MANTIA	A. MADDALONI	T. M. SAID
A1	17/10/03	ISSUED FOR CONSTRUCTION	CAREDDU	D. POLI	DORO D. LA MANTIA	A. MADDALONI	T. M. SAID
P2	30/06/03	ISSUED FOR DESIGN	ALVESTAD	D. POLI	DORO D. LA MANTIA	A. MADDALONI	T. SAID
P1	18/02/03	ISSUED FOR REVIEW	MB	AL	GL		

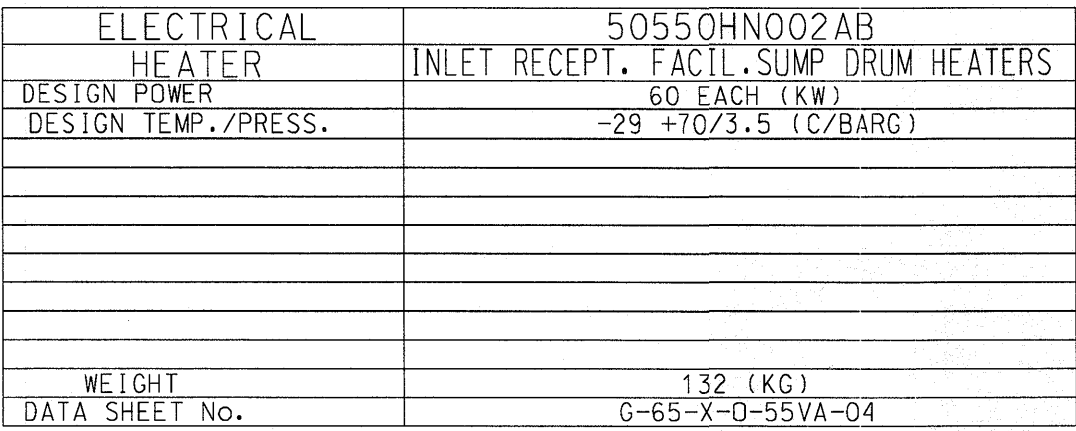
Agip Gas BV LIBYAN BRANCH	WESTERN LIBYA GAS PROJECT MELLITAH PLANT GSLPAJ	AGBV DWG No. G-65-D-0-1055-01 AGBV JOB No. 0177/NC02
Snamprogetti Lummus Gas Ltd Snamprogetti	ABB HYUNDAI	CTR DWG No. 51-550-6D-B-03890 CTR JOB No. 310300
PIPING & INSTRUMENT DIAGRAM UNIT 550 - CLOSED DRAIN GAS TRAINS AREA		Revision SCALE REPLACED BY

PUMPS	51550PH001
DIMENSIONS	WET HYDROCARBONS SUMP PUMP
FLOW RATE	815 DIA X 5055 (MMX MM)
PREVALENCE (DIFF. PRESS.)	13 (M3/H)
SPECIFIC WEIGHT	3 (BAR)
NPSH AVAILABLE/REQ.	0.6 / 1 MAX (KG/DM3)
DISCH. TEMP./PRESS.	SUBM. (M)
ABSORBED POWER & RATED FLOW	50/3.3 (C/BAR G)
	5.16 (KW)
DESIGN TEMP./PRESS.	120/10 (C/BAR G)
WEIGHT	417 (KG)
DATA SHEET No.	G-65-X-0-55PH-01

PRESSURE EQUIPMENT	51550VA001
DIMENSIONS	COLD HYDROCARB. DRAIN DRUM
DIAMETER	2200 (MM)
LENGTH TL-TL	5500 (MM)
CAPACITY	23.7 (M3)
OPERATING TEMP./PRESS.	45/0.3 (C/BAR G)
TRIM LINE	
DESIGN TEMP./PRESS.	51-550-401-CB-4"-AS002-N
	-100 +70/8+V (C/BAR G)
DATA SHEET No.	G-65-X-0-55VA-05

PRESSURE EQUIPMENT	51550VA002
DIMENSIONS	WET HYDROCARB. SUMP DRUM
DIAMETER	2500 (MM)
LENGTH TL-TL	6000 (MM)
CAPACITY	34.2 (M3)
OPERATING TEMP./PRESS.	50/0.3 (C/BAR G)
TRIM LINE	
DESIGN TEMP./PRESS.	51-550-402-SY-4"-AB603-P
	-29 +120/3.5 +FV (C/BAR G) 155°C 0.1 BAR G
DATA SHEET No.	G-65-X-0-55VA-02

ELECTRICAL	51550HN001AB
HEATER	WET HC SUMP DRUM HEATER
DESIGN POWER	60 EACH (KW)
DESIGN TEMP./PRESS.	-29 +120/3.5 (C/BAR G)
WEIGHT	132 (KG)
DATA SHEET No.	G-65-X-0-55VA-02



PUMPS	50550PH001
	COND. STABILIZATION AREA SUMP PUMP
DIMENSIONS	870 DIAx7265 (MM X MM)
FLOW RATE	29 (M3/H)
PREVALENCE (DIFF. PRESS.)	3.7 (BAR)
SPECIFIC WEIGHT	0.6 / 1 MAX (KG/DM3)
NPSH AVAILABLE /REQ.	0. SUBM. (M)
DISCH. TEMP./PRESS.	50/4 (C/BAR G)
ABSORBED POWER/RATED FLOW	12.71 (KW)
DESIGN TEMP./PRESS.	240/11.5 (C/BAR G)
WEIGHT	690 (KG)
DATA SHEET NO.	6-65-X-0-55PH-02

PUMPS		50550PH002
INLET RECEPT. FACIL. SUMP PUMP		
DIMENSIONS		870 DIAx7265 (MM X MM)
FLOW RATE		29 (M3/H)
PREVALENCE (DIFF. PRESS.)		3.7 (BAR)
SPECIFIC WEIGHT		0.6 / 1 MAX (KG/DM3)
NPSH AVAILABLE (REQ.)		SUBM.
DISCH. TEMP./PRESS.		45/4 (C/BAR G)
ABSORBED POWER/RATED FLOW		12.71 (KW)
DESIGN TEMP./PRESS.		85/11.5 (C/BAR G)
WEIGHT		690 (KG)
DATA SHEET No.		G-65-X-O-55PH-03

PRESSURE	50550VA001
EQUIPMENT	COND. STABILIZATION AREA SUMP DRUM
DIAMETER	3500 (MM)
LENGTH TL-TL	7000 (MM)
CAPACITY	79.3 (M3)
OPERATING TEMP./PRESS.	50/0.3 (C/BAR G)
TRIM LINE	50-550-406-SY-4"-AB603-7
DESIGN TEMP./PRESS.	-29 +240/3.5 + FT (C/BAR G) 155 @ 0.1 BARG
WEIGHT	19930 (KG)
DATA SHEET No.	C-65-X-0-55VA-03

PRESSURE EQUIPMENT	50550VA002
DIAMETER	3500 (MM)
LENGTH TL-TL	7000 (MM)
CAPACITY	79.5 (M3)
OPERATING TEMP./PRESS.	45/0.3 (C/BAR G)
TRIM LINE	50-550-407-SY-4"-AB603-7
DESIGN TEMP./PRESS.	-29 +70/3.5 (C/BAR G) H/15.0°C @ 0.1 BARG
WEIGHT	19320 (KG)
DATA SHEET No.	G-65-X-0-55VA-04

NOTES

- FOR SYMBOLS, LEGEND, TYPICAL AND GENERAL NOTES SEE P&ID'S:
G-00-D-0-1000-01/02/03/04/05/06/07/08/09/10/11/12/13/14/15/16/17/18/19/25.
- FOR GENERAL SHUTDOWN LOGIC SEE P&ID'S:G-65-D-0-1000-20/21/22/23/24.
- FOR PIPING ASSEMBLY (i.e. 4D) REFER TO DOC:G-65-S-P-0028-00

- 1-UNLESS OTHERWISE SPECIFIED, ALL INSTRUMENTS AND EQUIPMENTS SHOWN ON THIS DRAWING
ARE PREFIXED WITH THE UNIT IDENTIFICATION NUMBER "50-550"
- 2-DELETED.
- 3-L1 TO BE READABLE FROM THE LOCAL START/STOP PUSH BUTTON.
- 4-DRAINS FROM CONDENSATE STABILIZATION AREA.
- 5-DRAINS FROM SLUG CATCHER AREA.
- 6-ONE SAFETY PUMP COMMON TO PH001/PH002 IN WAREHOUSE.
- 7- DELETED.
- 8-OPERATING PROCEDURE IN ORDER TO ENSURE THAT THE DRUM IS SAFELY
OPERATED FOR MAINTENANCE CONDITION (IN/OUT CONNECTION BLOCKED IN
VENT CONNECTION OPEN, ETC). TO BE PROVIDED.
- 9-PIT TO BE SUITABLE FOR HN001 A/B AND HN002A/B MAINTENANCE.
- 10-UTILITIES REFERENCE DRAWING
FOR INLET RECEPTION FACILITIES AREA:

UN: G-65-D-0-1019-53
SY: G-65-D-0-1019-56
LF: G-65-D-0-1019-54

FDR CONDENSATE STABILIZATION AREA:

UN: G-65-D-0-1021-60
SY: G-65-D-0-1021-63
LF: G-65-D-0-1021-61

HEATER

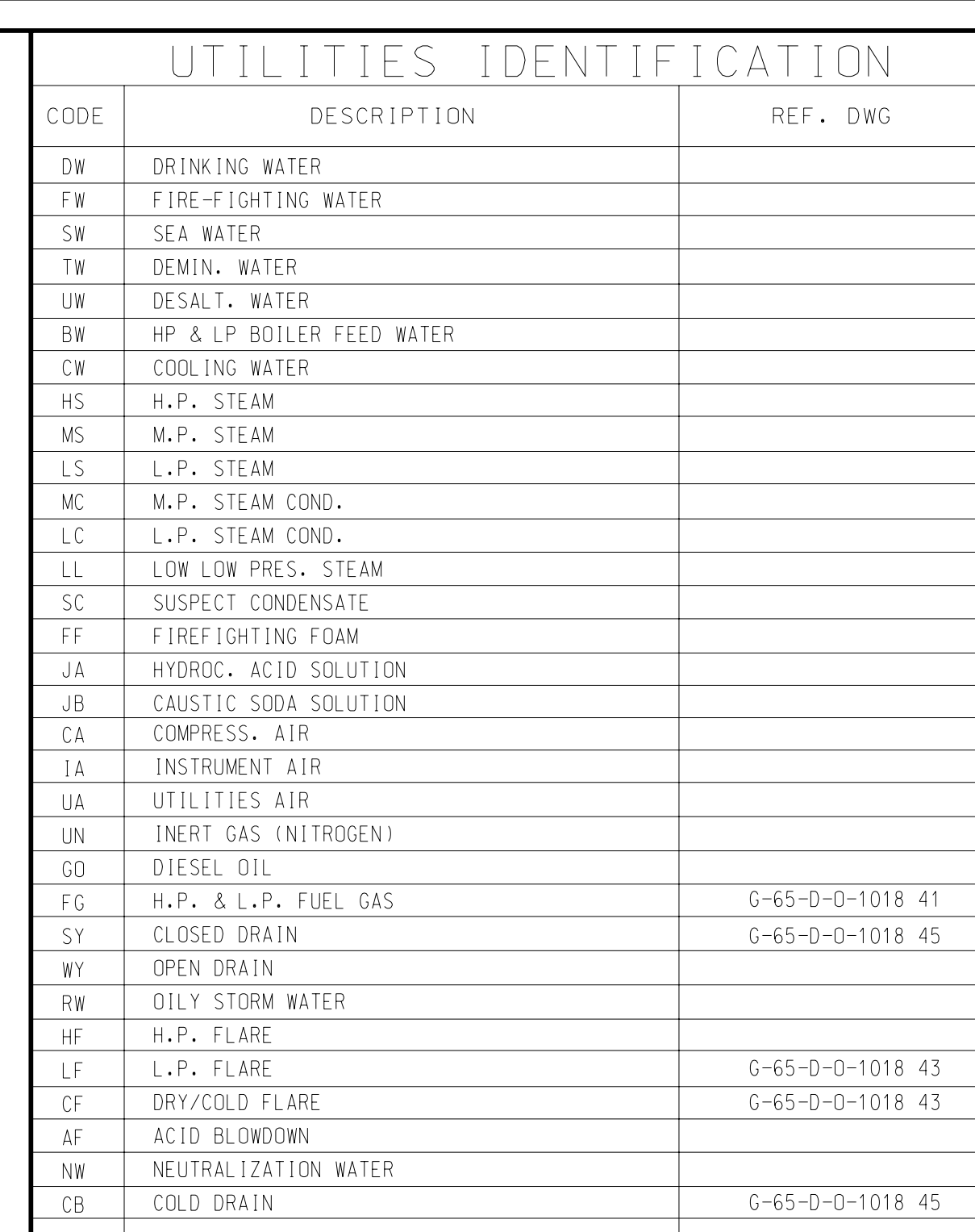
CONTRACTOR

VENDOR

[illegible]

E1	30/08/07	AS BUILT		CAREDO	D.POL	DIGRO	G.L.A	MANT	A.A	MADGALON	T.W.	SAID
A2	25/12/04	ISSUED FOR CONSTRUCTION REVISED WHERE INDICATED		CAREDO	D.POL	DIGRO	G.L.A	MANT	A.A	MADGALON	T.W.	SAID
A3	16/01/04	ISSUED FOR CONSTRUCTION		CAREDO	D.POL	DIGRO	G.L.A	MANT	A.A	MADGALON	T.W.	SAID
A1	17/10/03	ISSUED FOR CONSTRUCTION		CAREDO	D.POL	DIGRO	G.L.A	MANT	A.A	MADGALON	T.W.	SAID
P2	30/05/03	ISSUED FOR DESIGN		ALVESTAO	D.POL	DIGRO	G.L.A	MANT	A.A	MADGALON	T.W.	SAID
P1	18/02/03	ISSUED FOR REVIEW		MB	AL	DL						
Revision	DATE	DESCRIPTION		PREP.	CHECK.	APPR.	PMS CHECK.	ACBY	APPR.			

AgipGas BV LIBYAN BRANCH		WESTERN LIBYA GAS PROJECT MELIYAH PLANT GSPLAJ	AGBV DWG No. G-65-D-0-1055-02							
 Snamprogetti LUMMUS GAS LTD		 ABB LIBYAN BRANCH  HYUNDAI ENGINEERING & CONSTRUCTION CO., LTD.	AGBV JOB No. 17717NC02 CTR DWG No. 50-550-6D-B-03897 Sheet of							
TITLE: PIPING & INSTRUMENT DIAGRAM UNIT 50-550 - CLOSED DRAIN SLUG CATCHER AREA & CONDENSATE STAB. AREA			CTR JOB No. 310300 Revision: <table><tr><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td></tr></table>	1	2	3	4	5	6	7
1	2	3	4	5	6	7				
IN REPLACEMENT OF REPLACED BY:										

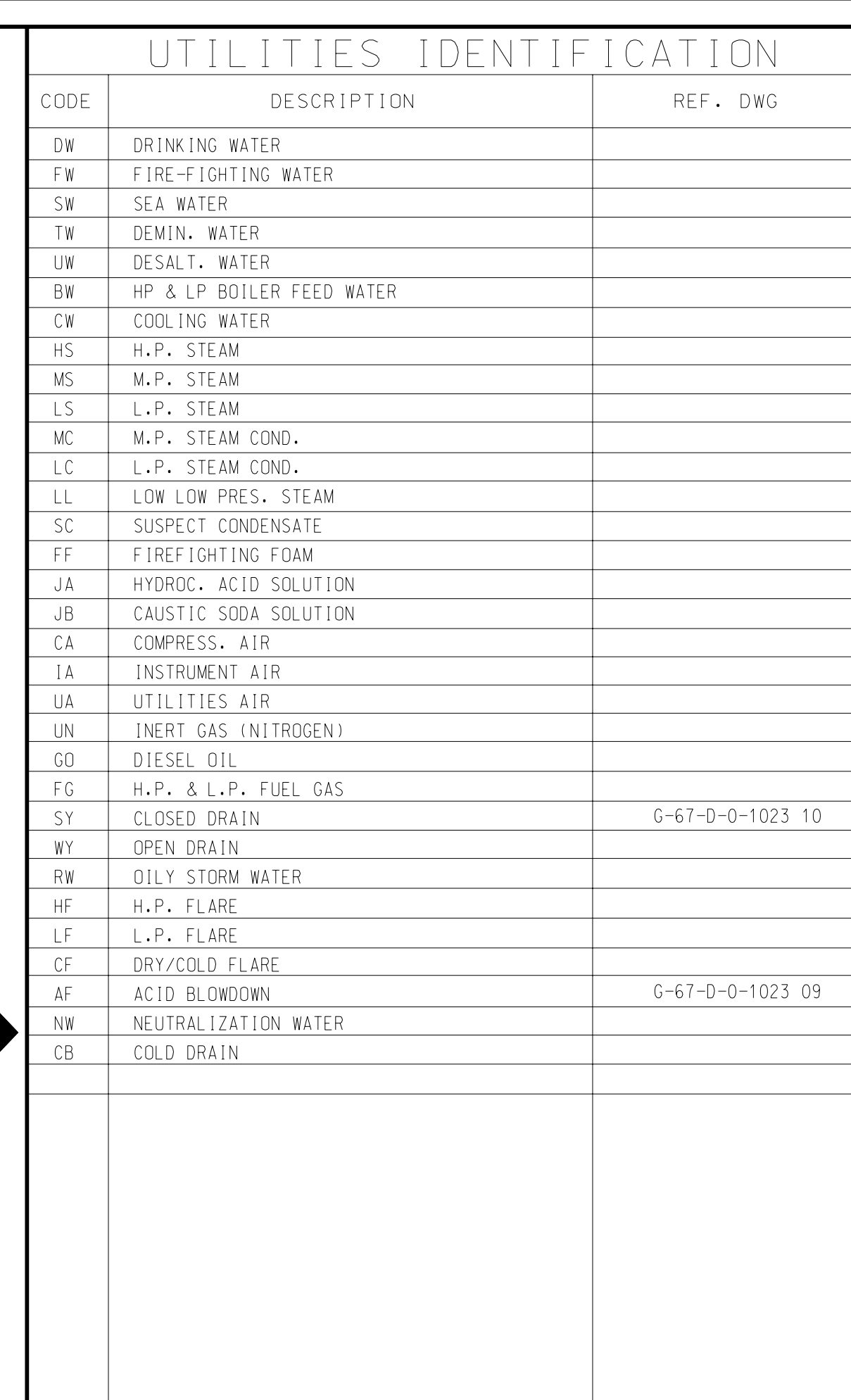


- FOR SYMBOLS, LEGEND, TYPICAL AND GENERAL NOTES SEE P & ID's:
G-00-0-0-1000-01/02/03/04/05/06/07/08/09/10/11/12/13/14/15/16/17/18/19/25.
- FOR GENERAL SHUTDOWN LOGIC SEE P & ID's G-65-0-0-1000-20/21/22/23/24.
- FOR PIPING ASSEMBLY (I.E. 40) REFER TO DOC No.: G-65-S-P-0028-00.

1. MINIMUM ELEVATION TO ALLOW FREE DRAINING TO 50-550-VA-004.
2. VALVE TO BE LOCATED MINIMUM 1 METER ABOVE 50-550-VA-003 HIGH LEVEL.
3. CLASS BREAK TO BE LOCATED MINIMUM 1 METER ABOVE 50-550-VA-003.
4. AT SAFE LOCATION.
5. LI TO BE READABLE FROM THE LOCAL START/STOP PUSH BUTTON.
6. FI TO BE READABLE FROM GLOBE VALVE.
7. DELETED.
8. DELETED.
9. DELETED.
10. OPERATING PROCEDURE IN ORDER TO ENSURE THAT THE DRUM IS SAFELY OPERATED FOR MAINTENANCE CONDITION (IN/OUT), CONNECTION BLOCKED IN, VENT CONNECTION OPEN, ETC.) TO BE PROVIDED.
11. HOT DRAINS FROM NGL FRACTIONATION AREAS.
12. COLD DRAINS FROM NGL FRACTIONATION AREAS.
13. ONE SPARE PUMP IN WAREHOUSE (50550PH0033).
14. TEMPERATURE WILL BE READABLE FROM THE BLOCK VALVE.
15. TWO ELECTRICAL HEATERS EACH OF 60 KW RATING TO BE PROVIDED.
16. OVER TEMPERATURE PROTECTION.
17. UNLESS OTHERWISE SPECIFIED, ALL INSTRUMENTS AND EQUIPMENT SHOWN ON THIS DRAWING ARE PREFIXED WITH THE UNIT IDENTIFICATION NUMBER "50-550".
18. VENT LINE TO FLARE FROM PUMP SEAL IS SHOWN ON P&ID
G-65-0-0-1018 4322.

E1	06-10-06	A5.BJ.W/L		QJJ	OS	WID	A1..MADALOS	I..W..SA
A2	29/04/04	ISSUED_FOR_CONSTRUCTION		..EVOGB..	..MID..	..MID..	A1..MADALOS	I..W..SA
A3	06/01/03	ISSUED_FOR_CONSTRUCTION		..EVOGB..	..SCBA..	..MID..	A1..MADALOS	I..W..SA
A3	28/10/04	ISSUED_FOR_CONSTRUCTION		..EVOGB..	..MID..	..MID..	A1..MADALOS	I..W..SA
E1	17/01/03	ISSUED_FOR_APPROVAL		..MWIT..	..EL..	..MWS..		
Revision	DATE	DESCRIPTION		PREP.	CHECK.	APPR.	PWS CHECK	AGBY APPR
AgipGas BV			WESTERN LIBYA GAS PROJECT			AGBY DWG NO		
LIBYAN BRANCH			MELITAH PLANT			G-65-D-01055_03		
			GSLPAJ			Sh 1 of 1		
						AGBY JOB NO.0177/NC/0		
Snamprogetti Lummus Gas Ltd						CTR DWG NO		
Snamprogetti			ABB			G6500105503-----		
400 Street West			HYUNDAI			Sh 1 of 1		
			ENGINEERING & CONSTRUCTION LTD, LLC					
PIPING & INSTRUMENT DIAGRAM						CTR JOB NO.04095		
CLOSED DRAIN						Revision		
NGL REACTIONATION AREA						P1 P2 A1 A2 A3 E1		
						SCALE		
						IN REPLACEMENT OF		
						REPLACED BY		

ELECTRICAL		50550HN003A/B	
HEATER		NGL FRACT. HOT HC SUMP DRUM HEAT.	
DESIGN POWER		60 EACH (KW)	
DESIGN TEMP./PRESS.		-29 +240/3.5 (°C/BAR G)	
DATA SHEET No.		G-65-X-0-55VA-06	



NOTES

HN004A/B	PH005	VA005		

PUMPS		50550PH005/S	
-		HP FLARE AREA SUMP PUMP	
DIMENSIONS		1MM X MM X MM)	
FLOW RATE		8 (M3/H)	
PREVALENCE (DIFF. PRESS.)		2,8 (BAR)	
SPECIFIC WEIGHT		0,71 (KG/DM3)	
NPSH AVAILABLE /REQ		47/2,2	
DISCH. TEMP./PRESS.		45/2,8 (C/BAR G)	
ABSORBED POWER		3,6 (KW)	
DESIGN TEMP./PRESS.		75/11,5 (C/BAR G)	
WEIGHT		414 (KG)	
DATA SHEET No.		T-65-X-D-55PH-04	

ELECTRICAL	50550HN004AB
HEATER	HP FLARE AREA SUMP DRUM HEATERS
DESIGN POWER	60 EACH (KW)
DESIGN TEMP./PRESS.	70/3.5 (C/BAR G)
DATA SHEET No.	T-65-X-0-55VA-07